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Random walks on decorated Galton-Watson trees

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Abstract

In this article, we study a simple random walk on a decorated Galton-Watson tree, obtained from a Galton-Watson tree by replacing each vertex of degree n with an independent copy of a graph G_n and gluing the inserted graphs along the tree structure. We assume that there exist constants $d, R \geq 1, v < \infty$ such that the diameter, effective resistance across and volume of G_n respectively grow like $n^{\frac{1}{d}}, n^{\frac{1}{R}}, n^v$ as $n \to \infty$. We also assume that the underlying Galton-Watson tree is critical with offspring tails decaying like $cx^{-\alpha}$ for some constant c and some $\alpha \in (1, 2)$. We establish the fractal dimension, spectral dimension, walk dimension and simple random walk displacement exponent for the resulting metric space as functions of α, d, R and v, along with bounds on the fluctuations of these quantities.

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1 Introduction

The purpose of this paper is to study random walks on a general class of so-called decorated Galton-Watson trees, informally obtained from classical Galton-Watson trees by replacing each vertex of degree n with a graph of boundary size n, and gluing these graphs along the underlying tree structure as indicated in Figure 1 below. In the planar case, we can alternatively construct these graphs from the looptrees of [CK14] by "filling in" the loops. Such decorated trees naturally describe the structure of a variety of critical statistical mechanics models on random planar maps; examples include critical percolation clusters [CK15, Ric18a], Fortuin-Kasteleyn models [BLR17], and quadrangulations with skewness [BR18]. More generally, this structure applies to critical Boltzmann maps in the dense phase [Ric18b], which are believed (and in several cases proved) to describe a wide range of statistical physics models on Brownian surfaces.

We are interested in the regime where the underlying Galton-Watson tree has a critical offspring distribution ξ satisfying $\xi([k, \infty)) \sim ck^{-\alpha}$ as $k \to \infty$, for some $\alpha \in (1, 2)$. The case $\alpha = 2$ is less interesting since in this case the metric space structure of the decorated tree does not differ substantially from that of the original undecorated tree (similarly for any finite variance offspring distribution). Analogous results will also hold on incorporating a slowly-varying function into the offspring tails.

In order to understand the long-term behaviour of the random walk, we will work with a critical Galton-Watson tree conditioned to survive, also known as Kesten's tree. Under the above assumptions on the offspring distribution, we denote this tree T_α^∞ and its root ρ. Given a sequence of connected, potentially random graphs $(G_n)_{n\geq 1}$, where each G_n is conditioned to have n boundary vertices, we construct our decorated tree from T_α^∞ by replacing each vertex $v \in T_\alpha^\infty$ with an independent copy of $G_{\deg v}$. We then uniquely identify each boundary vertex of G_n with an edge incident to v, in such a way as to respect any planarity restrictions, and then if $v \sim v'$ in T_α^∞ we glue their corresponding graphs at the two vertices identified with the edge (v, v'). G_n may be random (e.g. an Erdös-Rényi graph on n vertices) or deterministic

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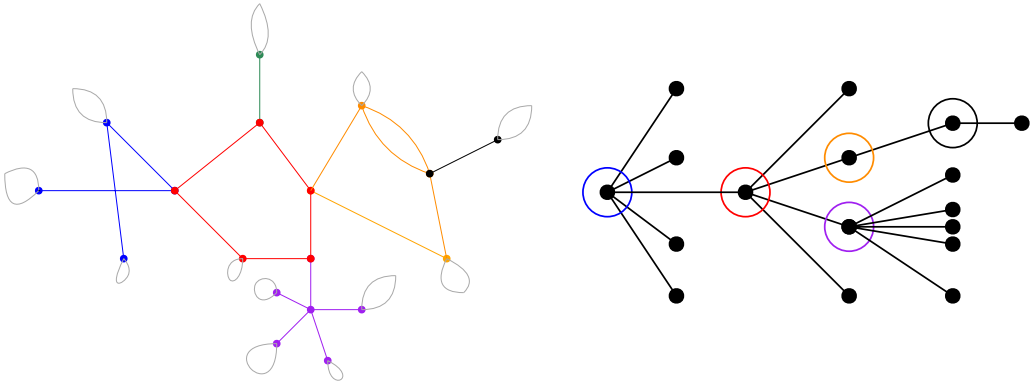


Figure 1: An example of a decorated tree, and its underlying Galton-Watson tree.

(e.g. the complete graph on n vertices); in the random case we sample independently for each vertex, conditional on the boundary size. We call the resulting structure $\mathcal{T}_\alpha^{\mathrm{dec}}$, and we let the vertex of $\mathcal{T}_\alpha^{\mathrm{dec}}$ that corresponds to the edge in T_α^∞ joining ρ to its leftmost child be the root of $\mathcal{T}_\alpha^{\mathrm{dec}}$, denoted $\rho_\alpha^{\mathrm{dec}}$. We further assume that the inserted graphs are each endowed with a measure and a metric, and extend these to a measure V^{dec} and a metric d_g^{dec} on $\mathcal{T}_\alpha^{\mathrm{dec}}$ by superposing the measures and adding distances along tree branches in the natural way. We will make the construction precise in Section 2.

Our aim is to establish the exponents governing the behaviour of a simple random walk $(X_n)_{n\geq 0}$ on $\mathcal{T}_\alpha^{\mathrm{dec}}$ in terms of α and the relevant exponents for the inserted graphs; in particular, to identify the following two quantities (and show that the limits exist):

(i) The spectral dimension, $d_S = -2\lim_{n\to\infty} \frac{\log p_{2n}(x,x)}{\log n}$,

(ii) The displacement exponent, $d_{\mathrm{dis}} = \lim_{n\to\infty} \frac{\log \sup_{k\leq n} d(X_0,X_k)}{\log n}$.

Here $p_n(x,y) = \frac{\mathbb{P}_x(X_n=y)}{\deg y}$ is the transition density of the simple random walk; it is easy to see that for recurrent graphs, the limits do not depend on the choice of x or X_0. The quantity $d_w = \frac{1}{d_{\mathrm{dis}}}$ is also known as the walk dimension, and can be naturally compared to the fractal dimension of $\mathcal{T}_\alpha^{\mathrm{dec}}$, denoted d_f, given by

$$d_f = \lim_{r\to\infty} \frac{\log\left(V^{\mathrm{dec}}(B(x,r))\right)}{\log r},$$

where $B(x,r)$ denotes the open ball of radius r centred at x, defined with respect to the metric d_g^{dec}. We will see that, as is commonly the case for sufficiently homogeneous graphs, the relation

$$d_S = \frac{2d_f}{d_w}$$

holds on $\mathcal{T}_\alpha^{\mathrm{dec}}$. This will hold as a consequence of verifying the conditions of [KM08].

It is well-known that these simple random walk exponents are determined by two key properties of the ambient graph: its effective resistance and volume growth. Most of the paper is therefore devoted to a study of these two properties of $\mathcal{T}_\alpha^{\mathrm{dec}}$, and as a result, we also determine d_f as defined above, which is of independent interest.

Given $n \in \mathbb{N}$, we define the following notation.

- G_n denotes a copy of the inserted graph conditioned on having n boundary vertices,
- $(U_n^{(1)}, U_n^{(2)})$ denotes a uniform pair of distinct points on the boundary of G_n,
- d_n is a given metric on G_n (e.g. the graph metric), and $d_n^U = d_n(U_n^{(1)}, U_n^{(2)})$,

- R_{eff} denotes effective resistance on G_n when each edge has conductance 1, and $R_n^U = R_{\text{eff}}(U_n^{(1)}, U_n^{(2)})$,
- $\mathsf{Diam}(G_n) = \sup_{x,y\in G_n} d_n(x,y)$ and $\mathsf{Diam}_{\text{res}}(G_n) = \sup_{x,y\in G_n} R_{\text{eff}}(x,y)$,
- $V_n^{\text{dec}}(G_n) = \sum_{x\in G_n} V_n^{\text{dec}}(x)$, where V_n^{dec} is a pre-defined measure on G_n (e.g. $V_n^{\text{dec}}(x) = \deg x$).

In what follows, we assume that the decorated tree $\mathcal{T}_\alpha^{\text{dec}}$ is defined on the probability space $(\mathbf{\Omega}, \mathcal{F}, \mathbf{P})$, and let $\mathbb{P}(\cdot)$ denote the (quenched) law of a simple random walk on $\mathcal{T}_\alpha^{\text{dec}}$, started from the root. For the volume results we will make the following assumption on these quantities.

Assumption 1.1.
(D) Metric growth. *There exists $d \geq 1$ and $\varepsilon > 0$ such that, as $\lambda \to \infty$,*

$$\liminf_{n\to\infty} \mathbf{P}\Big(d_n^U \geq n^{\frac{1}{d}}\Big) > 0, \qquad \limsup_{n\to\infty} \mathbf{P}\Big(\mathit{Diam}(G_n) \geq \lambda n^{\frac{1}{d}}\Big) = O(\lambda^{-([d(\alpha-1)\wedge 1]+\varepsilon)}).$$

(R) Resistance growth. *There exists $R \geq 1, \varepsilon > 0$ such that, as $\lambda \to \infty$.*

$$\liminf_{n\to\infty} \mathbf{P}\Big(R_n^U \geq n^{\frac{1}{R}}\Big) > 0, \qquad \limsup_{n\to\infty} \mathbf{P}\Big(\mathit{Diam}_{res}(G_n) \geq \lambda n^{\frac{1}{R}}\Big) = O(\lambda^{-([R(\alpha-1)\wedge 1]+\varepsilon)}).$$

(V) Volume growth. *There exists $v \geq 1, \varepsilon > 0$ such that, as $\lambda \to \infty$,*

$$\liminf_{n\to\infty} \mathbf{P}\big(V_n^{dec}(G_n) \geq n^v\big) > 0, \qquad \limsup_{n\to\infty} \mathbf{P}\big(V_n^{dec}(G_n) \geq \lambda n^v\big) = O(\lambda^{-(\left[\frac{\alpha}{v}\wedge 1\right]+\varepsilon)}).$$

Under Assumption 1.1, we define the following key exponents.

$$s_\alpha^d = d(\alpha-1), s_\alpha^R = R(\alpha-1), s_\alpha^v = \tfrac{\alpha-1}{v} \text{ and } f_\alpha^v = \tfrac{\alpha}{v}.$$

Remark 1.2. *Assumption 1.1 is essentially what is required to ensure that inserted graphs of high volume or diameter correlate with high degree vertices in T_α^∞. (We could technically weaken it very slightly to an integrability condition, but have chosen to keep things simple). If this is not satisfied, then the exponents of interest for $\mathcal{T}_\alpha^{dec}$ additionally depend on the precise tail decay of these probabilities, and not just on α, v, R and d.*

Under Assumption 1.1, we will see in Section 4 that, roughly speaking, for a typical vertex v on the backbone of T_α^∞ and a typical vertex u not on the backbone of T_α^∞, with corresponding graphs $G(v), G(u)$, we have for large x that $cx^{-s_\alpha^d} \leq \mathbf{P}(\mathsf{Diam}(G(v)) \geq x) \leq Cx^{-(s_\alpha^d \wedge 1)}$ and $cx^{-f_\alpha^v} \leq \mathbf{P}\big(V^{\text{dec}}(G(u)) \geq x\big) \leq Cx^{-(f_\alpha^v \wedge 1)}$.

We define the decorated volume exponent as

$$d_\alpha^{\text{vol}} = \frac{\alpha(s_\alpha^d \wedge 1)}{(\alpha-1)(f_\alpha^v \wedge 1)}. \tag{1}$$

We also let $B_{\mathcal{T}_\alpha^{\text{dec}}}(\rho_\alpha^{\text{dec}}, r))$ denote a ball of radius r around the root of $\mathcal{T}^{\text{dec}}$ with respect to the metric d_g^{dec}. The main volume growth theorem is as follows.

Theorem 1.3 (Volume growth)**.** *Under Assumption 1.1, $d_f(\mathcal{T}_\alpha^{dec}) = d_\alpha^{vol}$, $\mathbf{P}$-almost surely. Moreover, define the constant*

$$b_1 = \begin{cases} 0 & \text{if } s_\alpha^d, f_\alpha^v \neq 1, \\ -d_\alpha^{vol} & \text{if } s_\alpha^d = 1, f_\alpha^v \neq 1, \\ 1 & \text{if } s_\alpha^d \neq 1, f_\alpha^v = 1, \\ 1 - d_\alpha^{vol} & \text{if } s_\alpha^d = f_\alpha^v = 1. \end{cases}$$

Then:

(i) There exists $c_1 < \infty, a_1 > 0$ such that for all $r, \lambda > 1$,

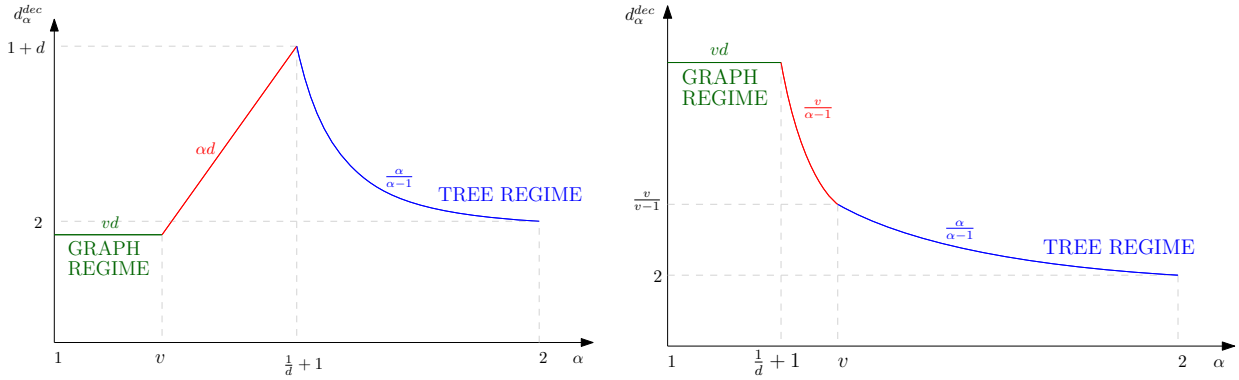
$$\mathbf{P}\Big(V^{dec}(B_{\mathcal{T}_\alpha^{dec}}(\rho_\alpha^{dec}, r)) \notin [r^{d_\alpha^{vol}}(\log r)^{b_1}\lambda^{-1}, r^{d_\alpha^{vol}}(\log r)^{b_1}\lambda]\Big) \leq c_1\lambda^{-a_1}.$$

(ii) There exists a finite, explicit constant $\beta_1 = \beta_1(d, \alpha, v)$, such that $\mathbf{P}$-almost surely,

$$r^{d_\alpha^{vol}}(\log r)^{-\beta_1} \leq V^{dec}(B_{\mathcal{T}_\alpha^{dec}}(\rho_\alpha^{dec}, r)) \leq r^{d_\alpha^{vol}}(\log r)^{\beta_1}$$

for all sufficiently large r.

These logarithmic fluctuations in part (ii) are not optimal in many cases, and in particular can often be improved to log-logarithmic when inserting deterministic graphs (although not always, T_α^∞ being an example where the upper fluctuations are genuinely logarithmic). However, although we make some comments on how the arguments can potentially be fine-tuned at appropriate parts of the proof, our emphasis here is on determining the correct leading terms for the volume growth, rather than the optimal fluctuations.



(a) If $\frac{1}{d}+1 \geq v$.

(b) If $\frac{1}{d}+1 \leq v$.

Figure 2: Different phases of the decorated volume exponent. Note that we do not necessarily see all three phases for one given model.

Note that the factor $\frac{\alpha}{\alpha-1}$ appearing in (1) corresponds to the volume growth exponent for the underlying tree T_α^∞. The factors of $s_\alpha^d \wedge 1$ and $f_\alpha^v \wedge 1$ appearing in (1) reflect the way that distances and volumes add up differently along branches and in subtrees in the decorated version $\mathcal{T}_\alpha^{\mathrm{dec}}$.

We have graphed the volume exponent in Figure 1, viewing v and d as fixed and varying α. There are two cases for the graph, depending on which of s_α^d and f_α^v exceeds one "first". In both cases, there are up to three regimes. The case where both of the exponents exceed 1 can be thought of as the "tree regime": in this case the relevant tails on the inserted graphs are not heavy enough to impact the exponents, so we see the same exponent appearing as for an undecorated tree. The case where both of the exponents are less than 1 can be thought of as the "graph regime", and we lose the dependence on α. This reflects the fact that as the offspring tails get heavier (i.e. as $\alpha \downarrow 1$), it is easier for a finite critical Galton-Watson tree to be "large" by having one vertex of macroscopic degree (cf [CK14, Proposition 3.6]), so that we essentially just see one macroscopic copy of the inserted graph in the decorated tree. In the case of a decorated Kesten's tree, we essentially just see a one-dimensional sequence of graphs glued along the backbone of T_α^∞. As $\alpha \uparrow 2$, however, the vertex degrees become more balanced, and the contribution from any one single vertex is less significant, so we regain some tree structure and eventually recover it entirely once the distance and volumes across typical inserted graphs have finite expectation.

We can also fix α and consider different sequences of graphs with differing values of v and d. As can be seen in (1), d_α^{vol} increases linearly with d (respectively v) up to the point at which the expected diameter (respectively volume) of a graph inserted at a typical vertex becomes finite, after which point we lose the dependence on d (respectively v).

We can use similar considerations to those discussed above to either establish the volume growth exponents with respect to the resistance metric, or otherwise add up resistance contributions along the backbone and

along paths in subtrees to compare resistance to the graph distance. Again, there are different regimes depending on whether resistance across a typical spinal vertex has finite expectation or not: as result, we will also see a factor of $s_\alpha^R \wedge 1$ in the exponents below. We can then combine the resistance and volume estimates using results of [KM08] to identify the random walk exponents.

In order to prove the volume results in Theorem 1.3, it is enough to be able to control distances to typical points in the decorated tree. In order to bound effective resistance in the decorated tree, we will need to bound the number of escape routes a random walk can take to exit a ball, so it will be necessary to simultaneously control distances along many branches of T_α^∞, for which we need a stronger assumption. For the purpose of understanding a simple random walk on $\mathcal{T}_\alpha^{\mathrm{dec}}$, we are also specifically interested in the measure on $\mathcal{T}_\alpha^{\mathrm{dec}}$ given by $\mu(x) = \deg x$ for all $x \in \mathcal{T}_\alpha^{\mathrm{dec}}$.

We therefore prove the main random walk results under the following, stronger, assumption.

Assumption 1.4.
(D′) Metric growth. *There exists $d \geq 1$ and $\varepsilon > 0$ such that*

$$\liminf_{n\to\infty} \mathbf{P}\Big(d_n^U \geq n^{\frac{1}{d}}\Big) > 0, \qquad \limsup_{n\to\infty} \mathbf{P}\Big(\mathsf{Diam}(G_n) \geq \lambda n^{\frac{1}{d}}\Big) = o(\lambda^{\frac{-\alpha}{\alpha-1}}).$$

(R′) Resistance growth. *There exists $R \geq 1, \varepsilon > 0$ such that*

$$\liminf_{n\to\infty} \mathbf{P}\Big(R_n^U \geq n^{\frac{1}{R}}\Big) > 0, \qquad \limsup_{n\to\infty} \mathbf{P}\Big(\mathsf{Diam}_{res}(G_n) \geq \lambda n^{\frac{1}{R}}\Big) = o(\lambda^{\frac{-\alpha}{\alpha-1}}).$$

(V′) Volume growth. *There exists $v \geq 1, \varepsilon > 0$ such that*

$$\liminf_{n\to\infty} \mathbf{P}(\mu(G_n) \geq n^v) > 0, \qquad \limsup_{n\to\infty} \mathbf{P}(\mu(G_n) \geq \lambda n^v) = O(\lambda^{\frac{-(\alpha+\varepsilon)}{v}}).$$

Remark 1.5. *Assumption 1.4(D′) is enough to prove that, if T_n is a Galton-Watson tree conditioned to have n vertices, with decorated version $\mathcal{T}_n^{dec}$, then the sequence $\left(n^{-\left[(1-\frac{1}{\alpha})\vee\frac{1}{\alpha d}\right]}\mathsf{Diam}(\mathcal{T}_n^{dec})\right)_{n\geq 1}$ is a tight sequence (with an extra scaling factor of $\log n$ when $d(\alpha-1)=1$). We do not expect this to be optimal in the regime where $d(\alpha-1)<1$, in which case tail decay of $O(\lambda^{-(\alpha d+\varepsilon)})$ is probably sufficient, as this is enough to ensure that the maximal diameter of any individual graph in $\mathcal{T}_n^{dec}$ is of order $n^{\frac{1}{\alpha d}}$, with high probability. However, the assumption is sharp in the regime $d(\alpha-1)>1$.*

We also set

$$\begin{aligned} d_\alpha^{\mathrm{spec}} &= \frac{2\alpha(s_\alpha^R \wedge 1)}{\alpha(s_\alpha^R \wedge 1) + (\alpha-1)(f_\alpha^v \wedge 1)}, \\ d_\alpha^{\mathrm{dis}} &= \frac{(s_\alpha^R \wedge 1)(\alpha-1)(f_\alpha^v \wedge 1)}{(\alpha-1)(f_\alpha^v \wedge 1)(s_\alpha^d \wedge 1) + \alpha(s_\alpha^R \wedge 1)(s_\alpha^d \wedge 1)}. \end{aligned} \tag{2}$$

The exponents may look complicated, but similarly to d_α^{vol} they can be obtained from the corresponding exponents for T_α^∞ by adding extra terms which reflect how distance, volume and resistance add up along branches of $\mathcal{T}_\alpha^{\mathrm{dec}}$, and in smaller subtrees. We will see below that they respectively give the quenched spectral dimension of $\mathcal{T}_\alpha^{\mathrm{dec}}$, and the quenched displacement exponent. On T_α^∞, these quantities are respectively $\frac{2\alpha}{2\alpha-1}$ and $\frac{\alpha-1}{2\alpha-1}$.

In what follows, we let $(X_n)_{n\geq 0}$ denote a simple random walk on $\mathcal{T}_\alpha^{\mathrm{dec}}$ started at $\rho_\alpha^{\mathrm{dec}}$, τ_r its exit time from a ball of radius r with respect to the metric d_g^{dec}, and $p_n(x,y) = \frac{\mathbb{P}_x(X_n=y)}{\deg y}$ its transition density. Our first result shows that $(X_n)_{n\geq 0}$ is $\mathbf{P}$-almost surely recurrent; as for discrete critical Galton-Watson trees, which essentially follows from the fact that the underlying tree T_α^∞ has a unique path to infinity.

Theorem 1.6. *$\mathbf{P}$-almost surely, $(X_n)_{n\geq 0}$ is recurrent.*

We establish the following quenched results on the random walk exponents using [KM08, Proposition 1.3].

Theorem 1.7 (Quenched random walk results)**.** *Under Assumption 1.4,* $\mathbf{P}$*-almost surely,*

$$d_S(\mathcal{T}_\alpha^{dec}) = d_\alpha^{spec},$$
$$d_{dis}(\mathcal{T}_\alpha^{dec}) = d_\alpha^{dis}.$$

Moreover, there exist finite constants $\beta_2, \dots, \beta_5$ *such that,* $\mathbf{P}$*-almost surely,*

a) There exists $N < \infty$ *such that*

$$n^{\frac{-d_\alpha^{spec}}{2}} (\log n)^{-\beta_2} \le p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec}) \le n^{\frac{-d_\alpha^{spec}}{2}} (\log n)^{\beta_2}$$

for all $n \ge N$*.*

b) Let τ_r *denote the exit time of a ball of radius* r *with respect to the metric* d_g^{dec}*. Then, for all* $R \ge 1$*,*

$$r^{\frac{1}{d_\alpha^{dis}}} (\log r)^{-\beta_3} \le \mathbb{E}[\tau_r] \le r^{\frac{1}{d_\alpha^{dis}}} (\log r)^{\beta_3}.$$

Moreover, there $\mathbb{P}$*-almost surely exists* $R < \infty$ *such that, for all* $r \ge R$*,*

$$r^{\frac{1}{d_\alpha^{dis}}} (\log r)^{-\beta_4} \le \tau_r \le r^{\frac{1}{d_\alpha^{dis}}} (\log r)^{\beta_4}.$$

c) $\mathbb{P}$*-almost surely, there exists* $N < \infty$ *such that*

$$n^{d_\alpha^{dis}} (\log n)^{-\beta_5} \le \sup_{k \le n} d_g^{dec}(\rho_\alpha^{dec}, X_k) \le n^{d_\alpha^{dis}} (\log n)^{\beta_5}$$

for all $n \ge N$*.*

d) Let $W_n = \{X_0, X_1, \dots, X_n\}$ *and* $S_n = V^{dec}(W_n)$*. Then,* $\mathbb{P}$*-almost surely,*

$$\lim_{n\to\infty} \frac{\log S_n}{\log n} = \frac{d_\alpha^{spec}}{2}.$$

The annealed results follow similarly from [KM08, Proposition 1.4].

Theorem 1.8 (Annealed random walk results)**.** *There exist* $\beta_\alpha^{(1)}, \beta_\alpha^{(2)}, \beta_\alpha^{(3)} < \infty$*, deterministic functions of* α, d, R, v*, such that under Assumption 1.4, we have that:*

a) There exist constants $c_1 > 0, c_2 < \infty, \gamma_1 > 0$ *such that for all* $r \ge 1$*,* $c_1 r^{\frac{1}{d_\alpha^{dis}}} (\log r)^{\beta_\alpha^{(1)}} \le \mathbf{E}\big[\mathbb{E}_{\rho_\alpha^{dec}}[\tau_r]\,\big]$*, and* $\mathbf{E}\big[\big(\mathbb{E}_{\rho_\alpha^{dec}}[\tau_r]\big)^{\gamma_1}\big] \le c_2 r^{\frac{\gamma_1}{d_\alpha^{dis}}} (\log r)^{\gamma_1 \beta_\alpha^{(1)}}$*.*

b) There exist constants $c_3 > 0, c_4 < \infty, \gamma_2 > 0$ *such that* $c_3 n^{\frac{-d_\alpha^{spec}}{2}} (\log n)^{\beta_\alpha^{(2)}} \le \mathbf{E}\big[p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec})\big]$ *for all* $n \ge 1$*, and* $\mathbf{E}\big[\big(p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec})\big)^{\gamma_2}\big] \le c_3 n^{\frac{-\gamma_2 d_\alpha^{spec}}{2}} (\log n)^{\gamma_2 \beta_\alpha^{(2)}}$*.*

c) There exist constants $c_5 > 0, c_6 < \infty, \gamma_3 > 0$ *such that* $c_5 n^{d_\alpha^{dis}} (\log n)^{\beta_\alpha^{(3)}} \le \mathbf{E}\big[\mathbb{E}_{\rho_\alpha^{dec}}\big[d_g^{dec}(\rho_\alpha^{dec}, X_n)\big]\,\big]$*, and* $\mathbf{E}\big[\big(\mathbb{E}_{\rho_\alpha^{dec}}\big[d_g^{dec}(\rho_\alpha^{dec}, X_n)\big]\,\big)^{\gamma_3}\big] \le c_6 n^{\gamma_3 d_\alpha^{dis}} (\log r)^{\gamma_3 \beta_\alpha^{(3)}}$*.*

If $s_\alpha^R, s_\alpha^d, f_\alpha^v \ne 1$*, then* $\beta_\alpha^{(1)} = \beta_\alpha^{(2)} = \beta_\alpha^{(3)} = 0$*. Otherwise, we will give the precise values in Section 7.*

Remark 1.9. *In general it is not possible to get annealed upper bounds without taking lower powers, since these quantities are infinite on the underlying tree* T_α^∞*. This is because the expected volume of a unit ball is infinite in this case, as established by Croydon and Kumagai [CK08]. This does not mean that it is always infinite in the decorated case (e.g. see [BS15, Theorem 1.2] for the corresponding result for discrete looptrees), but one must insert graphs that sufficiently "spread out" different branches of the tree.*

Finally, since Assumption 1.4 may not always apply, we also give some weaker results under Assumption 1.1. In this case, we are not able to control effective resistance so well on $\mathcal{T}_\alpha^{\mathrm{dec}}$, which means that we cannot control $\sup_{k\le n} d_g^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, X_k)$. In particular, there may be some exceptional branches of $\mathcal{T}_\alpha^{\mathrm{dec}}$ that are unexpectedly long, so that there may be some exceptional times when the random walk has travelled further than is typical. However, since Kesten's tree only has one path to infinity, the exceptional branches essentially don't lead anywhere, so we are still able to control $d_g^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, X_n)$ with high probability as $n \to \infty$. Note that there are no non-trivial quenched lower bounds on $d_g^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, X_n)$ due to recurrence.

Under Assumption 1.1 only, we are able to give the following (weaker) result.

Theorem 1.10. *Take $\beta_\alpha^{(1)}, \beta_\alpha^{(2)}, \beta_\alpha^{(3)}$ as in Theorem 1.8. Under Assumption 1.1, we have that*

(i) a) **P**-*almost surely, there exists $\beta_6, N < \infty$ such that $n^{\frac{-d_\alpha^{spec}}{2}}(\log n)^{-\beta_5} \le p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec}) \le n^{\frac{-d_\alpha^{spec}}{2}}(\log n)^{\beta_5}$ for all $n \ge N$. Consequently, $d_s(\mathcal{T}_\alpha^{dec}) = 2s_\alpha$.*

b) **P**-*almost surely, there exists $N, \beta_7 < \infty$ such that $n^{d_\alpha^{dis}}(\log n)^{-\beta_6} \le \sup_{k\le n} d_g^{dec}(\rho_\alpha^{dec}, X_n)$.*

c) $\limsup_{n\to\infty} \mathbf{P}\Big(d_g^{dec}(\rho_\alpha^{dec}, X_n) \ge n^{d_\alpha^{dis}}(\log n)^{\beta_\alpha^{(3)}}\lambda\Big) \to 0$ *as* $\lambda \to \infty$.

(ii) a) There exist constants $c_3 > 0, c_4 < \infty, \gamma_2 > 0$ such that $\mathbf{E}\big[\big(p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec})\big)^{\gamma_2}\big] \le c_3 n^{\frac{-\gamma_2 d_\alpha^{spec}}{2}}(\log n)^{\gamma_2\beta_\alpha^{(2)}}$ and $c_3 n^{\frac{-d_\alpha^{spec}}{2}}(\log n)^{\beta_\alpha^{(2)}} \le \mathbf{E}\big[p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec})\big]$ for all $n \ge 1$.

b) There exist constants $c_5 > 0, c_6 < \infty, \gamma_3 > 0$ such that $c_5 n^{d_\alpha^{dis}}(\log n)^{\beta_\alpha^{(3)}} \le \mathbf{E}\big[\mathbb{E}_{\rho_\alpha^{dec}}\big[d_g^{dec}(\rho_\alpha^{dec}, X_n)\big]\big]$, and $\mathbf{E}\big[\big(\mathbb{E}_{\rho_\alpha^{dec}}\big[d_g^{dec}(\rho_\alpha^{dec}, X_n)\big]\big)^{\gamma_3}\big] \le c_6 n^{\gamma_3 d_\alpha^{dis}}(\log n)^{\gamma_3\beta_\alpha^{(3)}}$.

Since scaling limits of critical trees and looptrees are now well-understood [Ald93, LG05, Duq03, CK14], it is natural to wonder if one can construct general scaling limits for decorated Galton-Watson trees. In the regime where $d(\alpha-1) < 1$, it will be clear from Proposition 4.11 that large graphs persist in the scaling limit and in this regime it will be shown in a forthcoming work of Sénizergues, Stefànsson and Stufler [SSS20] that the Gromov-Hausdorff-Prokhorov scaling limit can be constructed by gluing scaling limits of the inserted graphs along an appropriate tree structure. In the case $d(\alpha-1) > 1$, Proposition 4.11 indicates that the Gromov-Hausdorff-Prokhorov scaling limit will be a stable tree, endowed with a measure which may be supported on either the hubs or the leaves of the tree (or both), depending on the value of the parameter v. Moreover, we anticipate that in this regime, the techniques of [Cro18] will apply to show that the simple random walk on $\mathcal{T}_\alpha^{\mathrm{dec}}$ rescales to a limiting diffusion on the stable tree with this speed measure. Interestingly, if the hubs play a role the limiting measure will be singular with respect to the canonical mass measure on the stable tree, which is supported on the leaves, and the limiting diffusion will experience trapping at hubs, which contrasts with Brownian motion and classically studied FIN diffusions on stable trees, which spend almost all their time in the leaves.

We conclude the introduction by commenting briefly on the case $\alpha = 2$ (note that this may be finite or infinite variance). It was shown in [KR20, Theorem 2] and [CHK15, Theorem 13] that the metric space scaling limit of any discrete looptree with offspring distribution in the domain of attraction of a Gaussian law is the Brownian CRT, meaning that the loops do not persist in the scaling limit. Assuming that $R, d \ge 1$ this would therefore also be the case for our decorated tree model (since then distances are stochastically no bigger than those in looptrees). However, if there is polynomial tail decay in the offspring distribution it is always possible to choose the volume exponent v large enough that larger volumes persist in the scaling limit; in the same spirit, if $\alpha = 2$ and we were to repeat the arguments of this paper we expect that we would obtain a volume growth exponent

$$d_2^{\mathrm{dec}} = \frac{2}{f_2^v \wedge 1},$$

with f_α^v defined as above. We have not pursued this line of enquiry in this paper. In the infinite variance case when the offspring tails decay exactly like cx^{-2}, the arguments of this paper apply almost directly (one simple tweak is required) and we can simply take $\alpha = 2$ in (1) and (2) to give the correct exponents, but there are further subcases to consider for the logarithmic corrections for the precise gauge functions.

This paper is organised as follows. In Section 2 we formally define the model of a simple random walk on a decorated Galton-Watson tree. In Section 3 we give some lemmas on sums of stable random variables (the proofs are delayed to the appendix). The work starts Section 4, where we study the volume and distance asymptotics for finite decorated Galton-Watson trees. In particular, we show that if T_n is Galton-Watson tree conditioned to have n vertices, and $\mathcal{T}_n^{\mathrm{dec}}$ its decorated version, then the rescaled diameter and volume of $\mathcal{T}_n^{\mathrm{dec}}$ are tight sequences under Assumption 1.4. In Section 5 we apply these results to obtain the volume results of Theorem 1.3. In Section 6 we study resistance on $\mathcal{T}_\alpha^{\mathrm{dec}}$, and in Section 7 we bring everything together to prove Theorems 1.6, 1.7, 1.8 and 1.10. We conclude in Section 8 with some examples.

Notation. Throughout, c and C will denote constants bounded from above and below, but values may change on each appearance.

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2 Definition of the model

Before defining stable trees and looptrees, we briefly recap the Ulam-Harris labelling notation for discrete trees, following the formalism of [Nev86]. Firstly, introduce the set

$$\mathcal{U} = \cup_{n=0}^{\infty} \mathbb{N}^n.$$

By convention, $\mathbb{N}^0 = \{\emptyset\}$. If $u = (u_1, \dots, u_n)$ and $v = (v_1, \dots, v_m) \in \mathcal{U}$, we let $uv = (u_1, \dots, u_n, v_1, \dots, v_m)$ be the concatenation of u and v.

Definition 2.1. *A plane tree T is a finite subset of $\mathcal{U}$ such that*

(i) $\emptyset \in T$,

(ii) If $v \in T$ *and* $v = uj$ *for some* $j \in \mathbb{N}$, *then* $u \in T$,

(iii) For every $u \in T$, *there exists a number* $k_u(T) \geq 0$ *such that* $uj \in T$ *if and only if* $1 \leq j \leq k_u(T)$.

We let $\mathbb{T}$ denote the set of all plane trees. A plane tree T on n vertices can be coded by a random walk $(W_m^T)_{0 \leq m \leq n}$ called the Lukasiewicz path; this is defined by setting $W_0^T = 0$, then for $m \geq 1$ listing the vertices $u_0, u_1, \dots, u_n$ in lexicographical order and setting $W_{m+1}^T = W_m^T + k_{u_m}(T) - 1$. It is not too hard to see that $W_m^T \geq 0$ for all $0 \leq m \leq n-1$, and $W_n^T = -1$.

Given a finite Galton-Watson tree T, we let $v_h = \arg\max_{v \in T} d(\rho, v)$, and we call the path from ρ to v_h the Williams' spine. If v_h is not unique, we take the leftmost path. If $u, v \in T$, we let $[[u, v]]$ denote the path from u to v, inclusive of endpoints, and $((u, v]] = [[u, v]] \setminus \{u\}$.

2.1 Infinite critical trees and looptrees

In this section we introduce Kesten's tree T_∞ for a given critical offspring distribution ξ.

Definition 2.2. *[Kes86]. Let ξ be a critical offspring distribution, and define its size biased version ξ^* by*

$$\xi^*(n) = n\xi(n).$$

*The **Kesten's tree** T_∞ associated to the probability distribution ξ is a two-type Galton-Watson tree distributed as follows:*

- *Individuals are either normal or special.*

- *The root of T_∞ is special.*
- *A normal individual produces only normal individuals according to ξ.*
- *A special individual produces individuals according to the size-biased distribution ξ^*. Of these, one of them is chosen uniformly at random to be special, and the rest are normal.*

Almost surely, the special vertices form a unique infinite backbone of T_∞. Note that this is one-ended. Aldous in [Ald91] coined the term *sin-trees* for such trees, since they have a single infinite spine. Although a critical Galton-Watson tree is almost surely finite, Kesten [Kes86, Lemma 1.14] showed that T_∞ arises as the local limit of a critical Galton-Watson tree with offspring distribution ξ as its height goes to infinity.

2.2 Decorated tree

Formally, we let T_α^∞ denote Kesten's tree with critical (aperiodic) offspring distribution ξ satisfying

$$\xi(k) \sim ck^{-\alpha} \tag{3}$$

as $k \to \infty$, for some $\alpha \in (1,2), c \in (0,\infty)$, and let d_α^T denote the graph distance on T_α^∞.

In line with the literature, we also let $p_k = \xi(k)$. By standard theory, (e.g. [BGT89, Chapter VIII]), if $(X_i)_{i=1}^\infty$ are i.i.d. distributed according to ξ, it follows that $a_n^{-1}\sum_{i=1}^\infty X_i \to Z_\alpha$, where Z_α is an α-stable random variable and $a_n = (c|\Gamma(-\alpha)|n)^{\frac{1}{\alpha}}$.

To construct the decorated model, we will suppose that $((G_n, d_n, V_n, \ell_n))_{n\geq 1}$ is a sequence of random graphs with some pre-specified distribution, such that for all $n \geq 1$, G_n is almost surely connected, d_n is a metric on G_n, V_n is a measure on the vertices of G_n, and G_n and has n pre-specified "boundary" vertices (for example, a tree with n leaves, or a dissection of the n-gon). Moreover, we assume that the boundary vertices ∂G_n come pre-equipped with a labelling $\ell_n : \partial G_n \to \{0,1,\dots,n-1\}$ (in the case of a random planar map, this would represent the clockwise ordering of the boundary vertices when G_n is embedded in the plane; for graph such as the complete graph where the ordering is not important, ℓ_n can be a uniform bijection from ∂G_n to $\{1,2,\dots,n\}$).

To construct $\mathcal{T}_\alpha^{\mathrm{dec}}$, we work on the probability space $(\mathbf{\Omega}, \mathcal{F}, \mathbf{P})$ and do the following:

1. Sample a pair $(T_\alpha^\infty, ((G(u), d(u), V(u), \ell(u)))_{u\in T_\alpha^\infty})$, where T_α^∞ is distributed according to Definition 2.2 of Kesten's tree, and each $G(u)$ is independently sampled according to the law of $(G_{\deg u}, d_{\deg u}, V_{\deg u}, \ell_{\deg u})$.
2. For each $u \in T_\alpha^\infty$, independently choose $U_u \sim \mathsf{Uniform}\{0,\dots \deg u - 1\}$. Define a bijection $\ell_U(u) : \partial G(u) \to \{0,1,\dots,\deg u - 1\}$ by $\ell_U(u)(v) \mapsto (\ell(v) + U) \mod (\deg u - 1)$, and then a bijection $b(u) : \partial G(u) \to E(T_\alpha^\infty)$ such that $b(u)(v) = (u, \mathsf{par}(u))$ if $\ell_U(u)(v) = 0$, and $b(u)(v) = (u, u(\ell_U(u)(v)))$ otherwise.
3. Given a vertex $x \in \bigcup_{u\in T_\alpha^\infty} \partial G(u)$, if $x \in \partial G(v)$ we say that $v = V_T(x)$. We then define an equivalence relation $\overset{e}{\sim}$ on the vertices of $\bigcup_{u\in T_\alpha^\infty} \partial G(u)$ by saying that $x \overset{e}{\sim} y$ if and only if $b(V_T(x))(x) = b(V_T(y))(y)$, in other words that they are both in bijection with the same edge of T_α^∞.
4. We then set
$$\mathcal{T}_\alpha^{\mathrm{dec}} = \bigcup_{v\in T_\alpha^\infty} G(v)/\overset{e}{\sim}.$$
If ρ is the root of T_α^∞, we also define the root of $\mathcal{T}_\alpha^{\mathrm{dec}}$ to be the vertex $x \in G(\rho)$ such that $b(\rho)(x)$ is equal to the edge joining ρ to its first child, and denote this vertex by $\rho_\alpha^{\mathrm{dec}}$.
5. We define a metric d_g^{dec} on $\mathcal{T}_\alpha^{\mathrm{dec}}$ as follows. If $x, y \in \mathcal{T}_\alpha^{\mathrm{dec}}$, we let $[[V_T(x), V_T(y)]] = v_0, v_1, \dots, v_n$ denote the path of (underlying tree) vertices between $V_T(x)$ and $V_T(y)$ in T_α^∞; this can be constructed by setting
$$\begin{aligned} d_g^{\mathrm{dec}}(x,y) =& d_{G(v_0)}\left(x, b(v_0)^{-1}((v_0,v_1))\right) + \sum_{1\leq i\leq n-1} d_{G(v_i)}\left(b(v_i)^{-1}((v_{i-1},v_i)), b(v_i)^{-1}((v_i,v_{i+1}))\right) \\ &+ d_{G(v_n)}\left(b(v_n)^{-1}((v_{n-1},v_n)), y\right), \end{aligned} \tag{4}$$

where $d_{G(v_i)}$ represents the metric inherited from $G(v_i)$.

6. We define a measure $V^{\mathrm{dec}} = \sum_{u \in T_\alpha^\infty} V(u)$.

If $v \in T_\alpha^\infty$, let T_v be the subtree of T_α^∞ rooted at v. This corresponds to a subgraph of $\mathcal{T}_\alpha^{\mathrm{dec}}$ consisting of the graph $\bigcup_{u \in T_v} G(u)$. We denote this graph by $\mathcal{T}_\alpha^{\mathrm{dec}}(v)$.

To study a simple random walk on $\mathcal{T}_\alpha^{\mathrm{dec}}$, we will be most interested in the measure μ such that $\mu(x) = \deg x$ for all x in $\mathcal{T}_\alpha^{\mathrm{dec}}$. The assumptions on the measure $\mu(u)$ then correspond to assumptions on the number of edges in $G(u)$.

2.3 Simple random walk on $\mathcal{T}_\alpha^{\mathbf{dec}}$

The simple random walk $(X_n)_{n \geq 0}$ is the discrete-time Markov chain on $\mathcal{T}_\alpha^{\mathrm{dec}}$ with $X_0 = \rho_\alpha^{\mathrm{dec}}$ and

$$\mathbb{P}(X_{n+1} = y \mid X_n = x) = \frac{1}{\deg x} \mathbb{1}\{y \sim x\}.$$

We define the (consequently symmetric) transition density by

$$p_n(x, y) = \frac{\mathbb{P}_x(X_n = y)}{\deg y}.$$

In this paper, we will be interested in two metrics on $\mathcal{T}_\alpha^{\mathrm{dec}}$: the metric d_g^{dec} constructed above, and the effective resistance metric. We let τ_r denote the exit time of $(X_n)_{n \geq 0}$ from a ball of radius r defined with respect to the metric d_g^{dec}, and τ_r^R denote its exit time from a ball of radius r defined with respect to the effective resistance metric.

We will assume that T_α^∞ and the set $\left((G(v), d(v), V(v), \ell(v))\right)_{v \in T_\alpha^\infty}$ are defined on the probability space $(\mathbf{\Omega}, \mathcal{F}, \mathbf{P})$. We denote the law of a random walk on $\mathcal{T}_\alpha^{\mathrm{dec}}$ by $\mathbb{P}$: this law is therefore also a random variable on the probability space $(\mathbf{\Omega}, \mathcal{F}, \mathbf{P})$.

Throughout this paper, if T is a finite tree, its "decorated version" describes the metric measure space obtained by replacing T_α^∞ with T in the above construction.

3 Technical lemmas

In this section we collect some technical lemmas regarding sums of stable random variables. We give the proofs in the appendix.

Lemma 3.1. *Let $(X_i)_{i \geq 1}$ be i.i.d. and such that $\mathbb{P}(X_i \geq x) \sim c x^{-\beta}$ for some $\beta \in (0, 1]$. Let $S_n = \sum_{i=1}^n X_i$.*

(i) If $\beta < 1$, then there exists a constant $c' < \infty$ such that for each $n \geq 2$,

$$\mathbb{P}\Big(S_n \geq n^{\frac{1}{\beta}} \lambda\Big) \leq c' \lambda^{-\beta}$$

as $n \to \infty$.

(ii) If $\beta = 1$, then there exists a constant $c' < \infty$ such that for each $n \geq 2$,

$$\mathbb{P}\Big(S_n \geq \lambda n^{\frac{1}{\beta}} \log n\Big) \leq c' \lambda^{-\beta}$$

as $n \to \infty$.

(iii) If $\beta > 1$, then for any $\varepsilon > 0$ there exists a constant $c' < \infty$ such that for each $n \geq 2$,

$$\mathbb{P}(S_n \geq \lambda n) \leq c' \lambda^{-(\beta - \varepsilon)}$$

as $n \to \infty$. This bound still holds if the $(X_i)_{i \geq 1}$ are not independent.

We also have a bound for the lower tails.

Lemma 3.2. *Let $(X_i)_{i\geq 1}$ be i.i.d., non-negative and such that $\mathbb{P}(X_i \geq x) \sim cx^{-\beta}$ for some $\beta \in (0,\infty)$. Let $S_n = \sum_{i=1}^n X_i$.*

(i) If $\beta < 1$, then there exists a constant $c > 0$ such that for each $n \geq 2$,

$$\mathbb{P}\Big(S_n \leq n^{\frac{1}{\beta}}\lambda^{-1}\Big) \leq e^{-c\lambda^{\beta}}.$$

(ii) If $\beta = 1$, then there exists a constant $c > 0$ such that for each $n \geq 2$,

$$\mathbb{P}\big(S_n \leq \lambda^{-1} n \log n\big) \leq e^{-c\lambda \log \lambda}.$$

(iii) If $\beta > 1$, then there exists a constant $c > 0$ such that for each $n \geq 2$,

$$\mathbb{P}\big(S_n \leq \lambda^{-1} n\big) \leq e^{-c\lambda}.$$

Lemma 3.3. *Let X be a random variable, and suppose that $\mathbb{P}(X \geq x) \sim cx^{-\beta}$ as $x \to \infty$ for some $\beta < 1$. Then there exists a constant $c \in (0,\infty)$ such that $1 - \mathbb{E}\big[e^{-\theta X}\big] \sim c\theta^{\beta}$ as $\theta \to 0$.*

Lemma 3.4. *Let $\beta > 0$, let X be a random variable with $\mathbb{P}(X > x) \sim cx^{-\beta}$ as $x \to \infty$, and let f be a function of X such that there exists $z > 0$ and a function $p(\lambda)$ such that $\mathbb{P}(f(X) \geq \lambda n^z \mid X = n) \sim p(\lambda)$ for all sufficiently large n, where $p(\lambda) = o(\lambda^{\frac{-\beta}{z}})$ as $\lambda \to \infty$. Then, there exist $c, C \in (0,\infty)$ such that*

$$ck^{-\frac{1}{z}\beta} \leq \mathbb{P}(f(X) \geq k) \leq Ck^{-\frac{1}{z}\beta}.$$

If instead $p(\lambda) \asymp \lambda^{-m}$ where $m < \frac{\beta}{z}$, then

$$ck^{-m} \leq \mathbb{P}(f(X) \geq k) \leq Ck^{-m}.$$

4 Exponents for finite decorated Galton-Watson trees

In this section we prove some results on volume and distance on finite decorated Galton-Watson trees. Since T_α^∞ is constructed to have one infinite backbone to which many finite fringe Galton-Watson trees are grafted, these estimates will be crucial when we prove the volume and resistance bounds for $\mathcal{T}_\alpha^{\mathrm{dec}}$ in Sections 5 and 6.

We will assume throughout that λ is implicitly a function of n, and that for all $\varepsilon > 0$, $\lambda = \lambda_n = o(n^\varepsilon)$ as $n \to \infty$.

In this section, we work under Assumption 1.1, unless explicitly stated otherwise (the main exception to this is in Section 4.2.1).

We then define a further set of modified exponents as follows. Under Assumption 1.1, we set

$$A_d = \frac{\log\Big(\limsup_{n\to\infty} \mathbf{P}\Big(\mathsf{Diam}(G_n) \geq \lambda n^{\frac{1}{d}}\Big)\Big)}{\log\lambda},$$
$$A_R = \frac{\log\Big(\limsup_{n\to\infty} \mathbf{P}\Big(\mathsf{Diam}_{\mathrm{res}}(G_n) \geq \lambda n^{\frac{1}{R}}\Big)\Big)}{\log\lambda},$$
$$A_v = \frac{\log\Big(\limsup_{n\to\infty} \mathbf{P}\Big(V_n^{\mathrm{dec}}(G_n) \geq \lambda n^{\frac{1}{R}}\Big)\Big)}{\log\lambda}.$$

We then define:

$$\tilde{s}^d_\alpha = d(\alpha-1)\wedge A_d,\ \tilde{s}^R_\alpha = R(\alpha-1)\wedge A_R \text{ and } \tilde{f}^v_\alpha = \tfrac{\alpha}{v}\wedge A_v.$$

Note that, under Assumption 1.1, $s^d_\alpha > 1 \Leftrightarrow \tilde{s}^d_\alpha > 1$, $s^d_\alpha = 1 \Leftrightarrow \tilde{s}^d_\alpha = 1$, and $s^d_\alpha < 1 \Leftrightarrow \tilde{s}^d_\alpha < 1$, so that in particular $s^d_\alpha \wedge 1 = \tilde{s}^d_\alpha \wedge 1$. Similarly for all the other exponents. Moreover, it follows from Lemma 3.4 that, if v is a backbone vertex of T^∞_α, then there exist constants $c, C \in (0,\infty)$ such that

$$cx^{-\tilde{s}^d_\alpha} \le \mathbf{P}(\mathsf{Diam}(G(v)) \ge x) \le Cx^{-\tilde{s}^d_\alpha}.$$

Therefore, if s_n is the n^{th} backbone vertex of T^∞_α, we expect that $\sum_{v\preceq s_n} \mathsf{Diam}(G(u)) \approx x^{\frac{1}{\tilde{s}^d_\alpha\wedge 1}} = x^{\frac{1}{s^d_\alpha\wedge 1}}$. Similar results respectively hold for $\tilde{s}^R_\alpha$, $\tilde{s}^v_\alpha$ and $\tilde{f}^v_\alpha$ on adding resistance and volumes of graphs inserted along the infinite backbone, and volumes of graphs inserted at vertices in subtrees.

4.1 Bounds for undecorated Galton-Watson trees

In this section, we let T denote a critical Galton-Watson tree with offspring distribution ξ, where ξ is aperiodic and satisfies (3). Note that T is almost surely finite. The strategy will be to first give the relevant exponents for undecorated trees (these are already known), and then use Lemma 3.4 to compute the corresponding exponents for decorated trees. We also let T_n denote a copy of T conditioned on having n vertices, and $\mathcal{T}^{\mathrm{dec}}_n$ denote a decorated version of T_n (since ξ is aperiodic, this is well-defined for all sufficiently large n).

4.1.1 Progeny bounds

Lemma 4.1. *There exists a constant $c \in (0,\infty)$ such that*

$$\mathbf{P}(|T| \ge k) \sim ck^{\frac{-1}{\alpha}}$$

as $k \to \infty$.

Proof. This is shown in [CK15, Proposition A.3(i)]. □

Lemma 4.2. *There exists a constant $q > 0$ such that $1 - \mathbf{E}\left[e^{-\theta|T|}\right] \sim q\theta^{\frac{1}{\alpha}}$ as $\theta \to 0$.*

Proof. This is a direct consequence of Lemma 3.3, with $\beta = \frac{1}{\alpha}$. □

4.1.2 Height bounds and associated spinal decomposition

At various points, we will also need to perform spinal decompositions along various choices of spine in T. For this, the following results will be useful. If T has root ρ, we let

$$\mathsf{Height}(T) = \sup_{v\in T} d(\rho, v).$$

Lemma 4.3. *As $n \to \infty$,*

$$\mathbf{P}(\mathit{Height}(T) \ge n) \sim cn^{-\frac{1}{\alpha-1}}.$$

Proof. This follows directly from [Sla68, Theorem 2]. □

We also give the following (elementary) result. We include the proof for completeness.

Lemma 4.4. *Let $H = \mathit{Height}(T)$. The function $h : \mathbb{N} \to [0,1]$, $x \mapsto \mathbb{P}(H = x)$ is non-increasing in x.*

Proof. Let $\xi^{(0)}$ denote the number of offspring of ρ, and conditional on $\{\xi^{(0)} = n\}$, let $T^{(i)}$ denote the subtree emanating from the i^{th} offspring of ρ. For $x \geq 0$, we then write:

$$\begin{aligned}
\mathbf{P}(H = x+1) &= \sum_{n=0}^{\infty} \mathbf{P}\Big(\xi^{(0)} = n\Big)\, \mathbf{P}\Big(H = x+1 \;\Big|\; \xi^{(0)} = n\Big) \\
&\leq \sum_{n=0}^{\infty} p_n \sum_{i=1}^{n} \mathbf{P}\Big(\mathsf{Height}(T^{(i)}) = x\Big)\, \mathbf{P}\Big(\mathsf{Height}(T^{(j)}) \leq x \forall j \neq i\Big) \\
&\leq \sum_{n=0}^{\infty} n p_n \mathbf{P}(H = x) \\
&= \mathbf{P}(H = x)\,,
\end{aligned}$$

where the final line follows since ξ is critical. □

At several points in this paper we will perform a spinal decomposition of a tree along its leftmost spine of maximal height. We denote the vertices of this spine by $s_0, s_1, \ldots, s_H$ where $s_0 = \rho$ and s_i is the parent of s_{i+1} for all $i < H$, and call this spine the Williams' spine. It is well-known [GK99, remark below Proposition 2.2] that the offspring distribution of spinal vertices is asymptotically size-biased as $H \to \infty$, but we will need the following precise formulation of this result.

Proposition 4.5. *Take the notation as above, and let $\xi^{(n)}$ denote the number of offspring of s_n. Then, for every $C, \varepsilon > 0$ there exists a constant $c \in (0, \infty)$, uniform in n and k, such that for all $n \in \mathbb{N}$ and all $k \leq Cn^{\frac{1}{\alpha-1}}$:*

$$\mathbf{P}\Big(\xi^{(n)} \geq k \;\Big|\; H \geq (1+\varepsilon)n\Big) \geq ck^{-(\alpha-1)}.$$

Proof. We first prove a corresponding result for $\mathbf{P}\Big(\xi^{(n)} = \tilde{k} \;\Big|\; H \geq (1+\varepsilon)n\Big)$, and then obtain the result by summing over $\tilde{k} \geq k$. In particular, letting

$$P^H_{j,k} = p_k \mathbf{P}(H < m-n)^{j-1}\, \mathbf{P}(H < m+1-n)^{k-j}\, \frac{\mathbf{P}(H = m-n)}{\mathbf{P}(H = m)},$$

it follows from [GK99, Lemma 2.1] and then Lemma 4.4 that

$$\begin{aligned}
\mathbf{P}\Big(\xi^{(n)} = k \;\Big|\; H \geq (1+\varepsilon)n\Big) &= \sum_{m \geq (1+\varepsilon)n} \mathbf{P}(H = m \mid H \geq (1+\varepsilon)n) \sum_{j=1}^{k} P^H_{j,k} \\
&\geq \sum_{m \geq (1+\varepsilon)n} \mathbf{P}(H = m \mid H \geq (1+\varepsilon)n)\, k p_k \mathbf{P}(H < \varepsilon n)^{k-1} \\
&= k p_k \mathbf{P}(H < \varepsilon n)^{k-1}\,.
\end{aligned}$$

Then, by Lemma 4.3, we know that $\mathbf{P}(\text{Height}(T) \geq \varepsilon n) \leq 2(\varepsilon n)^{-\frac{1}{\alpha-1}}$, so if $k \leq Cn^{\frac{1}{\alpha-1}}$ then

$$\mathbf{P}(H < \varepsilon n)^{k-1} \geq (1 - 2(\varepsilon n)^{-\frac{1}{\alpha-1}})^{Cn^{\frac{1}{\alpha-1}}} \geq e^{-3C\varepsilon^{-\frac{1}{\alpha-1}}}$$

for all sufficiently large n, so that $\mathbf{P}\big(\xi^{(n)} = k \;\big|\; H \geq (1+\varepsilon)n\big) \geq e^{-3C\varepsilon^{-\frac{1}{\alpha-1}}} k p_k$.

To prove the result as stated, we then write

$$\mathbf{P}\Big(\xi^{(n)} \geq k \;\Big|\; H \geq (1+\varepsilon)n\Big) \geq \sum_{k \leq \tilde{k} \leq 2k} \mathbf{P}\Big(\xi^{(n)} = \tilde{k} \;\Big|\; H \geq (1+\varepsilon)n\Big) \geq \sum_{k \leq \tilde{k} \leq 2k} e^{-3C_\varepsilon} k p_k \geq c_\varepsilon k^{-(\alpha-1)}.$$

□

We also note the following (unconditioned) probabilistic upper bound, which follows even more straightforwardly from [GK99, Lemma 2.1].

Proposition 4.6. *Take the notation as above, and let $\xi^{(n)}$ denote the number of offspring of s_n. Then there exists $c < \infty$ such that*

$$\mathbf{P}\Big(\xi^{(n)} \geq K\Big) \leq cK^{-(\alpha-1)}.$$

Proof. Given $\xi^{(n)} = k$, let $T^{(1)}, T^{(2)}, \ldots, T^{(k)}$ denote the subtrees rooted at each of the offspring of s_n, listed in lexicographical order. Then

$$\mathbf{P}\Big(\xi^{(n)} \geq K\Big) \leq \sum_{k \geq K} p_k \sum_{j=1}^{k} \mathbf{P}\bigg(\mathsf{Height}(T^{(j)}) = \sup_{i \leq k} \mathsf{Height}(T^{(k)})\bigg) \leq \sum_{k \geq K} k p_k \leq cK^{-(\alpha-1)}.$$

□

4.1.3 Vertex degrees

We will also need the following result on the degree of a typical vertex.

Lemma 4.7. *Let T_n be a Galton-Watson tree with offspring distribution ξ but conditioned to have n vertices. Let v_U be a uniformly chosen vertex of T_n. Then there exists a constant c such that for all $n \geq 1$,*

$$\mathbf{P}(\deg(v_U) \geq k) \leq ck^{-\alpha}.$$

Proof. Recall from Section 2 that the vertex degrees of the vertices of T_n correspond to (one less than) each of the jump sizes of the Lukasiewicz path $W^{(n)}$, which is conditioned to first hit -1 at time n. Since v_U is uniform amongst the vertices of T_n, its label in the lexicographical ordering is uniform amongst $\{1, \ldots, n\}$. Letting U denote this label, it follows from the (discrete) Vervaat transform (e.g. see [Kor17, Proposition 10]) that the U^{th} cyclic shift of $W^{(n)}$, i.e. the random walk $\tilde{W}^{(n)}$ given by

$$\tilde{W}^{(n)}(t) = \begin{cases} W^{(n)}(U+t) - W^{(n)}(U) & \text{if } U + t \leq n \\ W^{(n)}(U+t-n) - W^{(n)}(U) & \text{if } U + t > n \end{cases}$$

is a random walk bridge from 0 to -1, by which we mean that, if W is a simple random walk with $\mathbf{P}(W_{m+1} - W_m = x - 1) = \xi(x)$ for all $m, x \in \mathbb{N}$, then $\Big(\tilde{W}^{(n)}\Big)_{m=0}^{n}$ has the law of $(W)_{m=0}^{n}$ but conditioned on $W_n = -1$. In particular, $\tilde{W}^{(n)}$ has a density with respect to the unconditioned walk W, and moreover $\deg v_U$ is equal to $\tilde{W}^{(n)}(1) - 1$. We deduce that

$$\mathbf{P}(\deg v_U \geq k) = \mathbf{P}\Big(\tilde{W}^{(n)}(1) > k\Big) = \mathbf{E}\left[\mathbb{1}\{\tilde{W}^{(n)}(1) > k\} \frac{p_{n-1}(-\tilde{W}^{(n)}(1) - 1)}{p_n(-1)}\right].$$

Moreover, by the local limit theorem on p.236 of [GK54], we have that

$$a_n \mathbf{P}(W_n = k) - \tilde{p}_\alpha\Big(\frac{k}{a_n}\Big) \to 0$$

uniformly in k, where $\tilde{p}_\alpha$ is the density of a centred stable random variable Y_α (different to Z_α from before, which is almost surely non-negative), and such that $\tilde{p}_\alpha(0) = |\Gamma(\frac{-1}{\alpha})|^{-1} > 0$. It follows from this that the ratio $\frac{p_{n-1}(-\tilde{W}^{(n)}(1)-1)}{p_n(-1)}$ is bounded uniformly in k, which gives the result. □

Lemma 4.8. *For all $k \geq 1$, there exist constants $c, C > 0$ such that*

$$ck^{-1} \leq \mathbf{P}\bigg(\sup_{v \in T}(\deg v) \geq k\bigg) \leq Ck^{-1}$$

Proof. The proof is essentially the same as above: given $|T| = n$, and applying the Vervaat transform and absolute continuity relation on the first $\lfloor \frac{n}{2} \rfloor$ vertices as in the previous proof, we get that

$$\mathbf{P}\Big(\sup_{v\in T_n} \deg v \geq n^{\frac{-1}{\alpha}}\lambda\Big) \geq 1 - (1 - cn^{-1}\lambda^{-\alpha})^{\frac{n-1}{2}} \geq 1 - e^{-c\lambda^{-\alpha}} \geq c\lambda^{-\alpha}$$
$$\mathbf{P}\Big(\sup_{v\in T_n} \deg v \geq n^{\frac{-1}{\alpha}}\lambda\Big) \leq \sum_{v\in T_n} \mathbf{P}\Big(\deg v \geq n^{\frac{-1}{\alpha}}\lambda\Big) = n\mathbf{P}\Big(\deg v_U \geq n^{\frac{-1}{\alpha}}\lambda\Big) \leq c\lambda^{-\alpha}.$$

Then, since $\mathbf{P}(|T| \geq n) \sim cn^{\frac{-1}{\alpha}}$, we can apply Lemma 3.4 with $\beta = z = \frac{1}{\alpha}, m = \alpha$ to deduce the result. □

4.2 Decorated height bounds

4.2.1 Spinal decomposition

If $\mathcal{T}^{\mathrm{dec}}$ is the decorated version of a finite Galton-Watson tree T, we define the *decorated height* of $\mathcal{T}^{\mathrm{dec}}$ by

$$\mathsf{Height}^{\mathrm{dec}}(\mathcal{T}^{\mathrm{dec}}) = \sup_{x\in\mathcal{T}^{\mathrm{dec}}} d_g^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, x).$$

At some points in this paper it will be necessary to decompose along a path achieving maximal decorated height, rather than the maximal tree height, and we give similar results for the offspring distribution along this spine below.

The path in $\mathcal{T}^{\mathrm{dec}}$ joining $\rho_\alpha^{\mathrm{dec}}$ to the point achieving maximal decorated height corresponds in a natural way to a path in T joining ρ to a leaf (if this point is not unique, we will take the leftmost path). Analogously with the notation above we call this path the decorated spine and denote this by $s_0^{\mathrm{dec}}, s_1^{\mathrm{dec}}, \ldots, s_{H^{\mathrm{dec}}}^{\mathrm{dec}}$, where H^{dec} denotes the length of this decorated spine. We also let ξ_n^{dec} denote the number of offspring of s_n^{dec}.

Note in particular that H^{dec} gives the length of the decorated spine in the underlying tree, rather than the length with respect to the decorated metric, so $H^{\mathrm{dec}} \leq H$.

We start with a lower bound on the decorated height.

Proposition 4.9. *Under Assumption 1.1, if $s_\alpha^d \neq 1$, then there exists a constant $c \in (0,\infty)$ such that*

$$\mathbf{P}\Big(\mathit{Height}^{dec}(\mathcal{T}^{dec}) \geq n\Big) \geq cn^{\frac{-(s_\alpha^d \wedge 1)}{\alpha-1}}$$

If $s_\alpha^d = 1$, then there exists a constant $c \in (0,\infty)$ such that

$$\mathbf{P}\Big(\mathit{Height}^{dec}(\mathcal{T}^{dec}) \geq n\log n\Big) \geq cn^{\frac{-(s_\alpha^d \wedge 1)}{\alpha-1}}$$

Before we give the proof of the lower bound, we will need the following result on distances across the graphs corresponding to spinal vertices along the decorated spine.

Lemma 4.10. *For an arbitrary vertex s_n on the backbone of T_α^∞, let $d^U(s_n)$ denote the distance between the two boundary vertices of $G(s_n)$ that correspond to edges joining s_n to neighbouring spinal vertices (so these are two distinct uniform vertices in $G(s_n)$). There exist $C, c, \in (0,\infty)$ such that*

(i) For all $k \geq 1$,

$$\mathbf{P}\big(d^U(s_n) \geq k\big) \leq Ck^{-\tilde{s}_\alpha^d}$$

(ii) For all $1 \leq k \leq c' n^{\frac{1}{s_\alpha^d}}$:

$$\mathbf{P}\big(d^U(s_n) \geq k \;\big|\; H \geq 2n\big) \geq ck^{-\tilde{s}_\alpha^d}$$

Proof. This follows from Proposition 4.5 and Lemma 3.4 with $\beta = \alpha - 1$ and $z = \frac{1}{d}$. □

Proof of Proposition 4.9. $\mathsf{Height}^{\mathrm{dec}}(\mathcal{T}^{\mathrm{dec}})$ stochastically dominates $\sum_{n=1}^{\frac{H}{2}} d^U(s_n)$, so we will bound this latter quantity.

To do this, first note that by Lemma 4.3, $\mathbf{P}\Big(H \geq n^{s_\alpha^d \wedge 1}\Big) \geq cn^{\frac{-(s_\alpha^d \wedge 1)}{\alpha - 1}}$. Then, if $s_\alpha^d < 1$, we have by Lemma 4.10 that there exists a constant $c > 0$ (deterministic), such that

$$\mathbf{P}\left(\sum_{n=1}^{\frac{H}{2}} d^U(s_n) \geq cn \;\middle|\; H \geq n^{s_\alpha^d \wedge 1}\right) \geq \mathbf{P}\left(\exists n \leq \frac{H}{2} : d^U(s_n) \geq cn \;\middle|\; H \geq n^{s_\alpha^d}\right) \geq \frac{1}{2}.$$

If $s_\alpha^d > 1$ then the same result holds by the Law of Large Numbers. If $s_\alpha^d = 1$, recall that by standard results on asymmetric distributions in the domain of attraction of a Cauchy distribution (e.g. see [Ber19, Equation (1.3)]) that if $(X_i)_{i=1}^\infty$ are i.i.d. with $\mathbb{P}(X_1 > x) \sim cx^{-1}$ for some $c > 0$, then there exists $C_1 \in (0, \infty)$ such that $n^{-1}\left(\sum_{i=1}^n X_i - C_1 n \log n\right)$ has a non-trivial scaling limit, so that there exists another $c > 0$ such that

$$\mathbf{P}\left(\sum_{n=1}^{\frac{H}{2}} d^U(s_n) \geq cn \log n \;\middle|\; H \geq n\right) \geq \frac{1}{2}.$$

This proves the result by Lemma 4.3. □

We now turn to proving an upper bound for the decorated height of a finite Galton-Watson tree. Note that, since $\mathsf{Height}(T_n)$ is of order $n^{1-\frac{1}{\alpha}}$ with high probability, the proof of Proposition 4.9 gives the complementary lower bounds.

Proposition 4.11. *Assume that Assumption 1.4 holds and let $\mathcal{T}_n^{dec}$ be as above (i.e. the decorated version of a Galton-Watson tree with n vertices). Given $\varepsilon > 0$, set*

$$t_\alpha^d = \frac{\alpha - \varepsilon}{\alpha + 1 + [1 - s_\alpha^d]\alpha(\alpha - 1)} \wedge \frac{\alpha^2 - \varepsilon}{\alpha(\alpha - 1) + \alpha(s_\alpha^d + \alpha)}.$$

Then, for all $\varepsilon > 0$:

- *If $s_\alpha^d < 1$, $\mathbf{P}\Big(\mathsf{Height}^{dec}(\mathcal{T}_n^{dec}) \geq n^{\frac{1}{\alpha d}}\lambda\Big) \leq \lambda^{-t_\alpha^d}$.*
- *If instead $s_\alpha^d = 1$, $\mathbf{P}\Big(\mathsf{Height}^{dec}(\mathcal{T}_n^{dec}) \geq \lambda n^{\frac{1}{\alpha d}} \log n\Big) \leq \lambda^{-t_\alpha^d}$.*
- *If $s_\alpha^d > 1$, $\mathbf{P}\Big(\mathsf{Height}^{dec}(\mathcal{T}_n^{dec}) \geq n^{1-\frac{1}{\alpha}}\lambda\Big) \leq \lambda^{\frac{-(1-\varepsilon)}{\alpha}}$.*

Remark 4.12.

1. *These tail bounds are not good enough to plug into Lemma 3.4 and get an overall tail bound for the unconditioned tree complementing Proposition 4.9, but are nevertheless sufficient for our purposes, and in particular show that the decorated height of $\mathcal{T}_n^{dec}$ forms a tight sequence.*
2. *When $s_\alpha^d \geq 1$, it is certainly necessary that $\limsup_{n\to\infty} \mathbf{P}\Big(\mathsf{Diam}(G_n) \geq \lambda n^{\frac{1}{d}}\Big) = o(\lambda^{\frac{-\alpha(1-\varepsilon)}{\alpha-1}})$ for all $\varepsilon > 0$. Otherwise, there exists $\delta > 0$ such that*

$$\limsup_{n\to\infty} \mathbf{P}\Big(\mathsf{Diam}(\mathcal{T}_n^{dec}) \geq n^{(1-\frac{1}{\alpha})(1+\delta)}\Big) \geq \limsup_{n\to\infty} \mathbf{P}\left(\exists u \in T_n : \frac{\mathsf{Diam}(G(u))}{(\deg u)^{\frac{1}{d}}} \geq n^{(1-\frac{1}{\alpha})(1+\delta)}\right) > 0.$$

3. *When $s_\alpha^d \leq 1$, it is certainly necessary that $\limsup_{n\to\infty} \mathbf{P}\Big(\mathsf{Diam}(G_n) \geq \lambda n^{\frac{1}{d}}\Big) = o(\lambda^{-d\alpha(1-\varepsilon)})$ for all $\varepsilon > 0$. Otherwise, there exists $\delta > 0$ such that*

$$\limsup_{n\to\infty} \mathbf{P}\Big(\mathsf{Diam}(\mathcal{T}_n^{dec}) \geq n^{\frac{1}{\alpha d}(1+\delta)}\Big) \geq \limsup_{n\to\infty} \mathbf{P}\left(\exists u \in T_n : \frac{\mathsf{Diam}(G(u))}{(\deg u)^{\frac{1}{d}}} \geq n^{\frac{1}{\alpha d}(1+\delta)}\right) > 0.$$

However, the condition of $o(\lambda^{\frac{-\alpha}{\alpha-1}})$ is probably not sharp in this regime.

The idea of the following proof is to write $\mathsf{Diam}(G(u)) = Y_u(\deg u)^{\frac{1}{d}}$ for each $u \in T_n$, and consider the behaviour of $(Y_u)_{u\in T_n}$ and $\left((\deg u)^{\frac{1}{d}}\right)_{u\in T_n}$ separately. Under Assumption 1.4(D′), $(Y_u)_{u\in T_n}$ satisfy the assumptions of [Mar20, Theorem 2]. Moreover, the same arguments as in Proposition 4.6 show that the degree distribution is at most size-biased along the decorated spine. Therefore, since $H^{\mathrm{dec}}(\mathcal{T}_n^{\mathrm{dec}}) \leq H(T_n) \leq \lambda n^{1-\frac{1}{\alpha}}$ with high probability as $\lambda \to \infty$, we deduce that (if $s_\alpha^d \neq 1$, otherwise we include an extra $\log n$ term), then

$$\begin{aligned}\lim_{\lambda\to\infty}\limsup_{n\to\infty}\mathbf{P}\left(\sum_{i=0}^{H^{\mathrm{dec}}(\mathcal{T}_n^{\mathrm{dec}})}(\deg s_i^{\mathrm{dec}})^{\frac{1}{d}} \geq \lambda n^{\frac{1}{\alpha d}\vee\left(1-\frac{1}{\alpha}\right)}\right) = 0\\ \lim_{\lambda\to\infty}\limsup_{n\to\infty}\mathbf{P}\left(\sum_{i=0}^{H^{\mathrm{dec}}(\mathcal{T}_n^{\mathrm{dec}})}\tilde{Y}_i \geq \lambda n^{1-\frac{1}{\alpha}}\right) = 0.\end{aligned} \tag{5}$$

Here we have written $\tilde{Y}_i$ in place of Y_i to emphasise the fact that the law of Y_i is biased by the fact that we are specifically considering the spine maximising the decorated height (the bias may also depend on n, but for clarity we suppress this notationally). We note the following.

1. By the same arguments as in Proposition 4.6, a size-biased bound on the tails of $\deg s_i^{\mathrm{dec}}$ holds independently of the values of $\left(\deg s_i^{\mathrm{dec}}\right)_{i\neq j}$;
2. This size-biased bound on the tails of $\deg s_i^{\mathrm{dec}}$ also holds independently of the value of $\tilde{Y}_i$;

It is therefore reasonable to expect in light of Lemma 3.4 that, when considering the sum $\sum_{i=0}^{H^{\mathrm{dec}}(\mathcal{T}_n^{\mathrm{dec}})}\tilde{Y}_i(\deg s_i^{\mathrm{dec}})^{\frac{1}{d}}$, the heaviest tails above will dominate, i.e. that

$$\lim_{\lambda\to\infty}\limsup_{n\to\infty}\mathbf{P}\left(\sum_{i=0}^{H^{\mathrm{dec}}(\mathcal{T}_n^{\mathrm{dec}})}(\deg s_i^{\mathrm{dec}})^{\frac{1}{d}}\tilde{Y}_i \geq \lambda n^{\frac{1}{\alpha d}\vee\left(1-\frac{1}{\alpha}\right)}\right) = 0.$$

If $s_\alpha^d = 1$ we similarly expect

$$\lim_{\lambda\to\infty}\limsup_{n\to\infty}\mathbf{P}\left(\sum_{i=0}^{H^{\mathrm{dec}}(\mathcal{T}_n^{\mathrm{dec}})}(\deg s_i^{\mathrm{dec}})^{\frac{1}{d}}\tilde{Y}_i \geq \lambda n^{\frac{1}{\alpha d}\vee\left(1-\frac{1}{\alpha}\right)}\log n\right) = 0.$$

We now prove this formally. We consider a sequence of i.i.d. random variables $\left(\left(Y_u^{(n)}\right)_{u\in T_n}\right)_{n\geq 1}$ satisfying the tail bound of Assumption 1.4 (D'). For each $n \geq 1, u, v \in T_n$ we let $Y_u^{(n,c)} = Y_u^{(n)} - \mathbf{E}\left[Y_{\rho_n}^{(n)}\right]$ denote the recentred random variables, and let $Z_v^{(n)} = \sum_{u\preceq v} Y_u^{(n)}, Z_v^{(n,c)} = \sum_{u\preceq v} Y_u^{(n,c)}$ denote the corresponding snake processes.

We first give a couple of lemmas which allow us to control $\max_{v\in T_n}\sum_{u\preceq v} Y_u$. The arguments use the same principles as those employed in [Mar20], but we just tweak some arguments as our aims here are slightly different (in particular, we would like to obtain tail bound that decays in λ uniformly in n). Similarly to the notation used there, if $u \in T_n$ we let $\mathsf{par}(u)$ denote its parent, and χ_u denote its relative position amongst the offspring of $\mathsf{par}(u)$. We also let $|u - v|$ denote the *lexicographical* distance between u and v: that is, the number of vertices explored by the height function between visiting u and v.

Lemma 4.13. *Fix some $k > 0$. There exists an event $E_{n,\lambda,k}$ with $\mathbf{P}\left(E_{n,\lambda,k}^c\right) \leq c\lambda^{-k}$ such that, for all $\gamma < 1 - \frac{1}{\alpha}, q > 0$*

$$\mathbf{E}\left[\left(\sup_{u,v\in T_n}\frac{n^{-(1-\frac{1}{\alpha})}d(u,v)}{(\log\lambda)\left(n^{-1}|u-v|\right)^\gamma}\right)^q \,\middle|\, E_{n,\lambda,k}\right] < \infty.$$

Proof. This follows from a minor modification of the arguments of [Mar20, proof of Lemma 1], by taking $C = \frac{10}{\xi(0)^2}$ and defining the event (cf [Mar20, Lemma 2])

$$E_{n,\lambda,k} = \left\{ \frac{|\{w \in ((u,v]] : \chi_x = 1\}|}{d_{T_n}(u,v)} \leq 1 - \frac{\xi(0)}{2} \text{ for all } u, v \in T_n \text{ such that } u \preceq v \text{ and } d_{T_n}(u,v) \geq C \log(n\lambda^k) \right\}.$$

By repeating the proof of [Mar20, Lemma 2], we have that $\mathbf{P}(E_{n,\lambda,k}) \leq c\lambda^{-k}$. Moreover, by exactly the same arguments as in [Mar20, proof of Lemma 1], if $u \preceq v$, $d(u,v) \geq C \log(n\lambda^k)$ and $\lambda > 1$ we have that

$$\mathbf{P}\left(\frac{n^{-(1-\frac{1}{\alpha})} d(u,v)}{(n^{-1}|u-v|)^\gamma} > x \;\middle|\; E_{n,\lambda,k} \right) \leq Ce^{-cx}$$

(cf [Mar20, Equation (6)]). If instead $u \preceq v$, $\lambda > 1$ but $d(u,v) < C \log(n\lambda^k)$ we instead have, since $|u-v| \geq 1$, that

$$\frac{n^{-(1-\frac{1}{\alpha})} d(u,v)}{(n^{-1}|u-v|)^\gamma} \leq n^{\gamma-(1-\frac{1}{\alpha})} \left[\log n + k \log \lambda\right] \leq k \log \lambda.$$

Either way, by breaking paths at the most recent common ancestor we certainly have for all $u, v \in T_n$ that

$$\mathbf{P}\left(\frac{n^{-(1-\frac{1}{\alpha})} d(u,v)}{k \log \lambda \, (n^{-1}|u-v|)^\gamma} > x \;\middle|\; E_{n,\lambda,k} \right) \leq Ce^{-cx},$$

so that Kolmogorov's continuity criterion (applied to the rescaled height function of T_n, which is indexed by $[0,1]$) gives that, for all $q > 0$,

$$\mathbf{E}\left[\sup_{u,v \in T_n} \left(\frac{n^{-(1-\frac{1}{\alpha})} d(u,v)}{k \log \lambda \, (n^{-1}|u-v|)^\gamma} \right)^q \;\middle|\; E_{n,\lambda,k} \right] < \infty.$$

□

Fix $\gamma < \frac{\alpha-1}{\alpha}$. We define the random variable κ_n by

$$\kappa_n = \sup_{u,v \in T_n} \frac{n^{-(1-\frac{1}{\alpha})} d(u,v)}{(\log \lambda) \, (n^{-1}|u-v|)^\gamma}. \tag{6}$$

Lemma 4.14. *Choose $p \in [2, \frac{\alpha}{\alpha-1})$ (recall that $\alpha < 2$) and take any $\theta < \frac{p}{2(p+2)}$. Then there exists $C_{p,\theta} < \infty$ such that*

$$\mathbf{E}\left[\sup_{u,v \in T_n} \left(\frac{n^{-\frac{1}{2}(1-\frac{1}{\alpha})} |Z_u^{(n,c)} - Z_v^{(n,c)}|}{\left(n^{-(1-\frac{1}{\alpha})} d(u,v) \right)^\theta} \right)^p \;\middle|\; E_{n,\lambda,k} \right]^{\frac{1}{p}} \leq C_{p,\theta} (\log \lambda)^{\frac{2\theta}{p}}.$$

Proof. a) The proof essentially follows the same steps as [Mar20, Theorem 2], though we have to be a bit careful about applying Kolmogorov's continuity criterion directly to the tree-indexed process (rather than considering $Z^{(n,c)}$ to be indexed by an interval of $\mathbb{R}$). Similarly to there, we first note that for all $u, v \in T_n$, we have by [HH80, Theorem 2.10] and the triangle inequality that for all $p \geq 2$, there exists $c_p < \infty$ such that

$$\mathbf{E}\left[|Z_u^{(n,c)} - Z_v^{(n,c)}|^p \right] \leq c_p \mathbb{E}\left[Y_\rho^p \right] d(u,v)^{\frac{p}{2}}. \tag{7}$$

We now essentially repeat the proof of Kolmogorov's continuity criterion for the tree-indexed process. In particular, we do the following:

- We turn T_n into a length space endowed with a continuous function by interpolating both $\left(Z_u^{(n,c)} \right)_{u \in T_n}$ and $(d(\rho,u))_{u \in T_n}$ along the edges of T_n. For notational convenience we still denote the resulting metric space by (T_n, d).

- For each $m \geq 0$, we let $\mathbb{D}_m = \left\{u \in T_n : 2^m n^{-\left(1-\frac{1}{\alpha}\right)} d(\rho_n, u) \in \mathbb{N}\right\}$ be the set of points at rescaled dyadic distance from the root.
- Conditioning on $E_{n,\lambda,k}$ and taking κ_n and γ as in (6), and setting $\theta = \frac{1}{\gamma}$ we have for all $a > 0$ that for all $u, v \in T_n$ with $d(u,v) \geq n^{1-\frac{1}{\alpha}} a$, it also holds that $|u - v| \geq n a^\theta \kappa_n^{-\theta} (\log \lambda)^{-\theta}$.
- In particular, the previous point implies that $|D_m| \leq 2^{m\theta} \kappa_n^\theta (\log \lambda)^\theta$ for all $m \geq 0$, and that $\mathsf{Height}(T_n) \leq \kappa_n n^{1-\frac{1}{\alpha}} \log \lambda$.
- If $u, v \in \mathbb{D}_m$, we write $u \overset{m}{\sim} v$ if there is a direct path from u to v such that the interior of the path is contained within a single connected component of $T_n \setminus \mathbb{D}_m$. Note that this implies that $d(u,v) \leq n^{1-\frac{1}{\alpha}} 2^{-(m-1)}$.
- Set $K_{n,m} = \sup_{u,v \in T_n : u \overset{m}{\sim} v} |Z_u^{(n,c)} - Z_v^{(n,c)}|$. By (7), we then have for any $2 \leq p < \frac{\alpha}{\alpha - 1}$ that

$$\mathbf{E}\left[K_{n,m}^p \,\middle|\, T_n, E_{n,\lambda,k}\right] \leq |D_m|^2 \tilde{c}_p \sup_{u,v \in T_n : u \overset{m}{\sim} v} d(u,v)^{\frac{p}{2}} \leq 2^{2m\theta} \kappa_n^{2\theta} (\log \lambda)^{2\theta} \tilde{c}_p \left(n^{1-\frac{1}{\alpha}} 2^{-(m-1)}\right)^{\frac{p}{2}}. \quad (8)$$

Now if $u, v \in T_n$ and $d(u,v) \in [n^{1-\frac{1}{\alpha}} 2^{-(m+1)}, n^{1-\frac{1}{\alpha}} 2^{-m}]$ for some $m > 0$, by breaking the path $[[u, v]]$ at $u \wedge v$ we have that

$$|Z_u^{(n,c)} - Z_v^{(n,c)}| \leq 4 \sum_{l \geq m} K_{n,l},$$

so that, if $2 \leq p < \frac{\alpha}{\alpha - 1}$ and $\theta < \frac{p}{2(p+2)}$, we have from (8) that

$$\begin{aligned}
&\mathbf{E}\left[\sup_{u,v \in T_n} \left(\frac{n^{-\frac{1}{2}(1-\frac{1}{\alpha})} |Z_u^{(n,c)} - Z_v^{(n,c)}|}{\left(n^{-\left(1-\frac{1}{\alpha}\right)} d(u,v)\right)^\theta}\right)^p \,\middle|\, T_n, E_{n,\lambda,k}\right]^{\frac{1}{p}} \\
&\leq 4 n^{-\frac{1}{2}(1-\frac{1}{\alpha})} 2^{(m+1)\theta} \sum_{l \geq m} \mathbf{E}\left[K_{n,l}^p \,\middle|\, T_n, E_{n,\lambda,k}\right]^{\frac{1}{p}} \\
&\leq n^{-\frac{1}{2}(1-\frac{1}{\alpha})} 2^{(m+1)\theta} \sum_{l \geq m} \left(2^{2l\theta} \kappa_n^{2\theta} (\log \lambda)^{2\theta} \tilde{c}_p\right)^{\frac{1}{p}} \left(n^{1-\frac{1}{\alpha}} 2^{-(l-1)}\right)^{\frac{1}{2}} \\
&\leq \left(\kappa_n^{2\theta} (\log \lambda)^{2\theta} \tilde{c}_p\right)^{\frac{1}{p}} 2^{(m+1)\theta} \sum_{l \geq m} 2^{\left(\frac{2\theta}{p} - \frac{1}{2}\right) l} \\
&\leq c_{p,\theta} \left(\kappa_n^{2\theta} (\log \lambda)^{2\theta}\right)^{\frac{1}{p}}.
\end{aligned}$$

(Note this also works if $m < 0$). By the tower property, we therefore deduce from Lemma 4.13 that

$$\begin{aligned}
&\mathbf{E}\left[\sup_{u,v \in T_n} \left(\frac{n^{-\frac{1}{2}(1-\frac{1}{\alpha})} |Z_u^{(n,c)} - Z_v^{(n,c)}|}{\left(n^{-\left(1-\frac{1}{\alpha}\right)} d(u,v)\right)^\theta}\right)^p \,\middle|\, E_{n,\lambda,k}\right] \\
&= \mathbf{E}\left[\mathbf{E}\left[\sup_{u,v \in T_n} \left(\frac{n^{-\frac{1}{2}(1-\frac{1}{\alpha})} |Z_u^{(n,c)} - Z_v^{(n,c)}|}{\left(n^{-\left(1-\frac{1}{\alpha}\right)} d(u,v)\right)^\theta}\right)^p \,\middle|\, T_n, E_{n,\lambda,k}\right] \,\middle|\, E_{n,\lambda,k}\right] \\
&\leq c_{p,\theta} (\log \lambda)^{\frac{2\theta}{p}} \mathbf{E}\left[\kappa_n^{\frac{2\theta}{p}} \,\middle|\, E_{n,\lambda,k}\right] \\
&= c'_{p,\theta} (\log \lambda)^{\frac{2\theta}{p}}.
\end{aligned}$$

□

Remark 4.15. *For any $\varepsilon > 0$ we expect that we can replace the term $|D_m|^2$ in (8) with something of order $|D_m|^{1+\varepsilon}$.*

Given p, θ satisfying the assumptions of Lemma 4.14, and $\lambda > 1$, we define the random variable $\zeta_{n,\lambda,p}$ by

$$\zeta_{n,\lambda,p} = \sup_{u,v \in T_n} \frac{n^{-\frac{1}{2}(1-\frac{1}{\alpha})} |Z_u^{(n,c)} - Z_v^{(n,c)}|}{(\log \lambda)^{\frac{2\theta}{p}} \left(n^{-\left(1-\frac{1}{\alpha}\right)} d(u,v) \right)^{\theta}}. \tag{9}$$

Proof of Proposition 4.11. First assume that $s_\alpha^d = d(\alpha - 1) < 1$. Let Γ_n denote the set of branches of T_n, i.e. the set $\{[[\rho_n, x]] : \deg x = 1\}$. For now, we take some $x, h > 0$ (to be specified later) and work conditionally on $H^{\mathrm{dec}}(\mathcal{T}_n^{\mathrm{dec}}) \leq n^{1-\frac{1}{\alpha}} \lambda^h, \sup_{u \in \gamma_n} (\deg u)^{\frac{1}{d}} < n^{\frac{1}{\alpha d}} \lambda^x$ and on the event $E_{n,\lambda,k}$.

Let γ_n be the decorated spine, i.e. such that

$$\sum_{u \in \gamma_n} \mathsf{Diam}(G(u)) = \sup_{\gamma \in \Gamma_n} \sum_{u \in \gamma_n} \mathsf{Diam}(G(u))$$

For each $u \in T_n$, we write $\mathsf{Diam}(G(u)) = Y_u (\deg u)^{\frac{1}{d}}$. Note that $(Y_u)_{u \in T_n}$ is an i.i.d sequence satisfying the tail bound of Assumption 1.4(D′). Along the branch γ_n, we instead write $\mathsf{Diam}(G(u)) = \tilde{Y}_u (\deg u)^{\frac{1}{d}}$ to emphasise that the laws of $(Y_u)_{u \in \gamma_n}$, are biased by our choice of the branch γ_n, and therefore do not necessarily satisfy the tail bound of Assumption 1.4(D′). However, we nevertheless have that:

- For each $u \in \gamma_n$, $\mathbf{P}\left((\deg u)^{\frac{1}{d}} \geq x \right) \leq c x^{-s_\alpha^d}$ for all $x \geq 1$, independently of the value of $\tilde{Y}_u$, and independently of the values of $(\deg v)_{v \in \gamma_n \setminus \{u\}}$: this is essentially because, if u has k offspring, then this gives k possibilities for the next spinal vertex. More formally, this holds for similar reasons as it does for the Williams' spine on the tree. In particular, it is shown in the construction on [GK99, p.3] that the offspring distribution of each vertex on the Williams' spine of a critical Galton-Watson tree can be independently stochastically dominated by a size-biased version of the offspring distribution for that tree (this does not mean that the vertex degrees themselves are independent of each other, just that they can be independently stochastically dominated). This is achieved by building the Galton-Watson tree recursively, by starting at the tip of the Williams' spine and working backwards towards the root, and the size-biased distribution corresponds to a heaviest tail one could possibly obtain during this process, and specifically arises when conditioning the tree to survive forever beyond the current vertex.

  The same logic applies to the case when we are maximising the decorated height over all paths in the tree: once γ_n is identified, we can work backwards from its tip to construct the appropriate spine recursively, write down a similar expression to [GK99, Equation (1)] that determines the offspring distribution, which in turn implies that there exists a constant $c \leq \infty$ such that

  $$\mathbf{P}(\deg(u) \geq k) \leq ckp_k \tag{10}$$

  for all $u \in \gamma_n$. Moreover, the algorithm on [GK99, p.3] is insightful because at each step the only dependence on the previous steps is through the restriction that the heights of the subtrees sprouting from the non-spinal offspring of the next spinal vertex must not exceed the height of the Williams' spine, and the bound (10) holds for $\deg(s_i)$ regardless of the value of $\deg(s_{i+1})$. The same principle applies in this case, so that conditional on $|\gamma_n|$ and $\left(\tilde{Y}_u\right)_{u \in \gamma_n}$, we can therefore upper bound $(\deg u)_{u \in \gamma_n}$ by a sequence $(X_u)_{u \in \gamma_n}$, where X_u independently satisfy the tail bound of (10).

- For any $\varepsilon \in (0,1), \delta > 0$, and any p, θ satisfying the assumptions of Lemma 4.14, we have that

  $$\mathbf{P}\left(\sum_{u \in \gamma_n} \tilde{Y}_u \geq n^{1-\frac{1}{\alpha}} \lambda^q \,\middle|\, E_{n,\lambda,k} \right) \leq n^{-p(1-\varepsilon)(1-\frac{1}{\alpha})} \lambda^{-[(1-\varepsilon)pq - \delta]}.$$

  To see this, recall from Lemma 4.14 that conditionally on $E_{n,\lambda,k}$,

  $$\zeta_{n,\lambda,p} := \sup_{u,v \in T_n} \left(\frac{n^{-\frac{1}{2}(1-\frac{1}{\alpha})} |Z_u^{(n,c)} - Z_v^{(n,c)}|}{(\log \lambda)^{\frac{2\theta}{p}} \left(n^{-\left(1-\frac{1}{\alpha}\right)} d(u,v) \right)^{\theta}} \right)$$

is such that $\sup_{n,\lambda>1}\zeta_{n,\lambda,p}$ has a finite p^{th} moment. Moreover, by assumption,

$$\sum_{u\in\gamma_n}\tilde{Y}_u \le m|\gamma_n| + \sup_{u\in\gamma_n} Z_u^{(n,c)} \le m|\gamma_n| + \zeta_{n,\lambda,p}(\log\lambda)^{\frac{2\theta}{p}} n^{\frac{1}{2}(1-\frac{1}{\alpha})}\left(n^{-(1-\frac{1}{\alpha})}|\gamma_n|\right)^{\theta}. \tag{11}$$

Therefore, for any $\varepsilon\in(0,1)$ and any $\delta>0$, we have by [Kor17, Theorem 2] that

$$\begin{aligned}
\mathbf{P}\left(\sum_{u\in\gamma_n}\tilde{Y}_u \ge n^{1-\frac{1}{\alpha}}\lambda^q\right) &\le \mathbf{P}\left(\zeta_{n,\lambda,p}(\log\lambda)^{\frac{2\theta}{p}}\left(n^{-(1-\frac{1}{\alpha})}|\gamma_n|\right)^{\theta} \ge \frac{1}{2}n^{1-\frac{1}{\alpha}}\lambda^q\right)\\
&\le \mathbf{P}\left(\mathsf{Height}(T_n)^{\theta} \ge \frac{1}{2}n^{(\theta+\varepsilon)(1-\frac{1}{\alpha})}\lambda^{\varepsilon q}\right) + \mathbf{P}\left(\zeta_{n,\lambda,p} \ge n^{(1-\varepsilon)(1-\frac{1}{\alpha})}\lambda^{(1-\varepsilon)q}(\log\lambda)^{\frac{-2\theta}{p}}\right)\\
&\le \exp\left\{-c\left(n^{\frac{\varepsilon}{\theta}}\lambda^{\frac{\varepsilon q}{\theta}}\right)^{\alpha-\delta}\right\} + n^{-p(1-\varepsilon)(1-\frac{1}{\alpha})}\lambda^{-[(1-\varepsilon)pq-\delta]}.
\end{aligned}$$

For notational convenience, we take some x,h as before, some $z>0$ (to be specified later) and define the event

$$F_{n,\lambda,k} = \left\{\sup_{u\in\gamma_n}(\deg u)^{\frac{1}{d}} < n^{\frac{1}{\alpha d}}\lambda^x, |\gamma_n| \le n^{1-\frac{1}{\alpha}}\lambda^h, \zeta_{n,\lambda,p}\le\lambda^z\right\}\cap E_{n,\lambda,k}.$$

If $s_\alpha^d<1$, note also that by the fact that (10) holds independently for each $u\in\gamma_n$, and also independently of $\tilde{Y}_u$, we have for each $u\in\gamma_n$ that

$$\mathbf{E}\left[(\deg u)^{\frac{1}{d}}\;\middle|\;\left(\tilde{Y}_u\right)_{u\in\gamma_n},\zeta_{n,\lambda,p},|\gamma_n|,F_{n,\lambda,k}\right] = \mathbf{E}\left[(\deg u)^{\frac{1}{d}}\;\middle|\;\sup_{u\in\gamma_n}(\deg u)^{\frac{1}{d}} < n^{\frac{1}{\alpha d}}\lambda^x\right] \le cn^{\frac{1}{\alpha d}(1-s_\alpha^d)}\lambda^{x(1-s_\alpha^d)}.$$

Using (11), we can therefore write

$$\begin{aligned}
&\mathbf{E}\left[\sum_{u\in\gamma_n}\tilde{Y}_u(\deg u)^{\frac{1}{d}}\;\middle|\;F_{n,\lambda,k}\right] = \mathbf{E}\left[\mathbf{E}\left[\sum_{u\in\gamma_n}\tilde{Y}_u(\deg u)^{\frac{1}{d}}\;\middle|\;\zeta_{n,\lambda,p},|\gamma_n|,F_{n,\lambda,k}\right]\;\middle|\;F_{n,\lambda,k}\right]\\
&= \mathbf{E}\left[\mathbf{E}\left[\mathbf{E}\left[\sum_{u\in\gamma_n}\tilde{Y}_u(\deg u)^{\frac{1}{d}}\;\middle|\;\left(\tilde{Y}_u\right)_{u\in\gamma_n},\zeta_{n,\lambda,p},|\gamma_n|,F_{n,\lambda,k}\right]\;\middle|\;\zeta_{n,\lambda,p},|\gamma_n|,F_{n,\lambda,k}\right]\;\middle|\;F_{n,\lambda,k}\right]\\
&= \mathbf{E}\left[\mathbf{E}\left[\sum_{u\in\gamma_n}\tilde{Y}_u\mathbf{E}\left[(\deg u)^{\frac{1}{d}}\;\middle|\;\left(\tilde{Y}_u\right)_{u\in\gamma_n},\zeta_{n,\lambda,p},|\gamma_n|,F_{n,\lambda,k}\right]\;\middle|\;\zeta_{n,\lambda,p},|\gamma_n|,F_{n,\lambda,k}\right]\;\middle|\;F_{n,\lambda,k}\right]\\
&\le \mathbf{E}\left[\mathbf{E}\left[cn^{\frac{1}{\alpha d}(1-s_\alpha^d)}\lambda^{x(1-s_\alpha^d)}\sum_{u\in\gamma_n}\tilde{Y}_u\;\middle|\;\zeta_{n,\lambda,p},|\gamma_n|,F_{n,\lambda,k}\right]\;\middle|\;F_{n,\lambda,k}\right]\\
&\le cn^{\frac{1}{\alpha d}(1-s_\alpha^d)}\lambda^{x(1-s_\alpha^d)}\mathbf{E}\left[\mathbf{E}\left[\left(m|\gamma_n|+\zeta_{n,\lambda,p}(\log\lambda)^{\frac{2\theta}{p}}n^{\frac{1}{2}(1-\frac{1}{\alpha})}\left(n^{-(1-\frac{1}{\alpha})}|\gamma_n|\right)^{\theta}\right)\;\middle|\;\zeta_{n,\lambda,p},|\gamma_n|,F_{n,\lambda,k}\right]\;\middle|\;F_{n,\lambda,k}\right]\\
&\le c\left(m\lambda^{h+x[1-d(\alpha-1)]}+\lambda^z\lambda^{\theta h+x[1-d(\alpha-1)])}n^{\frac{-1}{2}(1-\frac{1}{\alpha})}(\log\lambda)^{\frac{2\theta}{p}}\right)n^{\frac{1}{\alpha d}}.
\end{aligned}$$

We can treat the cases $s_\alpha^d=1$ and $s_\alpha^d>1$ (in which case $\mathbf{E}\left[(\deg u)^{\frac{1}{d}}\;\middle|\;\sup_{u\in\gamma_n}(\deg u)^{\frac{1}{d}}<n^{\frac{1}{\alpha d}}\lambda^x\right]$ is upper bounded by $c_x\log(n\lambda)$ and c respectively) by the same arguments and get that

$$\mathbf{E}\left[\sum_{u\in\gamma_n}\tilde{Y}_u(\deg u)^{\frac{1}{d}}\;\middle|\;F_{n,\lambda,k}\right] \le \begin{cases} c\left(m\lambda^{h+x[1-d(\alpha-1)]}+\lambda^z\lambda^{\theta h+x[1-d(\alpha-1)])}n^{\frac{-1}{2}(1-\frac{1}{\alpha})}(\log\lambda)^{2\theta}\right)n^{\frac{1}{\alpha d}} & \text{if } s_\alpha^d<1,\\ c\left(m\lambda^{h+x[1-d(\alpha-1)]}+\lambda^z\lambda^{\theta h+x[1-d(\alpha-1)])}n^{\frac{-1}{2}(1-\frac{1}{\alpha})}(\log\lambda)^{2\theta}\right)n^{\frac{1}{\alpha d}}(\log n\lambda) & \text{if } s_\alpha^d=1,\\ c\left(m\lambda^{h+x[1-d(\alpha-1)]}+\lambda^z\lambda^{\theta h+x[1-d(\alpha-1)])}n^{\frac{-1}{2}(1-\frac{1}{\alpha})}(\log\lambda)^{2\theta}\right)n^{1-\frac{1}{\alpha}} & \text{if } s_\alpha^d>1.\end{cases}$$

Continuing with the working for the case $s_\alpha^d<1$, it therefore follows by Markov's inequality that

$$\mathbf{P}\left(\sum_{u\in\gamma_n}\tilde{Y}_u(\deg u)^{\frac{1}{d}}\ge n^{\frac{1}{\alpha d}}\lambda\;\middle|\;F_{n,\lambda,k}\right) \le c\left(m\lambda^{-(1-x[1-d(\alpha-1)]-h)}+\lambda^{-(1-z-\theta h-x[1-d(\alpha-1)])}(\log\lambda)^{\frac{2\theta}{p}}n^{\frac{-1}{2}(1-\frac{1}{\alpha})}\right).$$

To bring this all together, we therefore take a union bound and write

$$\begin{aligned}
&\mathbf{P}\left(\sum_{u\in\gamma_n}\mathsf{Diam}(G(u)) \geq n^{\frac{1}{\alpha d}}\lambda \;\middle|\; E_{n,\lambda,k}\right)\\
&\leq \mathbf{P}\left(\sup_{u\in\gamma_n}(\deg u)^{\frac{1}{d}} > n^{\frac{1}{\alpha d}}\lambda^x\right) + \mathbf{P}\Big(\mathsf{Height}(T_n) \geq n^{1-\frac{1}{\alpha}}\lambda^h\Big) + \mathbf{P}(\zeta_{n,\lambda,p} > \lambda^z \mid E_{n,\lambda,k})\\
&\qquad + \mathbf{P}\left(\sum_{u\in\gamma_n}\tilde{Y}_u(\deg u)^{\frac{1}{d}} \geq n^{\frac{1}{\alpha d}}\lambda \;\middle|\; F_{n,\lambda,k}\right)\\
&\leq \lambda^{-\alpha x} + \lambda^{\frac{-h}{\alpha-1}} + \lambda^{-pz} + m\lambda^{-(1-x[1-d(\alpha-1)]-h)} + \lambda^{-(1-z-\theta h-x[1-d(\alpha-1)])}(\log\lambda)^{\frac{2\theta}{p}}.
\end{aligned}$$

For sufficiently small $\varepsilon > 0$, we can optimise this by taking $p = \frac{\alpha}{\alpha-1} - \varepsilon, \theta = \varepsilon, x = \frac{1}{\alpha+1+[1-s_\alpha^d]\alpha(\alpha-1)} \wedge \frac{p}{\alpha+p(s_\alpha^d+\alpha)}, h = \frac{\alpha(\alpha-1)x}{\alpha+1+[1-s_\alpha^d]\alpha(\alpha-1)}, z = \frac{\alpha}{\alpha+p(s_\alpha^d+\alpha)}$, which gives overall tail decay of λ^{-t}, where

$$t = \frac{\alpha-\varepsilon}{\alpha+1+[1-s_\alpha^d]\alpha(\alpha-1)} \wedge \frac{\alpha^2-\varepsilon}{\alpha(\alpha-1)+\alpha(s_\alpha^d+\alpha)}.$$

Choosing $k > t$, we also get the same upper bound for $\mathbf{P}(E_{n,\lambda,k})$ from Lemma 4.13 (note that this choice of k only affects the values of some of the constants appearing in the above calculation, and not the exponents for the tail decay). We get the same tail bound for $\mathbf{P}\Big(\sum_{u\in\gamma_n}\mathsf{Diam}(G(u)) \geq n^{\frac{1}{\alpha d}}(\log n)\lambda\Big)$ if $s_\alpha^d = 1$, in which case $t = \frac{\alpha-\varepsilon}{\alpha+1}$. If $s_\alpha^d > 1$, we don't need to condition on the maximal degree in T_n and we similarly get

$$\begin{aligned}
&\mathbf{P}\left(\sum_{u\in\gamma_n}\mathsf{Diam}(G(u)) \geq n^{1-\frac{1}{\alpha}}\lambda \;\middle|\; E_{n,\lambda,k}\right)\\
&\leq \mathbf{P}\Big(\mathsf{Height}(T_n) \geq n^{1-\frac{1}{\alpha}}\lambda^h\Big) + \mathbf{P}(\zeta_{n,\lambda,p} > \lambda^z \mid E_{n,\lambda,k})\\
&\qquad + \mathbf{P}\left(\sum_{u\in\gamma_n}\tilde{Y}_u(\deg u)^{\frac{1}{d}} \geq n^{1-\frac{1}{\alpha}}\lambda \;\middle|\; |\gamma_n| \leq n^{1-\frac{1}{\alpha}}\lambda^h, \zeta_{n,\lambda,p} \leq \lambda^z, E_{n,\lambda,k}\right)\\
&\leq \lambda^{\frac{-h}{\alpha-1}} + \lambda^{-pz} + m\lambda^{-(1-h)} + \lambda^{-(1-z-\theta h)}
\end{aligned}$$

and optimise by taking $\theta = \varepsilon, p = \frac{\alpha}{\alpha-1} - \varepsilon, z = \frac{1}{p+1}, h = 1 - \frac{1}{\alpha}$, giving an upper bound of $\lambda^{\frac{-(1-\varepsilon)}{\alpha}}$. □

4.3 Decorated volume bounds

We now give the asymptotics for the tail decay for the volume of $\mathcal{T}^{\mathrm{dec}}$. Recall the fragmental volume exponent $f_\alpha^v = \frac{\alpha}{v}$.

Proposition 4.16. *If $f_\alpha^v \neq 1$,*

$$\mathbf{P}\big(V^{dec}(\mathcal{T}^{dec}) \geq x\big) \leq x^{\frac{-(f_\alpha^v \wedge 1)}{\alpha}}$$

as $x \to \infty$. If $f_\alpha^v = 1$,

$$\mathbf{P}\big(V^{dec}(\mathcal{T}^{dec}) \geq x\log x\big) \leq x^{\frac{-(f_\alpha^v \wedge 1)}{\alpha}}$$

as $x \to \infty$.

Proof. Note that applying the Vervaat transform and then the absolute continuity relation as in Lemma 4.7 allows us to treat the different vertices in T as though they are independent. Therefore, if $f_\alpha^v \neq 1$ the result follows by two applications of Lemma 3.4: firstly, by similar arguments to those in Lemma 4.7, it follows that $\mathbf{P}(\deg(v) \geq k) \leq ck^{-\alpha}$ for all $v \in T_n$, so taking $\beta = \alpha, z = v, m = A_v$ in Lemma 3.4 gives $\mathbf{P}\big(V^{\mathrm{dec}}(G(v)) \geq x\big) \leq x^{-\tilde{f}_\alpha^v}$.

Then, since $V^{\mathrm{dec}}(\mathcal{T}_n^{\mathrm{dec}}) = \sum_{v\in T_n} V^{\mathrm{dec}}(G(v))$, we get from Lemma 3.1 that for any $\varepsilon > 0$,

$$\mathbf{P}\Big(V^{\mathrm{dec}}(\mathcal{T}_n^{\mathrm{dec}}) \geq \lambda n^{\frac{1}{f_\alpha^v \wedge 1}}\Big) \leq c\lambda^{-(\tilde{f}_\alpha^v - \varepsilon)}.$$

Then, using Lemma 4.1 and choosing ε sufficiently small, we can apply Lemma 3.4 again with $\beta = \frac{1}{\alpha}, z = \frac{1}{f_\alpha^v \wedge 1}, m = \tilde{f}_\alpha^v - \varepsilon$ to deduce the result.

If instead $f_\alpha^v = 1$, we have to incorporate the extra log term from Lemma 3.1, but the proof is the same on applying Lemma 3.4 and setting $f(n) = \frac{V^{\mathrm{dec}}(T_n)}{\log n}$ there. □

Proposition 4.17. *If $f_\alpha^v \leq 1$, then there exists $c, C \in (0,\infty)$ such that for all $x > 1$*

$$cx^{\frac{-1}{v}} \leq \mathbf{P}\Big(\sup_{v\in T} V^{dec}(G(v)) > x\Big) \leq Cx^{\frac{-1}{v}}.$$

Proof. First note that, by Assumption 1.1 (D) and Lemma 3.4 with $\beta = \alpha$ and $z = v$, we have that

$$cx^{\frac{-\alpha}{v}} \leq \mathbf{P}(\mathsf{Diam}(G(v_U)) \geq x) \leq Cx^{\frac{-\alpha}{v}}.$$

Then, similarly to the proof of Lemma 4.7, given $|T| = n$, and applying the Vervaat transform and absolute continuity relation on the first $\lfloor \frac{n}{2} \rfloor$ vertices of T_n, if $v_1, \ldots, v_n$ denote the vertices of T_n, and v_U is a uniform vertex of T_n, we have that

$$\mathbf{P}\Big(\sup_{v\in T_n} V^{\mathrm{dec}}(G(v)) > \lambda n^{\frac{v}{\alpha}}\Big) \leq \sum_{i=1}^{n} \mathbf{P}\big(V^{\mathrm{dec}}(G(v_i)) \geq \lambda n^{\frac{v}{\alpha}}\big) = n\mathbf{P}\big(V^{\mathrm{dec}}(G(v_U)) \geq \lambda n^{\frac{v}{\alpha}}\big) \leq C\lambda^{\frac{-\alpha}{v}}$$

$$\mathbf{P}\Big(\sup_{v\in T_n} V^{\mathrm{dec}}(G(v)) \geq n^{\frac{-v}{\alpha}}\lambda\Big) \geq 1 - (1 - cn^{-1}\lambda^{\frac{-\alpha}{v}})^{\frac{n-1}{2}} \geq 1 - e^{-c\lambda^{\frac{-\alpha}{v}}} \geq c\lambda^{\frac{-\alpha}{v}}.$$

Then, since $\mathbf{P}(|T| \geq x) \sim cx^{\frac{-1}{\alpha}}$ by Lemma 4.1, we can again apply Lemma 3.4 with $\beta = \frac{1}{\alpha}, z = \frac{v}{\alpha}$ to deduce the result. □

5 Volume bounds for $\mathcal{T}_\alpha^{\mathbf{dec}}$

In this section we prove volume bounds for $\mathcal{T}_\alpha^{\mathrm{dec}}$ under Assumption 1.1. The proof for the upper bounds follow a simplified version of the strategy for stable looptrees in [Arc20a, Section 5.2], but due to the variability of the inserted graphs we do not optimise the argument as precisely as we do in [Arc20a].

Recall that

$$d_\alpha^{\mathrm{vol}} = \frac{\alpha(s_\alpha^d \wedge 1)}{(\alpha - 1)(f_\alpha^v \wedge 1)}.$$

In this section we will show that this is the correct volume growth exponent for $\mathcal{T}_\alpha^{\mathrm{dec}}$.

We have stated the volume bounds with respect to the decorated metric as defined by (4), since these are of independent interest aside from determining the random walk exponents. However, the construction in (4) holds when $d_{G(v_i)}$ is an arbitrary metric on $G(v_i)$, so in particular, on replacing s_α^d with s_α^R we obtain the exponent for volume growth with respect to the effective resistance metric.

For all of Section 5, we will work under Assumption 1.1.

5.1 Volume upper bounds

The main result is as follows.

Proposition 5.1. *Take b_1 as in Theorem 1.3. For any $\varepsilon > 0, r, \lambda > 1$, there exist constants $c, C \in (0, \infty)$ such that*

$$\mathbf{P}\left(V^{dec}(B_{\mathcal{T}_\alpha^{dec}}(\rho, r)) \geq r^{\frac{\alpha(s_\alpha^d \wedge 1)}{(\alpha-1)(f_\alpha^v \wedge 1)}} (\log r)^{b_1} \lambda \right) \leq C\lambda^{\frac{-(f_\alpha^v \wedge 1 - \varepsilon)(\alpha-1)}{\alpha^2}}.$$

Remark 5.2.

1. *The tail decay here is not optimal. In all of the propositions in the rest of the section, the precise tail decay is not important for our purpose, other than that it is of polynomial form.*
2. *In the case $f_\alpha^v > 1$ and $s_\alpha^d < 1$, if the graphs inserted are deterministic or do not have too much randomness, it should be possible to extend the arguments of this section to get stretched exponential decay for the upper volume bounds, similarly to how we did for stable looptrees in [Arc20a, Section 5.2]. This would involve defining an interative procedure on large subtrees that fall close to the root, analogously to that on [Arc20a, p. 23].*

This therefore gives half of Theorem 1.3(i). The quenched upper bound in Theorem 1.3(ii) follows from this proposition by applying Borel-Cantelli along the subsequence $r_n = 2^n$ with $\lambda_r = (\log r)^{\beta_1}$, and using monotonicity of $V^{\mathrm{dec}}(B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r))$.

5.1.1 Heuristics

Fix $r \geq 1$. Before starting the proof, we briefly outline the strategy, which has several steps.

1. Consider the vertices of the underlying tree T_α^∞ along its backbone in sequential order of their distance from the root, and label them in order as $\rho = s_0, s_1, \ldots$.
2. We will make an appropriate choice of a vertex s_{i_r} so that all of $B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)$ is completely contained within the inserted graphs corresponding to the segment of backbone from ρ to s_{i_r} and the decorated subtrees attached to these graphs.
3. We will first bound the quantity
$$N_r^f = \sum_{j \leq i_r} \deg(s_j),$$
which corresponds to the number of subtrees attached to the backbone within (decorated) distance r of the root. More specifically, we will show that, w.h.p. as $r, \lambda \to \infty$ appropriately, $N_r^f \leq r^{\frac{s_\alpha^d \wedge 1}{\alpha - 1}} \lambda$.
4. On this event, we proceed as follows. We first set
$$\mathrm{Spine}_r = \bigcup_{j \leq i_r} G(s_j) \cap B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho, r), \qquad \partial\mathrm{Spine}_r = \bigcup_{j \leq i_r} \partial G(s_j) \cap B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho, r),$$
where $\partial G(v)$ denotes the set of boundary vertices of $G(v)$. We then continue by analysing the tree structures of all the decorated subtrees that are grafted to a vertex $v \in \partial\mathrm{Spine}_r$. Recall that such a vertex v corresponds to an edge of the underlying tree T_α^∞ joining a backbone vertex to one of its offspring. Let this offspring vertex be w_v. By Definition 2.2, T_{w_v} is a critical Galton-Watson tree, again with offspring distribution ξ. We let $\mathcal{T}_v^{\mathrm{dec}}$ denote the corresponding decorated subtree.

   It is then the case that
$$V^{\mathrm{dec}}(B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho, r)) \leq \sum_{v \in \partial\mathrm{Spine}_r} V^{\mathrm{dec}}(\mathcal{T}_v^{\mathrm{dec}}) + \sum_{j \leq i_r} V^{\mathrm{dec}}(G(s_j)). \tag{12}$$

We bound each of these terms separately in the next subsection.

5.1.2 Main argument

In order to instigate the method outlined above, we need to bound the following quantities:

1. The number of subtrees grafted to the backbone of $\mathcal{T}_\alpha^{\mathrm{dec}}$ within decorated distance r of $\rho_\alpha^{\mathrm{dec}}$.
2. The sum of the volumes of these decorated subtrees.
3. The volume contribution from the graphs inserted along the backbone.

We consider these one by one in the next subsections. At many points, this will involve adding up sums of random variables with exponents corresponding to those we introduced in Assumption 1.1 and Section 4. In many cases, the relevant exponent may be more than or less than 1, so we will have to consider two regimes: one in which the sum follows law of large numbers type behaviour, and the other in which the tails are heavier and we see behaviour more like that of a stable subordinator. This will eventually give rise to several phase transitions in the value of d_α^{vol}, which can also be surmised from its expression as

$$d_\alpha^{\mathrm{vol}} = \frac{\alpha(s_\alpha^d \wedge 1)}{(\alpha-1)(f_\alpha^v \wedge 1)}.$$

5.1.3 Controlling the number of subtrees

We start with the case $s_\alpha^d < 1$. For $r > 0$, we let N_r^f denote the number of decorated subtrees that are grafted to the decorated backbone within decorated distance r of $\rho_\alpha^{\mathrm{dec}}$.

Proposition 5.3. *Suppose that $s_\alpha^d \neq 1$. Then there exists $C < \infty$ such that for all $r, \lambda > 1$,*

$$\mathbf{P}\Big(N_r^f \geq r^{\frac{s_\alpha^d}{\alpha-1}}\lambda\Big) \leq C\lambda^{-(\alpha-1-\varepsilon)}.$$

If $s_\alpha^d = 1$, then

$$\mathbf{P}\Big(N_r^f \geq \lambda r^{\frac{1}{\alpha-1}}(\log r)^{\frac{-1}{\alpha-1}}\Big) \leq C\lambda^{-(\alpha-1-\varepsilon)}.$$

Proof. We prove the case $s_\alpha^d < 1$, in which case $s_\alpha^d = \tilde{s}_\alpha^d$. Choose $\varepsilon > 0$, and set $N_r = r^{s_\alpha^d}\lambda^\varepsilon$. Consider the sequence of spinal vertices $s_0, s_1, \ldots, s_{N_r}$. We will show that, with high probability, Spine_r does not extend to graphs corresponding to spinal vertices beyond s_{N_r}, in which case N_r^f is bounded by $\sum_{i=1}^{N_r} \deg(s_i)$. By Lemmas 4.10 and 3.2(i), we therefore have that

$$\mathbf{P}\left(\sum_{i\leq N_r} d^U(s_i) \leq r\right) \leq \mathbf{P}\big(\nexists i \leq N_r : d^U(s_i) \geq r\big) \leq (1 - cr^{-s_\alpha^d})^{r^{s_\alpha^d}\lambda^\varepsilon} \leq e^{-c\lambda^\varepsilon}.$$

Also, by Lemma 3.1(i),

$$\mathbf{P}\left(\sum_{i=0}^{N_r} \deg(s_i) \geq r^{\frac{s_\alpha^d}{\alpha-1}}\lambda\right) \leq \mathbf{P}\left(N_r^{\frac{-1}{\alpha-1}} \sum_{i=0}^{N_r} \deg(s_i) \geq \lambda^{1-\varepsilon'}\right) \leq C\lambda^{-(1-\varepsilon')(\alpha-1)}$$

Combining these in a union bound, we deduce that

$$\mathbf{P}\Big(N_r^f \geq r^{\frac{s_\alpha^d}{\alpha-1}}\lambda\Big) \leq Ce^{-c\lambda^{\varepsilon\beta}} + C\lambda^{-(1-\varepsilon')(\alpha-1)} \leq C'\lambda^{-(1-\varepsilon')(\alpha-1)}.$$

The proofs of the two other cases are identical, except that we respectively apply Lemma 3.2(ii) and 3.2(iii) in place of Lemma 3.2(i). □

5.1.4 Controlling the volumes of the decorated subtrees

We now turn to bounding the quantity

$$\sum_{v\in\partial\mathrm{Spine}_r(x)} V^{\mathrm{dec}}(\mathcal{T}_v^{\mathrm{dec}}).$$

In keeping with earlier notation, we set $N_r^f = |\partial\mathrm{Spine}_r|$. We will condition on $N_r^f < r^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda$, and recall from Proposition 4.16 that for $v\in\partial\mathrm{Spine}_r$, $\mathbf{P}\big(V^{\mathrm{dec}}(\mathcal{T}_v^{\mathrm{dec}}\geq x\big) \leq c'x^{\frac{-(f_\alpha^v\wedge 1)}{\alpha}}$ for all $x>1$. We deduce that this expression falls into the framework of Lemma 3.1, with $\beta = \frac{f_\alpha^v\wedge 1}{\alpha}$ and $n = N_r^f$; the next proposition is an immediate consequence.

Proposition 5.4. *Take b_1 as defined in Theorem 1.3. If $s_\alpha^d \neq 1$, we have for any $z < \frac{f_\alpha^v\wedge 1}{\alpha}$ that*

$$\mathbf{P}\left(\sum_{v\in\partial Spine_r} V^{dec}(\mathcal{T}_v^{dec}) \geq \lambda r^{d_\alpha^{vol}}(\log r)^{b_1}, N_r^f < r^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda^z\right) \leq c\lambda^{-\left(\frac{f_\alpha^v\wedge 1}{\alpha}-z\right)}.$$

If $s_\alpha^d = 1$, we have that

$$\mathbf{P}\left(\sum_{v\in\partial Spine_r} V^{dec}(\mathcal{T}_v^{dec}) \geq \lambda r^{d_\alpha^{vol}}(\log r)^{b_1}, N_r^f < \lambda^z r^{\frac{1}{\alpha-1}}(\log r)^{\frac{-1}{\alpha-1}}\right) \leq c\lambda^{-\left(\frac{f_\alpha^v\wedge 1}{\alpha}-z\right)}.$$

5.1.5 Controlling the spinal volume

Finally, we bound the spinal volume. Recall the spinal volume exponent defined on page 3 by $s_\alpha^v = \frac{\alpha-1}{v}$.

Proposition 5.5. *For all $r,\lambda>1$, and all $\varepsilon>0$, there exists $C<\infty$ such that, if $s_\alpha^d, s_\alpha^v \neq 1$,*

$$\mathbf{P}\Big(V^{dec}(Spine_r) \geq r^{\frac{s_\alpha^d\wedge 1}{s_\alpha^v}}\lambda\Big) \leq C\lambda^{-(s_\alpha^v-\varepsilon)}.$$

If instead $s_\alpha^d = 1, s_\alpha^v \neq 1$, we have that

$$\mathbf{P}\Big(V^{dec}(Spine_r) \geq \lambda r^{\frac{1}{s_\alpha^v}}(\log r)^{\frac{-1}{s_\alpha^v}}\Big) \leq C\lambda^{-(s_\alpha^v-\varepsilon)}.$$

Proof. This follows identically to the proof of Proposition 5.3, with $\alpha-1$ replaced by s_α^v (both are less than 1). □

5.1.6 Proof of Proposition 5.1

Proof of Proposition 5.1. We prove the case $f_\alpha^v, s_\alpha^d \neq 1$ for simplicity. The key to the proof is (12) which says that

$$V^{\mathrm{dec}}(B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho,r)) \leq \sum_{v\in\partial\mathrm{Spine}_r} V^{\mathrm{dec}}(\mathcal{T}_v^{\mathrm{dec}}) + \sum_{j\leq i_r} V^{\mathrm{dec}}(G(s_j)).$$

Applying the previous propositions and a union bound, we therefore deduce that

$$\begin{aligned}\mathbf{P}\Big(V^{\mathrm{dec}}(B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho,r)) \geq r^{d_\alpha^{\mathrm{vol}}}\lambda\Big) \leq{}& \mathbf{P}\left(N_r^f \geq r^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda^z\right) + \mathbf{P}\left(V^{\mathrm{dec}}(\mathrm{Spine}_r) \geq \frac{1}{2}r^{\frac{s_\alpha^d\wedge 1}{s_\alpha^v}}\lambda\right)\\ &+ \mathbf{P}\left(\sum_{v\in\partial\mathrm{Spine}_r} V^{\mathrm{dec}}(\mathcal{T}_v^{\mathrm{dec}}) \geq \frac{1}{2}r^{d_\alpha^{\mathrm{vol}}}\lambda, N_r^f < r^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda^z\right).\end{aligned}$$

Note that $\frac{s_\alpha^d \wedge 1}{s_\alpha^v} = \frac{v(s_\alpha^d \wedge 1)}{\alpha - 1} \leq \frac{v(s_\alpha^d \wedge 1)}{(\alpha-1)} \vee \frac{\alpha(s_\alpha^d \wedge 1)}{\alpha(\alpha-1)} = \frac{(s_\alpha^d \wedge 1)}{(\alpha-1)(f_\alpha^v \wedge 1)} = d_\alpha^{\mathrm{vol}}$, with equality if $\alpha \leq v$, which shows that the spinal volume can never dominate the volume contribution from the fragments. In the case $s_\alpha^d \neq 1$, it therefore follows from Propositions 5.3, 5.4 and 5.5 that for any $\varepsilon > 0$,

$$\mathbf{P}\Big(V^{\mathrm{dec}}(B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho, r)) \geq r^{d_\alpha^{\mathrm{vol}}}\lambda\Big) \leq c\lambda^{-z(\alpha-1-\varepsilon)} + \lambda^{-(s_\alpha^v - \varepsilon)} + c\lambda^{-\left(\frac{f_\alpha^v \wedge 1}{\alpha} - z\right)},$$

so we optimise by taking $z = \frac{f_\alpha^v \wedge 1}{\alpha^2}$, which gives

$$\mathbf{P}\left(V^{\mathrm{dec}}(B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho, r)) \geq r^{\frac{\alpha(s_\alpha^d \wedge 1)}{(\alpha-1)(f_\alpha^v \wedge 1)}}\lambda\right) \leq C\lambda^{\frac{-(f_\alpha^v \wedge 1 - \varepsilon)(\alpha-1)}{\alpha^2}} + C\lambda^{-(s_\alpha^v - \varepsilon)}.$$

The tail decay of the first term is always worse, so we deduce the result of Proposition 5.1.

The cases where $b_1 \neq 0$ follow in exactly the same way by applying the previous propositions and incorporating the extra log terms. □

5.2 Volume lower bounds

The main result is as follows.

Proposition 5.6. *Take b_1 as in Theorem 1.3.*

(i) If $f_\alpha^v < 1$, we have for any $\varepsilon > 0$ that there exist constants $c < \infty$ such that for all $r > 1, \lambda \in (1, r^{d_\alpha^{vol} - \varepsilon})$,

$$\mathbf{P}\Big(V^{dec}(B_{\mathcal{T}_\alpha^{dec}}(\rho, r) \leq \lambda^{-1} r^{d_\alpha^{vol}} (\log r)^{b_1}\Big) \leq c\lambda^{\frac{-(\tilde{s}_\alpha^d \vee 1)(\alpha-1-\varepsilon)}{v}}.$$

(ii) If $f_\alpha^v = 1$, then $\mathbf{P}\Big(V^{dec}(B_{\mathcal{T}_\alpha^{dec}}(\rho_\alpha^{dec}, r)) \leq \lambda^{-1} r^{d_\alpha^{vol}} (\log r)^{b_1}\Big) \leq c\lambda^{\frac{-(\alpha-1-\varepsilon)(1 \vee \tilde{s}_\alpha^d)}{2\alpha}}$.

(iii) If $f_\alpha^v > 1$, then $\mathbf{P}\Big(V^{dec}(B_{\mathcal{T}_\alpha^{dec}}(\rho_\alpha^{dec}, r)) \leq \lambda^{-1} r^{d_\alpha^{vol}} (\log r)^{b_1}\Big) \leq c\lambda^{\frac{-(\alpha-1-\varepsilon)(1 \vee \tilde{s}_\alpha^d)}{\alpha + (\alpha-1)[(\tilde{s}_\alpha^d - 1) \vee 0]}}$.

As before, we use the following result on the number of subtrees grafted to the decorated backbone.

Proposition 5.7. *For any $\varepsilon > 0$ there exists $c \in (0, \infty)$ such that, if $s_\alpha^d \neq 1$, then:*

$$\mathbf{P}\left(N_{\frac{r}{2}}^f \leq r^{\frac{s_\alpha^d \wedge 1}{\alpha - 1}}\lambda^{-1}\right) \leq c\lambda^{-(\tilde{s}_\alpha^d \vee 1)(\alpha-1-\varepsilon)}$$

If $s_\alpha^d = 1$, then

$$\mathbf{P}\left(N_{\frac{r}{2}}^f \leq r^{\frac{1}{\alpha-1}}(\log r)^{\frac{-1}{\alpha-1}}\lambda^{-1}\right) \leq c\lambda^{-(\alpha-1-\varepsilon')}$$

Proof. Similarly to previous proofs, suppose for now that $s_\alpha^d \neq 1$, let $N_r = r^{s_\alpha^d \wedge 1}\lambda^{-m}$, and let $\rho = s_0, s_1, \ldots, s_{N_r}$ denote the first N_r vertices on the backbone of T_α^∞. Then, by Lemmas 3.1 and 4.10, we have for any $\varepsilon > 0$ that

$$\mathbf{P}\left(\sum_{0=1}^{N_r} \mathsf{Diam}(G(s_i)) \geq \frac{r}{2}\right) \leq \begin{cases} c\lambda^{-m} & \text{if } s_\alpha^d < 1 \\ c\lambda^{-m(\tilde{s}_\alpha^d - \varepsilon)} & \text{if } s_\alpha^d > 1. \end{cases}$$

Also, note that, by Lemma 3.2(i),

$$\mathbf{P}\left(\sum_{0=1}^{N_r} \deg(G(s_i)) \leq r^{\frac{s_\alpha^d \wedge 1}{\alpha-1}}\lambda^{-1}\right) \leq e^{-c\lambda^{-(\alpha-1-m)}}.$$

Therefore, taking $m = \alpha - 1 - \varepsilon$ gives the result. If $s_\alpha^d = 1$, we instead take $N_r = r(\log r)^{-1}\lambda^{-m}$. □

Proof of Proposition 5.6. (i) We first deal with the case $f_\alpha^v < 1, s_\alpha^d \neq 1$. Take some $p > 0$. In this case the point is that there is likely to be an individual tree vertex v such that $G(v)$ has volume at least $r^{d_\alpha^{\mathrm{vol}}}\lambda^{-p}$, and $G(v)$ is completely contained within the ball $B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho, r)$. More precisely, since each of the subtrees grafted to the backbone are independent of each other, and letting $(T_i)_{i=1}^{N_{\frac{r}{2}}^f}$ denote these subtrees, we have from Lemmas 4.3, 4.8, and 4.17 that, if $q = \frac{p(\alpha-1)}{v}$, we can choose $C < \infty$ so that

$$\begin{aligned}
&\mathbf{P}\left(\nexists i : \sup_{v\in T_i} V^{\mathrm{dec}}(G(v)) \geq r^{d_\alpha^{\mathrm{vol}}}\lambda^{-p}, \mathsf{Height}(T_i) \leq Cr^{s_\alpha^d\wedge 1}\lambda^{-q}, \sup_{v\in T_i}\deg v \leq Cr^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda^{\frac{-q}{\alpha-1}} \;\middle|\; N_{\frac{r}{2}}^f \geq r^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda^{-h}\right)\\
&\leq (1 - cr^{\frac{-(s_\alpha^d\wedge 1)}{\alpha-1}}\lambda^{\frac{p}{v}} + \frac{c}{3}r^{\frac{-(s_\alpha^d\wedge 1)}{\alpha-1}}\lambda^{\frac{q}{\alpha-1}} + \frac{c}{3}r^{\frac{-(s_\alpha^d\wedge 1)}{\alpha-1}}\lambda^{\frac{q}{\alpha-1}})^{r^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda^{-h}}\\
&\leq e^{-c\lambda^{\frac{p}{v}-h}}.
\end{aligned}$$

Then, conditionally on the existence of such a tree T_i, and letting $v \in T_i$ be such that $V^{\mathrm{dec}}(G(v)) \geq r^{d_\alpha^{\mathrm{vol}}}\lambda^{-p}$, we have that $d_{T_i}(\rho_i, v) \leq Cr^{s_\alpha^d\wedge 1}\lambda^{-q}$, where d_{T_i} here denotes the graph distance in the undecorated tree T_i. Just as in the proof of Proposition 4.11, the degrees of vertices on the path to v are independently stochastically at most size-biased, and since the graphs $(G(u))_{u\prec v}$ are independent of $G(v)$, conditionally on their boundary size, it follows from Lemma 3.1 with $\beta = s_\alpha^d\wedge 1$, and Assumption 1.1(D) that (since for all $\varepsilon > 0$, $\lambda^{-qd} \leq \lambda^{-q(s_\alpha^d\wedge 1-\varepsilon)}$),

$$\begin{aligned}
&\mathbf{P}\left(\sum_{u\in T_i:u\preceq v}\mathsf{Diam}(G(u)) \geq \frac{r}{2} \;\middle|\; \sup_{u\in T_i}\deg u \leq Cr^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda^{\frac{-q}{\alpha-1}}, \mathsf{Height}(T_i) \leq Cr^{s_\alpha^d\wedge 1}\lambda^{-q}\right)\\
&\leq \mathbf{P}\left(\mathsf{Diam}(G(v)) \geq \frac{r}{4} \;\middle|\; \sup_{u\in T_i}\deg u \leq Cr^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda^{\frac{-q}{\alpha-1}}\right) + \mathbf{P}\left(\sum_{u\in T_i:u\prec v}\mathsf{Diam}(G(u)) \geq \frac{r}{4} \;\middle|\; \mathsf{Height}(T_i) \leq Cr^{s_\alpha^d\wedge 1}\lambda^{-q}\right)\\
&\leq \begin{cases}\lambda^{-q} & \text{if } s_\alpha^d < 1\\ \lambda^{-q(\tilde{s}_\alpha^d-\varepsilon)} & \text{if } s_\alpha^d > 1.\end{cases}
\end{aligned} \tag{13}$$

Moreover, on the event $\left\{V^{\mathrm{dec}}(G(v)) \geq r^{d_\alpha^{\mathrm{vol}}}\lambda^{-p}, \sum_{u\in T_i:u\preceq v}\mathsf{Diam}(G(u)) \leq \frac{r}{2}\right\}$ we have that

$$V^{\mathrm{dec}}(B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho, r) \geq V^{\mathrm{dec}}(G(v)) \geq r^{d_\alpha^{\mathrm{vol}}}\lambda^{-p},$$

so taking $p = 1, q = \frac{\alpha-1}{v}, h = \frac{1-\varepsilon}{v}$ we see that, if $s_\alpha^d < 1$,

$$\begin{aligned}
&\mathbf{P}\left(V^{\mathrm{dec}}(B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho, r) \leq r^{d_\alpha^{\mathrm{vol}}}\lambda^{-1}\right)\\
&\leq \mathbf{P}\left(N_{\frac{r}{2}}^f \leq r^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda^{-h}\right)\\
&\quad + \mathbf{P}\left(\nexists i : \sup_{v\in T_i} V^{\mathrm{dec}}(G(v)) \geq r^{d_\alpha^{\mathrm{vol}}}\lambda^{-1}, \mathsf{Height}(T_i) \leq Cr^{s_\alpha^d\wedge 1}\lambda^{-q}, \sup_{u\in T_i}\deg u \leq Cr^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda^{\frac{-q}{\alpha-1}} \;\middle|\; N_{\frac{r}{2}}^f \geq r^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda^{-h}\right)\\
&\quad + \mathbf{P}\left(\sum_{u\in T_i:u\preceq v}\mathsf{Diam}(G(u)) \geq \frac{r}{2} \;\middle|\; \sup_{u\in T_i}\deg u \leq Cr^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda^{\frac{-q}{\alpha-1}}, \mathsf{Height}(T_i) \leq Cr^{s_\alpha^d\wedge 1}\lambda^{-q}\right)\\
&\leq \lambda^{-h(\tilde{s}_\alpha^d\vee 1)(\alpha-1-\varepsilon)} + e^{-c\lambda^{\frac{1}{v}-h}} + \lambda^{-q(\tilde{s}_\alpha^d\vee 1-\varepsilon)}\\
&\leq \lambda^{\frac{-(\tilde{s}_\alpha^d\vee 1-\varepsilon)(\alpha-1-\varepsilon)}{v}} + e^{-c\lambda^{\varepsilon'}} + \lambda^{\frac{-(\tilde{s}_\alpha^d\vee 1-\varepsilon)(\alpha-1-\varepsilon)}{v}}.
\end{aligned}$$

If $s_\alpha^d > 1$, the same union bound gives an upper bound of $e^{-c\lambda^{\varepsilon'}} + \lambda^{\frac{-s_\alpha^d(\alpha-1)}{v}} + \lambda^{\frac{-s_\alpha^d(\alpha-1-\varepsilon)}{v}}$.

If $s_\alpha^d = 1$, the proof is similar but we instead consider the event

$$\{\sup_{v\in T_i} V^{\mathrm{dec}}(G(v)) \geq r^{d_\alpha^{\mathrm{vol}}}(\log r)^{-d_\alpha^{\mathrm{vol}}}\lambda^{-p},$$
$$\mathsf{Height}(T_i) \leq Cr(\log r)^{-1}\lambda^{-q}, \sup_{u\in T_i}\deg u \leq Cr^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}(\log r)^{\frac{-(s_\alpha^d\wedge 1)}{\alpha-1}}\lambda^{\frac{-q}{\alpha-1}}, N^f_{\frac{r}{2}} \geq r^{\frac{1}{\alpha-1}}(\log r)^{\frac{-1}{\alpha-1}}\lambda^{-h}\}.$$

(ii) We now treat the case $f_\alpha^v = 1$. In this case, if v_U is a uniform vertex of T, we can control $\sum_{u\preceq v_U}\mathsf{Diam}(G(v))$ with high probability. By definition of a uniform vertex, this in fact implies that we can control this sum for asymptotically almost all $v \in T$, so that if T_i is as above, the inserted graphs corresponding to "most" of the vertices in T_i are contained in a ball of radius r, which gives a large enough contribution to the volume.

We write the working for the case $s_\alpha^d \neq 1$. Similarly to part (i), we set $q = \frac{\alpha-1}{\alpha}$ and let A_r be the event

$$\{\exists i : r^{d_\alpha^{\mathrm{vol}}}\lambda^{-1} \leq |T_i| \leq 2r^{d_\alpha^{\mathrm{vol}}}\lambda^{-1}, \sup_{v\in T_i} V^{\mathrm{dec}}(G(v)) \leq Cr^{\frac{d_\alpha^{\mathrm{vol}}}{f_\alpha^v}}\lambda^{\frac{-1}{f_\alpha^v}},$$
$$\mathsf{Height}(T_i) \leq Cr^{s_\alpha^d\wedge 1}\lambda^{-q}, \sup_{u\in T_i}\deg u \leq r^{\frac{d_\alpha^{\mathrm{vol}}}{\alpha}}\lambda^{\frac{-1}{\alpha}}\}.$$

Again from Lemmas 4.3, 4.8 and 4.17, we can then write

$$\begin{aligned}\mathbf{P}\left(A_r^c \,\middle|\, N^f_{\frac{r}{2}} \geq r^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda^{-h}\right) &\leq (1 - cr^{\frac{-(s_\alpha^d\wedge 1)}{\alpha-1}}\lambda^{\frac{1}{\alpha}} + cr^{\frac{-(s_\alpha^d\wedge 1)}{\alpha-1}}\lambda^{\frac{1}{\alpha}} + \frac{c}{2}r^{\frac{-(s_\alpha^d\wedge 1)}{\alpha-1}}\lambda^{\frac{1}{\alpha(\alpha-1)}} + c'r^{-d})^{r^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda^{-h}}\\ &\leq e^{-c\lambda^{\frac{1}{\alpha}-h}}.\end{aligned}$$

Then, conditioning on existence of such a T_i, we let v_U be a uniform vertex in T_i. Along the path from the root of T_i to v_U, the vertex degrees are at most size-biased, again similarly to Proposition 4.11; we deduce from Lemma 3.1 similarly to the proof in part (i) that,

$$\begin{aligned}&\mathbf{P}\left(G(v_U) \not\subset B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r) \,\middle|\, \mathsf{Height}(T_i) \leq Cr^{s_\alpha^d\wedge 1}\lambda^{-q}\right)\\ &\leq \mathbf{P}\left(\mathsf{Diam}(G(v_U)) \geq \frac{r}{4} \,\middle|\, \sup_{u\in T_i}\deg u \leq Cr^{\frac{s_\alpha^d\wedge 1}{\alpha-1}}\lambda^{\frac{-q}{\alpha-1}}\right)\\ &\quad + \mathbf{P}\left(\sum_{u\in T_i: u\prec v_U}\mathsf{Diam}(G(u)) \geq \frac{r}{4} \,\middle|\, \mathsf{Height}(T_i) \leq Cr^{s_\alpha^d\wedge 1}\lambda^{-q}\right) \qquad (14)\\ &\leq \begin{cases}\lambda^{-q} & \text{if } s_\alpha^d < 1\\ \lambda^{-q(\tilde{s}_\alpha^d-\varepsilon)} & \text{if } s_\alpha^d > 1.\end{cases}\end{aligned}$$

In particular, by Fubini's theorem, this implies that $\mathbf{E}\Big[\frac{1}{|T_i|}\sum_{v\in T_i}\mathbb{1}\{G(v) \not\subset B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)\}\Big]$ enjoys the same upper bound. Now set

$$Y_{r,T_i} = \sum_{v\in T_i}\mathbb{1}\{G(v) \not\subset B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)\}.$$

Now choose $b, e > 0$ such that $b + e < q(s_\alpha^d \wedge 1)$. Since $r^{d_\alpha^{\mathrm{vol}}}\lambda^{-1} \leq |T_i| \leq 2r^{d_\alpha^{\mathrm{vol}}}\lambda^{-1}$, we therefore have by Markov's inequality that for any $\varepsilon > 0$ that

$$\mathbf{P}\Big(Y_{r,T_i} \geq 2r^{d_\alpha^{\mathrm{vol}}}\lambda^{-(1+q-b)}\Big) \leq \lambda^{-b} \text{ if } s_\alpha^d < 1,$$
$$\mathbf{P}\Big(Y_{r,T_i} \geq 2r^{d_\alpha^{\mathrm{vol}}}\lambda^{-(1+q[\tilde{s}_\alpha^d-\varepsilon]-b)}\Big) \leq \lambda^{-b} \text{ if } s_\alpha^d > 1.$$

We would like to proceed by arguing that since the inserted graphs at "most" of the vertices in T_i are contained in $B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)$, then "most" of the mass of T_i is also contained in $B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)$. However,

we have to be a bit careful because the volumes and diameters of the inserted graphs are correlated, so it is a priori more likely that the larger graphs in $\mathcal{T}_i^{\mathrm{dec}}$ will be missing from $B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)$. However, since we conditioned on $\sup_{u\in T_i} \deg u \le C r^{\frac{s_\alpha^d \wedge 1}{\alpha-1}} \lambda^{\frac{-q}{\alpha-1}}$, the bound in (14) holds independently of $V^{\mathrm{dec}}(G(v_U))$. Therefore, by taking a union bound with this tail bound, the "missing mass" is stochastically dominated by the mass lost on selecting the missing graphs uniformly from the vertices of T_i.

Therefore, continuing with the working for the case $s_\alpha^d < 1$, we have for each $v \in T_i$ that

$$\mathbf{E}\left[V^{\mathrm{dec}}(G(v)) \;\middle|\; V^{\mathrm{dec}}(G(v)) \le C r^{\frac{d_\alpha^{\mathrm{vol}}}{f_\alpha^u}}, Y_{r,T_i} < 2r^{d_\alpha^{\mathrm{vol}}} \lambda^{-(1+q-b)} \right] < C' \log r,$$

so that

$$\mathbf{E}\left[\sum_{v\in T_i} V^{\mathrm{dec}}(G(v)) \mathbb{1}\{G(v) \not\subset B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)\} \;\middle|\; Y_{r,T_i} < 2r^{d_\alpha^{\mathrm{vol}}} \lambda^{-(1+q-b)} \right] \le 2r^{d_\alpha^{\mathrm{vol}}} \lambda^{-(1+q-b)} C' \log r,$$

and in particular, for any $e > 0$ we have that

$$\mathbf{P}\left(\sum_{v\in T_i} V^{\mathrm{dec}}(G(v)) \mathbb{1}\{G(v) \not\subset B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)\} \ge r^{d_\alpha^{\mathrm{vol}}} \lambda^{-(1+q-b-e)} C' \log r \;\middle|\; Y_{r,T_i} < 2r^{d_\alpha^{\mathrm{vol}}} \lambda^{-(1+q-b)} \right) \le c\lambda^{-e}.$$

On the other hand, note from Lemma 3.2(ii) that, since $b + e < q$,

$$\mathbf{P}\left(\sum_{v\in T_i} V^{\mathrm{dec}}(G(v)) \le 2r^{d_\alpha^{\mathrm{vol}}} \lambda^{-(1+q-b-e)} C' \log r \;\middle|\; |T_i| \ge r^{d_\alpha^{\mathrm{vol}}} \lambda^{-1} \right) \le C e^{-c\lambda^{q-b-e} \log \lambda}.$$

To bring this all together, we note that on the complement of all the events described above, we have that

$$\begin{aligned} V^{\mathrm{dec}}(B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)) &\ge \sum_{v\in T_i} V^{\mathrm{dec}}(G(v)) - \sum_{v\in T_i} V^{\mathrm{dec}}(G(v)) \mathbb{1}\{G(v) \not\subset B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)\} \\ &\ge r^{d_\alpha^{\mathrm{vol}}} \lambda^{-(1+q-b-e)} C' \log r \end{aligned}$$

It therefore follows from a union bound that, if $s_\alpha^d < 1$,

$$\begin{aligned} &\mathbf{P}\Big(V^{\mathrm{dec}}(B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)) \le r^{d_\alpha^{\mathrm{vol}}} \lambda^{-(1+q-b-e)} C' \log r \Big) \\ &\le \mathbf{P}\left(N_{\frac{r}{2}}^f \le r^{\frac{s_\alpha^d \wedge 1}{\alpha-1}} \lambda^{-h} \right) + \mathbf{P}\left(A_r^c \;\middle|\; N_{\frac{r}{2}}^f \ge r^{\frac{s_\alpha^d \wedge 1}{\alpha-1}} \lambda^{-h} \right) + \mathbf{P}\Big(Y_{r,T_i} \ge 2r^{d_\alpha^{\mathrm{vol}}} \lambda^{-(1+q-b)} \Big) \\ &\quad + \mathbf{P}\left(\sum_{v\in T_i} V^{\mathrm{dec}}(G(v)) \mathbb{1}\{G(v) \not\subset B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)\} \ge r^{d_\alpha^{\mathrm{vol}}} \lambda^{-(1+q-b-e)} C' \log r \;\middle|\; Y_{r,T_i} < 2r^{d_\alpha^{\mathrm{vol}}} \lambda^{-(1+q-b)} \right) \\ &\quad + \mathbf{P}\left(\sum_{v\in T_i} V^{\mathrm{dec}}(G(v)) \le 2r^{d_\alpha^{\mathrm{vol}}} \lambda^{-(1+q-b-e)} C' \log r \;\middle|\; |T_i| \ge r^{d_\alpha^{\mathrm{vol}}} \lambda^{-1} \right) \\ &\le \lambda^{-h(\alpha-1-\varepsilon)} + e^{-c\lambda^{\frac{1}{\alpha}-h}} + \lambda^{-b} + c\lambda^{-e} + C e^{-c\lambda^{\frac{(\alpha-1)}{\alpha}-b-e} \log \lambda}. \end{aligned}$$

Taking $h = \frac{1}{\alpha} - \varepsilon, b = e = \frac{\alpha-1-\varepsilon}{2\alpha}$, we get that

$$\mathbf{P}\Big(V^{\mathrm{dec}}(B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)) \le r^{d_\alpha^{\mathrm{vol}}} \lambda^{-(1-\varepsilon)} C' \log r \Big) \le \lambda^{\frac{-(\alpha-1-\varepsilon)}{2\alpha}}.$$

If $s_\alpha^d > 1$ we write the same argument and again take $h = \frac{1}{\alpha} - \varepsilon$ but $b = e = \frac{(\alpha-1-\varepsilon)\bar{s}_\alpha^d}{2\alpha}$ to get tail decay of $\lambda^{\frac{-(\alpha-1-\varepsilon)\bar{s}_\alpha^d}{2\alpha}}$.

If $s_\alpha^d = 1$ as well, we carry through the extra log term from Proposition 5.7 and get that

$$\mathbf{P}\Big(V^{\mathrm{dec}}(B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)) \le r^{d_\alpha^{\mathrm{vol}}} \lambda^{-(1-\varepsilon)} C' (\log r)^{-(d_\alpha^{\mathrm{vol}}-1)} \Big) \le \lambda^{\frac{-(\alpha-1-\varepsilon)}{2\alpha}}.$$

(iii) If $f_\alpha^v > 1$, we use the same argument and don't need to incorporate the factor of $(\log r)^{d_\alpha^{\mathrm{vol}}}$. In particular, if $q = \frac{\alpha-1}{\alpha}$ and

$$A_r = \left\{ \exists i : r^{d_\alpha^{\mathrm{vol}}}\lambda^{-1} \le |T_i| \le 2r^{d_\alpha^{\mathrm{vol}}}\lambda^{-1}, \mathsf{Height}(T_i) \le C r^{s_\alpha^d \wedge 1}\lambda^{-q}, \sup_{u\in T_i} \deg u \le r^{\frac{d_\alpha^{\mathrm{vol}}}{\alpha}}\lambda^{\frac{-1}{\alpha}} \right\},$$

then $\mathbf{P}\left(A_r^c \,\middle|\, N_{\frac{r}{2}}^f \ge r^{\frac{s_\alpha^d \wedge 1}{\alpha-1}}\lambda^{-h} \right) \le e^{-c\lambda^{\frac{1}{\alpha}-h}}$, as above. Also set

$$Y_{r,T_i} = \sum_{v\in T_i} \mathbb{1}\{G(v) \not\subset B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)\},$$
$$Z_{r,T_i} = \sum_{v\in T_i} \mathbb{1}\{V^{\mathrm{dec}}(G(v)) \ge 1\}.$$

As in part (ii),

$$\mathbf{P}\Big(Y_{r,T_i} \ge 2r^{d_\alpha^{\mathrm{vol}}}\lambda^{-(1+q[\bar{s}_\alpha^d - \varepsilon]-b)} \Big) \le \lambda^{-b}$$

Also, by Assumption 1.1(iii), there exists $p > 0$ such that Z_{r,T_i} stochastically dominates a $\mathsf{Binomial}\Big(r^{d_\alpha^{\mathrm{vol}}}\lambda^{-1}, p\Big)$ random variable. Therefore, there exists $c > 0$ such that

$$\mathbf{P}\Big(Z_{r,T_i} \le 3r^{d_\alpha^{\mathrm{vol}}}\lambda^{-(1+q[\bar{s}_\alpha^d - \varepsilon]-b)} \Big) \le e^{-cr^{d_\alpha^{\mathrm{vol}}}\lambda^{-1}}.$$

Since $V^{\mathrm{dec}}(B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)) \ge Z_{r,T_i} - Y_{r,T_i}$, we deduce from a union bound that

$$\mathbf{P}\left(V^{\mathrm{dec}}(B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)) \le r^{d_\alpha^{\mathrm{vol}}}\lambda^{-(\frac{\alpha - b\alpha + (\alpha-1)[\bar{s}_\alpha^d - \varepsilon]}{\alpha})} \right) \le \lambda^{\frac{-(\alpha-1-\varepsilon)(1\vee \bar{s}_\alpha^d)}{\alpha}} + \lambda^{-b} + e^{-cr^{d_\alpha^{\mathrm{vol}}}\lambda^{-1}}.$$

Taking $b = \frac{(\alpha-1)(\bar{s}_\alpha^d \vee 1)}{\alpha}$ therefore gives a result equivalent to the stated one. If $s_\alpha^d = 1$, we have to incorporate the extra log term as usual.

□

Remark 5.8. *Under Assumption 1.4, which is stronger, we can improve the tail bound to*

$$\mathbf{P}\Big(V^{dec}(B_{\mathcal{T}_\alpha^{dec}}(\rho, r)) \le r^{d_\alpha^{vol}}\lambda^{-1} \Big) \le C\lambda^{\frac{-(\alpha-1)}{\alpha}+\varepsilon}$$

when $s_\alpha^d, f_\alpha^v \ne 1$, with appropriate logarithmic corrections in the boundary cases, by following the same strategy as that used in [BS15, Lemma 3.2]. However this requires the uniform bound on the decorated height obtained in Proposition 4.11. In the case of trees and looptrees where we have tighter control of $N_{\frac{r}{2}}^f$, this improves the tail decay to $e^{-\lambda^{\frac{\alpha-1}{\alpha}}}$ and $e^{-\lambda^{\frac{1}{\alpha}}}$ (the second of these was shown in [BS15, Equation (3.23)]).

6 Resistance on $\mathcal{T}_\alpha^{\mathbf{dec}}$

In order to apply results of [KM08] about random walk exponents, we also need to understand resistance on $\mathcal{T}_\alpha^{\mathrm{dec}}$.

In Assumption 1.4 we have assumed that the two-point function and diameters of the inserted graphs grow according to the same exponents. It would still be possible to get some kind of result if this was not the case, but we would need more information on the local geometry of the inserted graphs. When the two exponents are equal, it means that we are able to cut the decorated backbone at an appropriate cut point such that the cut point is roughly distance r from the root, and all vertices contained in the decorated graphs corresponding to ancestors of that cutpoint are also roughly within distance r of the root. The same holds for the resistance distance, so this gives a concrete way to separate $B_{res}^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, r)$ from $B_{res}^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, 2r)^c$, for

example. However, if the diameters of the inserted graphs grow differently to the two-point function, then we cannot separate balls just by exploiting the underlying tree structure.

Recall that the exponent s_α^R is defined so that, for a vertex s_i on the infinite backbone of T_α^∞, there exist constants c, c' such that

$$\begin{aligned}\mathbf{P}(\mathsf{Diam}_{res}(G(s_i)) \geq r) &\leq c r^{-s_\alpha^R} \\ \mathbf{P}\big(R^U(G(s_i)) \geq r\big) &\geq c' r^{-s_\alpha^R}\end{aligned}$$

for all $r > 1$.

In what follows, we will use the subscript "res" to denote that distances are defined with respect to the resistance metric. For example, $B_{res}^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, r)$ refers to a ball of radius r with respect to the effective resistance metric on $\mathcal{T}_\alpha^{\mathrm{dec}}$, and $B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)$ still refers to a ball with respect to the decorated metric d_g^{dec}. For a decorated tree $\mathcal{T}^{\mathrm{dec}}$, $\mathsf{Height}_{\mathrm{res}}^{\mathrm{dec}}(\mathcal{T}^{\mathrm{dec}}) = \sup_{x \in \mathcal{T}^{\mathrm{dec}}} R_{\mathrm{eff}}(\rho_\alpha^{\mathrm{dec}}, x)$.

For all of Section 6, we will work under Assumption 1.4.

Recall from Proposition 4.11 that

$$t_\alpha^d = \frac{\alpha - \varepsilon}{\alpha + 1 + [1 - s_\alpha^d]\alpha(\alpha - 1)} \wedge \frac{\alpha^2 - \varepsilon}{\alpha(\alpha - 1) + \alpha(s_\alpha^d + \alpha)}.$$

We analogously define

$$\begin{aligned}t_\alpha^R &= \frac{\alpha - \varepsilon}{\alpha + 1 + [1 - s_\alpha^R]\alpha(\alpha - 1)} \wedge \frac{\alpha^2 - \varepsilon}{\alpha(\alpha - 1) + \alpha(s_\alpha^R + \alpha)}, \\ \tilde{t}_\alpha^R &= \frac{\alpha(s_\alpha^R \wedge 1) - \varepsilon}{\left(\alpha - 1 + \alpha\left[\frac{1}{\alpha R} \vee (1 - \frac{1}{\alpha})\right] t_\alpha^R\right)} \wedge (s_\alpha^R \wedge 1)(\tilde{s}_\alpha^R \vee 1), \\ \tilde{t}_\alpha^d &= \frac{\alpha(s_\alpha^d \wedge 1) - \varepsilon}{\left(\alpha - 1 + \alpha\left[\frac{1}{\alpha d} \vee (1 - \frac{1}{\alpha})\right] t_\alpha^d\right)} \wedge (s_\alpha^d \wedge 1)(\tilde{s}_\alpha^d \vee 1).\end{aligned}$$

(These exponents are not optimal and their precise values are not important). We then have the following result on effective resistance in $\mathcal{T}_\alpha^{\mathrm{dec}}$.

Proposition 6.1. *For all $\varepsilon > 0$, there exists a constant $c < \infty$ such that for all $\lambda > 1$,*

$$\mathbf{P}\big(R_{\mathit{eff}}(\rho_\alpha^{dec}, B_{res}^{dec}(\rho_\alpha^{dec}, r)^c) \leq r\lambda^{-1}\big) \leq c\lambda^{-(\tilde{t}_\alpha^R - \varepsilon)}.$$

Proof. We write the proof for $s_\alpha^R \neq 1$. First fix some $k, l > 0$ (we will specify these precisely later). Recall that $\rho = s_0, s_1, \ldots$ denote the backbone vertice of T_α^∞. Similarly to previous proofs, we define a number N_r that corresponds to the index of the vertex where we "cut off" the infinite backbone. This time, we take $N_r = r^{s_\alpha^R \wedge 1}\lambda^{-k}$, and first observe that, by Lemma 3.1,

$$\mathbf{P}\left(\sum_{i \leq N_r} \mathsf{Diam}_{res}(s_i) \geq \frac{1}{2}r\right) \leq c\lambda^{-k[(\tilde{s}_\alpha^R \vee 1) - \varepsilon]}.$$

Additionally, by Lemma 3.2,

$$\mathbf{P}\left(\sum_{i \leq N_r} R^U(s_i) \leq r\lambda^{-k(s_\alpha^R \wedge 1)^{-1} - \varepsilon}\right) \leq Ce^{-c\lambda^{\varepsilon(\tilde{s}_\alpha^R \wedge 1)}}. \tag{15}$$

Moreover, similarly to the proof of Proposition 5.3, it follows from Lemma 3.1 that

$$\mathbf{P}\left(\sum_{i \leq N_r} \deg(s_i) \geq r^{\frac{s_\alpha^R \wedge 1}{\alpha - 1}}\lambda^{l - \frac{k}{\alpha - 1}}\right) \leq c\lambda^{-l(\alpha - 1)}.$$

The three probabilistic bounds above show that, with (quantified) high probability:

- all of the vertices contained in $\bigcup_{i\leq N_r} G(s_i)$ are within resistance distance $\frac{1}{2}r$ of the root;
- the number of subtrees joined to the decorated backbone at a vertex contained in $\bigcup_{i\leq N_r} G(s_i)$ is at most $N_r^{\frac{1}{\alpha-1}}\lambda^l$;
- the vertex joining $G(s_{N_r-1})$ to $G(s_{N_r})$ is at resistance at least $N_r^{(s_\alpha^R\wedge 1)^{-1}}\lambda^{-\varepsilon}$ from the root.

We will additionally show that, with high probability, all of the subtrees joined to the decorated backbone at a vertex contained in $\bigcup_{i\leq N_r} G(s_i)$ have decorated resistance height at most $\frac{1}{2}r$.

To do this, we first make the following observation. Recall from Proposition 4.11 that for a finite decorated tree $\mathcal{T}_n^{\mathrm{dec}}$, where the underlying Galton-Watson tree T_n has n vertices, there exists $t_\alpha^R > 0$ such that $\mathbf{P}\Big(\mathsf{Height}_{\mathrm{res}}^{\mathrm{dec}}(\mathcal{T}_n^{\mathrm{dec}}) \geq n^{\left(\frac{1}{\alpha R}\vee 1-\frac{1}{\alpha}\right)}\lambda\Big) \leq c\lambda^{-t_\alpha^R}$ (with an extra $\log n$ factor if $s_\alpha^R = 1$). We can also get a lower bound for the probability of this event as follows. Let A_n be the event that the number of individuals in generation $n^{1-\frac{1}{\alpha}}$ of T_n is at least $n^{\frac{1}{\alpha}}$, and that one of the subtrees emanating from level $n^{1-\frac{1}{\alpha}}$ has decorated resistance height at least $n^{\left(\frac{1}{\alpha R}\vee 1-\frac{1}{\alpha}\right)}\lambda$. Due to the Galton-Watson structure, all of the subtrees emanating from level $n^{1-\frac{1}{\alpha}}$ are independent Galton-Watson trees, so letting $(T^{(i)})_{i=1}^{n^{\frac{1}{\alpha}}}$ denote (a subset of) these Galton-Watson trees on the event described, and $((\mathcal{T}^{\mathrm{dec}})^{(i)})_{i=1}^{n^{\frac{1}{\alpha}}}$ their decorated versions, and $Z_{n^{1-\frac{1}{\alpha}}}$ denote the size of generation n we have by [Kor17, Theorem 5] that there exists a constant $c > 0$ such that

$$\begin{aligned}\mathbf{P}\Big(\mathsf{Height}_{\mathrm{res}}^{\mathrm{dec}}(\mathcal{T}_n^{\mathrm{dec}}) \geq n^{\left(\frac{1}{\alpha R}\vee 1-\frac{1}{\alpha}\right)}\lambda\Big) \geq \mathbf{P}(A_n) &\geq \mathbf{P}\Big(Z_{n^{1-\frac{1}{\alpha}}} \geq n^{\frac{1}{\alpha}}\Big)\,\mathbf{P}\left(\bigcup_{i=1}^{n^{\frac{1}{\alpha}}} \mathsf{Height}_{\mathrm{res}}^{\mathrm{dec}}((\mathcal{T}^{\mathrm{dec}})^{(i)}) \geq n^{\left(\frac{1}{\alpha R}\vee 1-\frac{1}{\alpha}\right)}\lambda\right)\\ &\geq c\mathbf{P}\left(\bigcup_{i=1}^{n^{\frac{1}{\alpha}}}\left\{\mathsf{Height}_{\mathrm{res}}^{\mathrm{dec}}((\mathcal{T}^{\mathrm{dec}})^{(i)}) \geq n^{\left(\frac{1}{\alpha R}\vee 1-\frac{1}{\alpha}\right)}\lambda\right\}\right).\end{aligned}$$

In particular, rearranging gives that for an i.i.d. sequence of decorated Galton-Watson trees $((\mathcal{T}^{\mathrm{dec}})^{(i)})_{i=1}^{n^{\frac{1}{\alpha}}}$,

$$\mathbf{P}\left(\bigcup_{i=1}^{n^{\frac{1}{\alpha}}}\left\{\mathsf{Height}_{\mathrm{res}}^{\mathrm{dec}}((\mathcal{T}^{\mathrm{dec}})^{(i)}) \geq n^{\left(\frac{1}{\alpha R}\vee 1-\frac{1}{\alpha}\right)}\lambda\right\}\right) \leq c\lambda^{-t_\alpha^R}. \tag{16}$$

We apply this now by taking $n^{\frac{1}{\alpha}} = N_r^{\frac{1}{\alpha-1}}\lambda^l = r^{\frac{s_\alpha^R\wedge 1}{\alpha-1}}\lambda^{l-\frac{k}{\alpha-1}}$. Letting C_r be the event that there exists a subtree joined to the decorated backbone of $\mathcal{T}_\alpha^{\mathrm{dec}}$ at a vertex in $\bigcup_{i\leq N_r} G(s_i)$ with decorated height at least $\frac{1}{2}r$, conditional on the number of such subtrees being upper bounded by $r^{\frac{s_\alpha^R\wedge 1}{\alpha-1}}\lambda^{l-\frac{k}{\alpha-1}}$, we have by independence of these subtrees that

$$\begin{aligned}\mathbf{P}(C_r) &\leq \mathbf{P}\left(\bigcup_{i=1}^{r^{\frac{s_\alpha^R\wedge 1}{\alpha-1}}\lambda^{l-\frac{k}{\alpha-1}}}\left\{\mathsf{Height}_{\mathrm{res}}^{\mathrm{dec}}((\mathcal{T}^{\mathrm{dec}})^{(i)}) \geq \frac{1}{2}r\right\}\right)\\ &= \mathbf{P}\left(\bigcup_{i=1}^{r^{\frac{s_\alpha^R\wedge 1}{\alpha-1}}\lambda^{l-\frac{k}{\alpha-1}}}\left\{\mathsf{Height}_{\mathrm{res}}^{\mathrm{dec}}((\mathcal{T}^{\mathrm{dec}})^{(i)}) \geq \left[\left(r^{\frac{s_\alpha^R\wedge 1}{\alpha-1}}\lambda^{(l-\frac{k}{\alpha-1})}\right)^\alpha\right]^{\left[\frac{1}{\alpha R}\vee(1-\frac{1}{\alpha})\right]}\lambda^{\alpha(\frac{k}{\alpha-1}-l)\left[\frac{1}{\alpha R}\vee(1-\frac{1}{\alpha})\right]}\right\}\right)\\ &\leq c\lambda^{-\alpha(\frac{k}{\alpha-1}-l)\left[\frac{1}{\alpha R}\vee(1-\frac{1}{\alpha})\right]t_\alpha^R}.\end{aligned}$$

To conclude, note that on the event C_r, the vertex v_r defined to be the vertex at which $G(s_{N_r})$ and $G(s_{N_r+1})$ intersect is a single point that separates the root from $B_{res}^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, r)^c$, and itself is at resistance distance at least $N_r^{(s_\alpha^R\wedge 1)^{-1}}\lambda^{-\varepsilon}$ from the root, so that

$$R_{\mathrm{eff}}(\rho_\alpha^{\mathrm{dec}}, B_{res}^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, r(\log r)^{1+\frac{1}{kr}})^c) \geq N_r^{(s_\alpha^R\wedge 1)^{-1}}\lambda^{-\varepsilon}$$

$$= r^{(s_\alpha^R\wedge 1)(s_\alpha^R\wedge 1)^{-1}}\lambda^{-k(s_\alpha^R\wedge 1)^{-1}}\lambda^{-\varepsilon}$$
$$= \lambda^{-k(s_\alpha^R\wedge 1)^{-1}-\varepsilon}.$$

Therefore, combining all the probabilistic bounds obtained with a union bound, we see that

$$\mathbf{P}\Big(R_{\mathrm{eff}}(\rho_\alpha^{\mathrm{dec}}, B_{res}^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, r)^c) \le r\lambda^{-k(s_\alpha^R\wedge 1)^{-1}-\varepsilon}\Big) \le c\lambda^{-k[(\tilde{s}_\alpha^R\vee 1)-\varepsilon]} + c\lambda^{-l(\alpha-1)} + c\lambda^{-\alpha\left(\frac{k}{\alpha-1}-l\right)\left[\frac{1}{\alpha R}\vee(1-\frac{1}{\alpha})\right]t_\alpha^R}$$

Recalling from Proposition 4.11 that

$$t_\alpha^R = \begin{cases} \frac{\alpha-\varepsilon}{\alpha+1+[1-s_\alpha^R]\alpha(\alpha-1)} \wedge \frac{\alpha^2-\varepsilon}{\alpha(\alpha-1)+\alpha(s_\alpha^R+\alpha)} & \text{if } R(\alpha-1)\le 1 \\ \frac{(1-\varepsilon)}{\alpha} & \text{if } R(\alpha-1)>1, \end{cases}$$

we can therefore optimise by setting $k = (s_\alpha^R\wedge 1)(1+\varepsilon), l = \frac{\alpha(s_\alpha^R\wedge 1)}{(\alpha-1)\left(\alpha-1+\alpha\left[\frac{1}{\alpha R}\vee(1-\frac{1}{\alpha})\right]t_\alpha^R\right)}$, and we obtain that

$$\mathbf{P}\big(R_{\mathrm{eff}}(\rho_\alpha^{\mathrm{dec}}, B_{res}^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, r)^c) \le r\lambda^{-1}\big) \le c\lambda^{-\left(\frac{\alpha(s_\alpha^R\wedge 1)-\varepsilon}{\left(\alpha-1+\alpha\left[\frac{1}{\alpha R}\vee(1-\frac{1}{\alpha})\right]t_\alpha^R\right)}\wedge(s_\alpha^R\wedge 1)(\tilde{s}_\alpha^R\vee 1)\right)},$$

which is the stated result.

If $s_\alpha^R = 1$, we instead take $N_r = r(\log r)^{-1}\lambda^{-k}$, and repeat the same proof. □

Remark 6.2. *Similarly to the volume bounds, in some cases where we have better control over the inserted graphs, we may be able to improve this to exponential decay by introducing an interative process that breaks up tall subtrees, similarly to how we did in [Arc20b, Proposition 6.4] in the stable looptree case.*

We will also need the following result in order to determine the random walk displacement exponent in terms of the intrinsic metric. To save space in the proposition below we write $B_r^R = B_{res}^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, r)$ and $B_r^d = B^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, r)$.

Proposition 6.3. **P**-*almost surely, for any* $\varepsilon > 0$ *there exists* $r_0 < \infty$ *such that for all* $r \ge r_0$, *if* $s_\alpha^d \ne 1, s_\alpha^R \ne 1$, *or* $s_\alpha^d = s_\alpha^R = 1$:

$$B^d_{r^{(s_\alpha^R\wedge 1)(s_\alpha^d\wedge 1)^{-1}}(\log r)^{\frac{-(1+\varepsilon)}{\tilde{t}_\alpha^R}}} \subset B_r^R \subset B^d_{r^{(s_\alpha^d\wedge 1)^{-1}(s_\alpha^R\wedge 1)}(\log r)^{\frac{1+\varepsilon}{\tilde{t}_\alpha^R}}}$$
$$B^R_{r^{(s_\alpha^d\wedge 1)(s_\alpha^R\wedge 1)^{-1}}(\log r)^{\frac{-(1+\varepsilon)}{\tilde{t}_\alpha^d}}} \subset B_r^d \subset B^R_{r^{(s_\alpha^R\wedge 1)^{-1}(s_\alpha^d\wedge 1)}(\log r)^{\frac{1+\varepsilon}{\tilde{t}_\alpha^d}}}$$

If $s_\alpha^d = 1$, *then the same result is true on multiplying the radius of the intrinsic metric ball by* $\log r$, *and if* $s_\alpha^R = 1$, *then the same result is true on multiplying the radius of the resistance metric ball by* $\log r$. *Moreover,*

- *If* $s_\alpha^d \ne 1, s_\alpha^R \ne 1$, *or* $s_\alpha^d = s_\alpha^R = 1$, *then* $\mathbf{P}\Big(B^R_{r^{(s_\alpha^d\wedge 1)(s_\alpha^R\wedge 1)^{-1}}\lambda^{-1}} \subset B_r^d \subset B^R_{r^{(s_\alpha^R\wedge 1)^{-1}(s_\alpha^d\wedge 1)}\lambda}\Big) \to 1$ *as* $\lambda\to\infty$,
- *If* $s_\alpha^d = 1, s_\alpha^R \ne 1$, $\mathbf{P}\Big(B^R_{r^{(s_\alpha^d\wedge 1)(s_\alpha^R\wedge 1)^{-1}}\lambda^{-1}} \subset B^d_{r\log r} \subset B^R_{r^{(s_\alpha^R\wedge 1)^{-1}(s_\alpha^d\wedge 1)}\lambda}\Big) \to 1$ *as* $\lambda\to\infty$,
- *If* $s_\alpha^d \ne 1, s_\alpha^R = 1$, $\mathbf{P}\Big(B^R_{\lambda^{-1}r^{(s_\alpha^d\wedge 1)(s_\alpha^R\wedge 1)^{-1}}\log r} \subset B_r^d \subset B^R_{\lambda r^{(s_\alpha^R\wedge 1)^{-1}(s_\alpha^d\wedge 1)}\log r}\Big) \to 1$ *as* $\lambda\to\infty$.

Proof. Assume for now that $s_\alpha^d, s_\alpha^R \ne 1$. By replacing R^U with d^U in (15), the same proof as in Proposition 6.1 shows that, with probability at least $1 - c\lambda^{-(\tilde{t}_\alpha^R-\varepsilon)}$, there exists a single vertex at distance at least $r^{(s_\alpha^R\wedge 1)(s_\alpha^d\wedge 1)^{-1}}\lambda^{-1}$ from the root, separating the root from $(B_r^R)^c$. Therefore, all vertices in $(B_r^R)^c$ lie at least distance $r^{(s_\alpha^R\wedge 1)(s_\alpha^d\wedge 1)^{-1}}\lambda^{-1}$ from the root, so that $B^d_{r^{(s_\alpha^R\wedge 1)(s_\alpha^d\wedge 1)^{-1}}\lambda^{-1}} \subset B_r^R$. Therefore, setting $r_n = 2^n$, $\lambda_n = (\frac{1}{2}\log r_n)^{\frac{1+\varepsilon}{\tilde{t}_\alpha^R}}$, applying Borel Cantelli and using monotonicity, we deduce that **P**-almost surely, there exists $r_0 < \infty$ such that

$$B^d_{r^{(s_\alpha^R\wedge 1)(s_\alpha^d\wedge 1)^{-1}}(\log r)^{\frac{-(1+\varepsilon)}{\tilde{t}_\alpha^R}}} \subset B_r^R$$

for all $r \geq r_0$. By symmetry, we can also use the same argument to go in the other direction, and also deduce that

$$B^R_{r^{(s^d_\alpha \wedge 1)(s^R_\alpha \wedge 1)^{-1}} (\log r)^{\frac{-(1+\varepsilon)}{\tilde{t}^d_\alpha}}} \subset B^d_r.$$

Note that, if $\tilde{r} = r^x (\log r)^{-y}$, then $\tilde{r}^{x^{-1}} (\log \tilde{r})^{x^{-1}y+\varepsilon} \geq r$ for all sufficiently large r, so that $B^R_{\tilde{r}} \subset B^d_r$ implies that $B^R_{\tilde{r}} \subset B^d_{\tilde{r}^{x^{-1}} (\log \tilde{r})^{x^{-1}y+\varepsilon}}$ for all sufficiently large r. The second inclusion above therefore gives the result as stated. The same proof works for the other values of s^d_α, s^R_α. □

7 Random walk exponents

The purpose of this section is to use the volume and resistance results of the previous sections to determine the exponents for a simple random walk on $\mathcal{T}^{\mathrm{dec}}_\alpha$. To do this, we will apply results of [KM08]. In order to fit in with the framework of [KM08], we will specifically consider the measure μ on $\mathcal{T}^{\mathrm{dec}}_\alpha$ defined by $\mu(x) = \deg x$ for all $x \in \mathcal{T}^{\mathrm{dec}}_\alpha$, rather than a general decorated measure V^{dec}. (This is because μ defined as such is the "natural speed measure" for the simple random walk).

To directly apply their results to get exponents for the decorated metric d, we would need to define deterministic functions v and r that govern the volume and resistance growth of the space, and for a given $\lambda > 1$ define the set

$$\begin{aligned} J(\lambda) = \big\{ R \in [1,\infty] : \lambda^{-1} v(R) \leq \mu(B^d_R) \leq \lambda v(R), R_{\mathrm{eff}}(\rho^{\mathrm{dec}}_\alpha, (B^d_R)^c) \geq \lambda^{-1} r(R) \big\} \\ \cap \big\{ R \in [1,\infty] : R_{\mathrm{eff}}(\rho^{\mathrm{dec}}_\alpha, y) \leq \lambda r(d(\rho^{\mathrm{dec}}_\alpha, y)) \forall y \in B^d_R \big\}, \end{aligned}$$

and show that $\mathbf{P}(R \in J(\lambda)) \to 1$ as $\lambda \to \infty$, uniformly in $R > 1$ (cf [KM08, Definition 1.1, Assumption 1.2(1)].

If we ignore the special cases with logarithmic corrections for a moment, by Propositions 5.1, 5.6 and 6.1 the appropriate volume function to take would be $v(R) = R^{\frac{\alpha(s^d_\alpha \wedge 1)}{(\alpha-1)(f^v_\alpha \wedge 1)}}$, and the appropriate resistance function would be $r(R) = R^{(s^d_\alpha \wedge 1)(s^R_\alpha \wedge 1)^{-1}}$. However, we encounter some technical difficulties with the final condition in the definition of $J(\lambda)$, in that it requires $R_{\mathrm{eff}}(\rho^{\mathrm{dec}}_\alpha, y) \leq \lambda r(d(\rho^{\mathrm{dec}}_\alpha, y))$ **for all** $y \in B^d_R$.

For a general graph, it is usually only possible to achieve this kind of control uniformly when there is some deterministic relation between the resistance metric and the intrinsic metric, for example as is the case for random trees and looptrees. In our decorated tree setting, this is probably still achievable in the case when we decorate the tree with deterministic graphs, but in the case when the inserted graphs are random we anticipate that there will be genuine multiplicative fluctuations in the relationship between the resistance metric and the intrinsic metric (for example these could be on the order of $\log R$ on the ball of radius R), so it is not possible to bound $\mathbf{P}\big(R_{\mathrm{eff}}(\rho^{\mathrm{dec}}_\alpha, y) \leq \lambda d(\rho^{\mathrm{dec}}_\alpha, y) \forall y \in B^d_R\big)$ uniformly in R.

Although this obstruction can probably be dealt with by tweaking the proofs of [KM08], it is easier to circumvent this problem by instead using the results of [KM08] to first estimate displacement with respect to the resistance metric, and then use Proposition 6.3 to account for the fluctuations and state the results in terms of the intrinsic metric. We therefore define the set

$$J^R(\lambda) = \big\{ r \in [1,\infty] : \lambda^{-1} v^R(r) \leq \mu(B^R_r) \leq \lambda v^R(r), R_{\mathrm{eff}}(\rho^{\mathrm{dec}}_\alpha, (B^R_r)^c) \geq \lambda^{-1} r \big\}, \tag{17}$$

where $v^R(r) = r^{\frac{\alpha(s^R_\alpha \wedge 1)}{(\alpha-1)(f^v_\alpha \wedge 1)}}$, with appropriate logarithmic corrections if $s^R_\alpha = 1$ or $f^v_\alpha = 1$ as in Theorem 1.3. It then follows from Propositions 5.1, 5.6 and 6.1 that there exists $c < \infty, \gamma > 0$ such that $\sup_{r \geq 1} \mathbf{P}\big(r \notin J^R(\lambda)\big) \leq c\lambda^{-\gamma}$, which allows us to directly apply [KM08, Proposition 1.3] to give exponents with respect to the resistance metric.

In what follows, we let $(X_n)_{n \geq 0}$ denote a simple random walk on $\mathcal{T}^{\mathrm{dec}}_\alpha$, started at $\rho^{\mathrm{dec}}_\alpha$, τ^d_r (respectively τ^R_r) denote the exit time of a simple random walk from B^d_r (respectively B^R_r), and let $p_{2n}(\rho^{\mathrm{dec}}_\alpha, \rho^{\mathrm{dec}}_\alpha)$ denote the transition density at the root. Also let I^R denote the inverse of the function $r \mapsto r v^R(r)$, so that $I^R(n) = n^{\frac{(\alpha-1)(f^v_\alpha \wedge 1)}{(\alpha-1)(f^v_\alpha \wedge 1) + \alpha(s^R_\alpha \wedge 1)}}$ when $s^R_\alpha, f^v_\alpha \neq 1$. The next result is a direct consequence of [KM08, Proposition 1.3].

Proposition 7.1 (Probabilistic results w.r.t. resistance metric)**.** *Assume that Assumption 1.4 holds, and let $n, r \geq 1$. Then, as $\theta \to \infty$, we have that*

$$\begin{aligned}
\liminf_{r\to\infty} \mathbf{P}\left(\theta^{-1} \leq \frac{\mathbb{E}_{\rho_\alpha^{dec}}[\tau_r^R]}{r v^R(r)} \leq \theta\right) &\to 1,\\
\liminf_{n\to\infty} \mathbf{P}\big(\theta^{-1} \leq v^R(I^R(n)) p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec}) \leq \theta\big) &\to 1,\\
\liminf_{n\to\infty} \mathbf{P} \times \mathbb{P}_{\rho_\alpha^{dec}}\left(\frac{R_{\mathit{eff}}(\rho_\alpha^{dec}, X_n)}{I^R(n)} \leq \theta\right) &\to 1,\\
\liminf_{n\to\infty} \mathbf{P} \times \mathbb{P}_{\rho_\alpha^{dec}}\left(\theta^{-1} \leq \frac{1 + R_{\mathit{eff}}(\rho_\alpha^{dec}, X_n)}{I^R(n)}\right) &\to 1.
\end{aligned}$$

While the definition of the transition density does not depend on the intrinsic metric, we can combine Proposition 7.1 with Proposition 6.3 to obtain displacement results in terms of the intrinsic metric d.

Before doing so, we recall from (2) that the (claimed) spectral dimension and displacement exponent for $\mathcal{T}_\alpha^{\mathrm{dec}}$ are respectively given by the following exponents:

$$\begin{aligned}
d_\alpha^{\mathrm{spec}} &= \frac{2\alpha(s_\alpha^R \wedge 1)}{\alpha(s_\alpha^R \wedge 1) + (\alpha - 1)(f_\alpha^v \wedge 1)},\\
d_\alpha^{\mathrm{dis}} &= \frac{(s_\alpha^R \wedge 1)(\alpha - 1)(f_\alpha^v \wedge 1)}{(\alpha - 1)(f_\alpha^v \wedge 1)(s_\alpha^d \wedge 1) + \alpha(s_\alpha^R \wedge 1)(s_\alpha^d \wedge 1)}.
\end{aligned}$$

To ease the notation in what follows, we further define:

1. The **walk dimension**, $w_\alpha = \frac{\alpha(s_\alpha^d \wedge 1)}{(\alpha-1)(f_\alpha^v \wedge 1)} + \frac{s_\alpha^d \wedge 1}{s_\alpha^R \wedge 1}$.
2. The **transition density exponent**, $s_\alpha = \frac{\alpha(s_\alpha^R \wedge 1)}{(\alpha-1)(f_\alpha^v \wedge 1)} \left(\frac{\alpha(s_\alpha^R \wedge 1)}{(\alpha-1)(f_\alpha^v \wedge 1)} + 1\right)^{-1}$.

Note that $d_\alpha^{\mathrm{dis}} = w_\alpha^{-1}$, and $d_\alpha^{\mathrm{spec}} = 2s_\alpha$.

The following proposition then follows from [KM08, Proposition 1.3] and Proposition 6.3.

Proposition 7.2 (Probabilistic results w.r.t. intrinsic metric)**.** *Assume that Assumption 1.4 holds, and let $n, r \geq 1$. Then, there exist (explicit) deterministic $\beta_\alpha^{(1)}, \beta_\alpha^{(2)}, \beta_\alpha^{(3)} > 0$ such that as $\theta \to \infty$,*

$$\begin{aligned}
\liminf_{r\to\infty} \mathbf{P} \times \mathbb{P}_{\rho_\alpha^{dec}}\left(\theta^{-1} \leq \frac{\tau_r^d}{r^{w_\alpha} (\log r)^{\beta_\alpha^{(1)}}} \leq \theta\right) &\to 1,\\
\liminf_{r\to\infty} \mathbf{P}\left(\theta^{-1} \leq \frac{\mathbb{E}_{\rho_\alpha^{dec}}[\tau_r^d]}{r^{w_\alpha} (\log r)^{\beta_\alpha^{(1)}}} \leq \theta\right) &\to 1,\\
\liminf_{n\to\infty} \mathbf{P}\Big(\theta^{-1} \leq n^{s_\alpha} (\log n)^{\beta_\alpha^{(2)}} p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec}) \leq \theta\Big) &\to 1,\\
\liminf_{n\to\infty} \mathbf{P} \times \mathbb{P}_{\rho_\alpha^{dec}}\left(\frac{\sup_{k\leq n} d_g^{dec}(\rho_\alpha^{dec}, X_k)}{n^{d_\alpha^{dis}} (\log n)^{\beta_\alpha^{(3)}}} \leq \theta\right) &\to 1,\\
\liminf_{n\to\infty} \mathbf{P} \times \mathbb{P}_{\rho_\alpha^{dec}}\left(\theta^{-1} \leq \frac{1 + d_g^{dec}(\rho_\alpha^{dec}, X_n)}{n^{d_\alpha^{dis}} (\log n)^{\beta_\alpha^{(3)}}}\right) &\to 1.
\end{aligned}$$

If $s_\alpha^R, s_\alpha^d, f_\alpha^v \neq 1$, then $\beta_\alpha^{(1)} = \beta_\alpha^{(2)} = \beta_\alpha^{(3)} = 0$. Otherwise they can be computed by carrying through the logarithmic terms from Theorem 1.3 and Proposition 6.3.

Proof. The first point does not follow directly from Proposition 7.1, which only gives a result for the *expected* exit time, but follows by exactly the same arguments used to prove the analogous result for trees in [CK08,

Proof of Theorem 1.1]. The second, third and fifth points are all a direct consequence of Proposition 7.1 and Proposition 6.3.

We therefore just need to prove the fourth point (the difference with the result in Proposition 7.1 is that we are considering the supremal distance up until time n, rather than the distance just at time n). We treat the case $s_\alpha^d, s_\alpha^R, f_\alpha^v \neq 1$ for simplicity, since this entails that $\beta_\alpha^{(3)} = 0$, but the other cases follow similarly. Set $R = n^{\frac{1}{w_\alpha}}$, and note that

$$\mathbf{P} \times \mathbb{P}_{\rho_\alpha^{\mathrm{dec}}} \left(\frac{\sup_{k \leq n} d_g^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, X_k)}{n^{d_\alpha^{\mathrm{dis}}} (\log n)^{\beta_\alpha^{(3)}}} > \theta \right) = \mathbf{P} \times \mathbb{P}_{\rho_\alpha^{\mathrm{dec}}} (\tau_{\theta R} < n) \, .$$

Now choose some $k \in (0, s_\alpha^d \wedge 1)$, and let $N_R = R^{s_\alpha^d \wedge 1} \theta^{(s_\alpha^d \wedge 1) - k}$. Let $\tilde{\tau}_R = \inf_{m \geq 0} X_m \in G(s_{N_R})$, where we recall that s_{N_R} is the N_R^{th} backbone vertex of T_α^∞. Then,

$$\mathbf{P} \times \mathbb{P}_{\rho_\alpha^{\mathrm{dec}}} (\tau_{\theta R} < n) \leq \mathbf{P} \times \mathbb{P}_{\rho_\alpha^{\mathrm{dec}}} (\tilde{\tau}_{N_R} < n) + \mathbf{P} \left(\sum_{i < N_R} \mathsf{Diam}(G(s_i)) \geq \frac{\theta}{2} R \right) + \mathbf{P} \left(A(N_R), \frac{\theta}{2} R \right), \tag{18}$$

where for $K \in \mathbb{N}, r > 0$, $A(K, r)$ is the event that there exists a decorated subtree $\mathcal{T}^{\mathrm{dec}}$ attached to the decorated backbone at some vertex $v \in \cup_{i \leq K} G(s_i)$, such that $\mathsf{Height}_{\mathrm{res}}^{\mathrm{dec}}(\mathcal{T}^{\mathrm{dec}}) \geq r$.

Now, note that under the annealed law $\mathbf{P} \times \mathbb{P}_{\rho_\alpha^{\mathrm{dec}}}$, the sequence of times for the random walk to get from $G(s_{(j-1)R^{(s_\alpha^d \wedge 1)}})$ to $G(s_{(j+1)R^{(s_\alpha^d \wedge 1)}})$ is an i.i.d. sequence. Therefore, for any $\delta \in (0, (s_\alpha^d \wedge 1) - k)$, we have that

$$\mathbf{P} \times \mathbb{P}_{\rho_\alpha^{\mathrm{dec}}} (\tilde{\tau}_{N_R} < n) \leq \mathbf{P} \times \mathbb{P}_{\rho_\alpha^{\mathrm{dec}}} \left(\tilde{\tau}_{R^{s_\alpha^d \wedge 1} \theta^\delta} < n \right)^{\theta^{(s_\alpha^d \wedge 1) - k - \delta}}. \tag{19}$$

Moreover,

$$\mathbf{P} \times \mathbb{P}_{\rho_\alpha^{\mathrm{dec}}} \left(\tilde{\tau}_{R^{s_\alpha^d \wedge 1} \theta^\delta} < n \right) \leq \mathbf{P} \left(\tau_{R\theta^{\frac{\delta}{2}(s_\alpha^d \wedge 1)^{-1}}} < n \right) + \mathbf{P} \left(\sum_{i \leq R^{s_\alpha^d \wedge 1} \theta^\delta} d^U(G(s_i)) \leq R \theta^{\frac{\delta}{2}(s_\alpha^d \wedge 1)^{-1}} \right). \tag{20}$$

Recall that $n = R^{w_\alpha}$. It therefore follows from the first point of this theorem that the first probability in (20) above tends to 0 as $n \to \infty$, uniformly in n. In particular, we have that $\mathbf{P} \times \mathbb{P}_{\rho_\alpha^{\mathrm{dec}}} \left(\tilde{\tau}_{R^{s_\alpha^d \wedge 1} \theta^\delta} < n \right) < \frac{1}{e}$ for all $\theta > \theta_c$, so it follows from (19) in this case that

$$\mathbf{P} \times \mathbb{P}_{\rho_\alpha^{\mathrm{dec}}} (\tilde{\tau}_{N_R} < n) \leq e^{-\theta^{(s_\alpha^d \wedge 1) - k - \delta}}.$$

Moreover, the second probability in (20) can be upper bounded by $e^{-c\theta^{\frac{\delta}{2}(s_\alpha^d \wedge 1)^{-1}}}$, by Lemma 3.2.

The second probability on the right hand side of (18) can be upper bounded by $\theta^{-(k-\varepsilon)(s_\alpha^d \vee 1)}$ by Lemma 3.1. We can deal with the third term in (18) using exactly the same arguments as in Proposition 6.1: we do not write the details, but we get for any $q > 0$ that

$$\mathbf{P} \left(A(N_R), \frac{\theta}{2} R \right) \leq c\theta^{-q(\alpha-1)} + c\theta^{-(\frac{k\alpha}{\alpha-1} - \alpha q)(\frac{\alpha-1}{\alpha} \vee \frac{1}{\alpha d}) t_\alpha^R}.$$

Therefore, by choosing $k > (s_\alpha^d \wedge 1) - \varepsilon, q \in (0, \frac{k}{\alpha - 1})$ and applying a union bound, we deduce from (18) that there exists some $A > 0$ such that

$$\mathbf{P} \times \mathbb{P}_{\rho_\alpha^{\mathrm{dec}}} \left(\frac{\sup_{k \leq n} d_g^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, X_k)}{n^{d_\alpha^{\mathrm{dis}}} (\log n)^{\beta_\alpha^{(3)}}} > \theta \right) = \mathbf{P} \times \mathbb{P}_{\rho_\alpha^{\mathrm{dec}}} (\tau_{\theta R} < n) \leq \theta^{-A}, \tag{21}$$

for all $n, R \geq 1$, which proves the result. □

With appropriate control, we can also get quenched and annealed results for these exponents. We give the quenched result first: again this follows directly from [KM08, Proposition 1.5] and Proposition 6.3.

Theorem 7.3 (Quenched random walk results)**.** *Under Assumption 1.4,* $\mathbf{P}$*-almost surely,*

a) There exist constants $\beta_1, \beta_2, \beta_3, \beta_4 \in (0, \infty)$ *such that*

(i) There exists $R < \infty$ *such that* $r^{w_\alpha}(\log r)^{-\beta_2} \le \mathbb{E}_{\rho_\alpha^{dec}}[\tau_r^d] \le r^{w_\alpha}(\log r)^{\beta_2}$ *for all* $r \ge R$.

(ii) There exists $N < \infty$ *such that* $n^{-s_\alpha}(\log n)^{-\beta_1} \le p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec}) \le n^{-s_\alpha}(\log n)^{\beta_1}$ *for all* $n \ge N$.

(iii) $\mathbb{P}$*-almost surely, there exist* $N, R < \infty$ *such that, conditional on* $X_0 = \rho_\alpha^{dec}$,

$$r^{w_\alpha}(\log r)^{-\beta_3} \le \tau_r^d \le r^{w_\alpha}(\log r)^{\beta_3} \qquad \forall r \ge R$$
$$n^{d_\alpha^{dis}}(\log n)^{-\beta_4} \le \sup_{k \le n} d_g^{dec}(\rho_\alpha^{dec}, X_k) \le n^{d_\alpha^{dis}}(\log n)^{\beta_4} \qquad \forall n \ge N$$

b) $d_S(\mathcal{T}_\alpha^{dec}) := -2\lim_{n\to\infty} \frac{\log p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec})}{\log n} = 2s_\alpha$.

c) $\lim_{n\to\infty} \frac{\log\left(\mathbb{E}_{\rho_\alpha^{dec}}[\tau_r^d]\right)}{\log r} = w_\alpha$.

d) Let $W_n = \{X_0, X_1, \dots, X_n^{exc}\}$*, and let* $S_n = \sum_{x \in W_n} \deg x$*. Then* $\mathbb{P}$*-almost surely,* $\lim_{n\to\infty} \frac{\log S_n}{\log n} = s_\alpha$.

Theorem 1.7 therefore follows directly. The annealed results also follow from similar arguments to those of [KM08, Proposition 1.4], but require some adaptation since [KM08, Assumption 1.2(2)] is not in general satisfied.

Theorem 7.4 (Annealed random walk results)**.** *Take* $\beta_\alpha^{(1)}, \beta_\alpha^{(2)}, \beta_\alpha^{(3)}$ *as in Proposition 7.2. Under Assumption 1.4, we have that:*

a) There exist constants $c_1 > 0, c_2 < \infty, \gamma_1 > 0$ *such that, for all* $r \ge 1$, $c_1 r^{w_\alpha}(\log r)^{\beta_\alpha^{(1)}} \le \mathbf{E}\big[\mathbb{E}_{\rho_\alpha^{dec}}[\tau_r^d]\,\big]$, *and* $\mathbf{E}\big[\big(\mathbb{E}_{\rho_\alpha^{dec}}[\tau_r^d]\,\big)^{\gamma_1}\big] \le c_2 r^{\gamma w_\alpha}(\log r)^{\gamma_1 \beta_\alpha^{(1)}}$.

b) There exist constants $c_3 > 0, c_4 < \infty, \gamma_2 > 0$ *such that* $c_3 n^{-s_\alpha}(\log n)^{\beta_\alpha^{(2)}} \le \mathbf{E}\big[p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec})\big]$ *for all* $n \ge 1$, *and* $\mathbf{E}\big[\big(p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec})\big)^{\gamma_2}\big] \le c_3 n^{-\gamma_2 s_\alpha}(\log n)^{\gamma_2 \beta_\alpha^{(2)}}$.

c) There exist constants $c_5 > 0, c_6 < \infty, \gamma_3 > 0$ *such that* $c_5 r^{d_\alpha^{dis}}(\log r)^{\beta_\alpha^{(3)}} \le \mathbf{E}\big[\mathbb{E}_{\rho_\alpha^{dec}}\big[d_g^{dec}(\rho_\alpha^{dec}, X_n)\big]\,\big]$, *and* $\mathbf{E}\big[\big(\mathbb{E}_{\rho_\alpha^{dec}}\big[d_g^{dec}(\rho_\alpha^{dec}, X_n)\big]\,\big)^{\gamma_3}\big] \le c_6 r^{\gamma_3 d_\alpha^{dis}}(\log r)^{\gamma_3 \beta_\alpha^{(3)}}$.

Proof. We give the proof since not all the statements are immediate from [KM08, Proposition 1.4]. The strategy is similar to the proof of [CK08, Theorem 1.2]. The lower bounds all follow directly from Proposition 7.2, since we can choose $c_1, c_3, c_5 > 0$ and $\delta > 0$ so that the required lower bounds hold with probability at least δ.

For the upper bound in part a), we recall from [KM08, Equation (3.7)] that

$$\mathbb{E}_{\rho_\alpha^{\mathrm{dec}}}\left[\tau_r^d\right]^{\gamma_1} < R_{\mathrm{eff}}(\rho_\alpha^{\mathrm{dec}}, B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r)^c)^{\gamma_1}\ \mu(B_{\mathcal{T}_\alpha^{\mathrm{dec}}}(\rho_\alpha^{\mathrm{dec}}, r))^{\gamma_1}.$$

Therefore, using Propositions 5.1, 6.3 and Cauchy-Schwarz we can find γ_1 positive but small enough that the required expectation is finite.

For the upper bound in part b) we can work directly with the resistance metric. As in the proof of [KM08, Remark 1.6(1)], we can rearrange the third line of [BCK05, Proposition 3.3] to deduce that there exists $c < \infty$ such that

$$p_{2n}(\rho_\alpha^{\mathrm{dec}}, \rho_\alpha^{\mathrm{dec}}) \le c\left(\frac{1}{\mu(B_{\mathrm{res}}^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, r))} \vee \frac{r}{n}\right).$$

We deduce that

$$\mathbf{E}\big[p_{2n}(\rho_\alpha^{\mathrm{dec}}, \rho_\alpha^{\mathrm{dec}})^{\gamma_2}\big] \le 2\left[\mathbf{E}\left[\left(\frac{c}{\mu(B_{\mathrm{res}}^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, r))}\right)^{\gamma_2}\right] \vee \left(\frac{cr}{n}\right)^{\gamma_2}\right],$$

so that if $n \in [rv^R(r), 2rv^R(r)]$ we have by Proposition 5.6 that we can choose γ_2 positive but small enough that both terms are upper bounded by $cn^{-\gamma_2 s_\alpha}(\log n)^{\gamma_2 \beta_\alpha^{(2)}}$.

The upper bound in part c) follows from the tail bound of (21). □

In particular, this gives all the results of Theorem 1.8. Note that it should not in general be possible to obtain all of the same results without taking lower powers, since it is known that the expected volume of a ball in T_α^∞ is infinite, which has similar consequences for some of the quantities below. See [CK08, Theorem 1.2] for analogous results for T_α^∞.

Finally, we establish some weaker results under Assumption 1.1 in place of Assumption 1.4. In particular we recall that in this case we are able to control decorated distances along most but not all the branches in the underlying tree. We therefore define the set A_r^d to be the graph obtained from $\mathcal{T}_\alpha^{\mathrm{dec}}$ by first removing all vertices contained in $G(u)$ for some u on the backbone of T_α^∞ that also lie at decorated distance greater than r from the root, and then retaining the connected component of containing $\rho_\alpha^{\mathrm{dec}}$. Note that $B_r^d \subset A_r^d$. We define A_r^R analogously but with respect to the resistance metric.

Since all of our proofs in Section 5 involve approximating B_r^d (respectively B_r^R) by cutting at an appropriate point on the decorated backbone, the volume results of Propositions 5.1 and 5.6 also hold for A_r^d (respectively A_r^R). Also, resistance to the boundary of A_r^R is really of order r with high probability, since the only way for the random walk to escape A_r^R is via the infinite decorated backbone. This motivates the definition of the set

$$J^{R,A}(\lambda) = \left\{ r \in [1,\infty) : \lambda^{-1} v^R(r) \le \mu(A_r^R) \le \lambda v^R(r), R_{\mathrm{eff}}(\rho_\alpha^{\mathrm{dec}}, (A_r^R)^c) \ge \lambda^{-1} r \right\}, \tag{22}$$

The only difference with the set $J^R(\lambda)$ in (17) is the use of A_r^R in place of B_r^R. By repeating many of the arguments of [KM08], we are therefore able to control exit times from A_r^R, with high probability. Moreover, although A_r^R contains points that are atypically far from the root, the proof of Proposition 5.6 shows that "most" points in A_r^R are also within B_r^R, so that at any given time, a random walk conditioned to be in A_r^R is also within B_r^R, with high probability. This enables us to control $d_g^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, X_n)$ with high probability, but not $\sup_{k\le n} d_g^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, X_k)$. We therefore deduce the following theorem, from which Theorem 1.10 follows.

Theorem 7.5. *Take $\beta_\alpha^{(1)}, \beta_\alpha^{(2)}, \beta_\alpha^{(3)}$ as in Proposition 7.1. Under Assumption 1.1, we have that*

(i) (cf Proposition 7.1). There exists $k < \infty$ such that for all $\theta > 1$,

$$\limsup_{n\to\infty} \mathbf{P}\left(\theta^{-1} \le n^{s_\alpha} (\log n)^{\beta_\alpha^{(2)}} p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec}) \le \theta \right) \ge 1 - \theta^{-k} \tag{23}$$

$$\limsup_{n\to\infty} \mathbf{P} \times \mathbb{P}_{\rho_\alpha^{dec}}\left(\frac{d_g^{dec}(\rho_\alpha^{dec}, X_n)}{n^{d_\alpha^{dis}} (\log n)^{\beta_\alpha^{(3)}}} \le \theta \right) \ge 1 - \theta^{-k}, \tag{24}$$

$$\limsup_{n\to\infty} \mathbf{P} \times \mathbb{P}_{\rho_\alpha^{dec}}\left(\theta^{-1} \le \frac{1 + d_g^{dec}(\rho_\alpha^{dec}, X_n)}{n^{d_\alpha^{dis}} (\log n)^{\beta_\alpha^{(3)}}} \right) \ge 1 - \theta^{-k}. \tag{25}$$

(ii) (cf Theorem 7.3).

a) $\mathbf{P}$-almost surely, there exists $\beta_5, N < \infty$ such that $n^{-s_\alpha} (\log n)^{-\beta_5} \le p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec}) \le n^{-s_\alpha} (\log n)^{\beta_5}$ for all $n \ge N$. Consequently, $d_s(\mathcal{T}_\alpha^{dec}) = 2s_\alpha$.

b) (Quenched displacement exponent, lower bound). $\mathbf{P}$-almost surely, there exists $N, \beta_6 < \infty$ such that $n^{d_\alpha^{dis}} (\log n)^{-\beta_6} \le \sup_{k\le n} d_g^{dec}(\rho_\alpha^{dec}, X_n)$.

(iii) (cf Theorem 7.4).

a) There exist constants $c_3 > 0, c_4 < \infty, \gamma_2 > 0$ such that $\mathbf{E}\left[\left(p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec}) \right)^{\gamma_2} \right] \le c_3 n^{-\gamma_2 s_\alpha} (\log n)^{\gamma_2 \beta_\alpha^{(2)}}$ and $c_3 n^{-s_\alpha} (\log n)^{\beta_\alpha^{(2)}} \le \mathbf{E}\left[p_{2n}(\rho_\alpha^{dec}, \rho_\alpha^{dec}) \right]$ for all $n \ge 1$.

b) There exist constants $c_5 > 0, c_6 < \infty, \gamma_3 > 0$ such that $c_5 n^{d_\alpha^{dis}} (\log n)^{\beta_\alpha^{(3)}} \le \mathbf{E}\left[\mathbb{E}_{\rho_\alpha^{dec}}\left[d_g^{dec}(\rho_\alpha^{dec}, X_n) \right] \right]$, and $\mathbf{E}\left[\left(\mathbb{E}_{\rho_\alpha^{dec}}\left[d_g^{dec}(\rho_\alpha^{dec}, X_n) \right] \right)^{\gamma_3} \right] \le c_6 n^{\gamma_3 d_\alpha^{dis}} (\log n)^{\gamma_3 \beta_\alpha^{(3)}}$.

Proof. By the same arguments as in Proposition 7.2, we deduce that there exists $k' < \infty$ such that

$$\begin{aligned} \limsup_{\theta\to\infty} \mathbf{P} \times \mathbb{P}_{\rho_\alpha^{\mathrm{dec}}}\left(\theta^{-1} \le \frac{\tau^d_{A^d_r}}{r^{w_\alpha}(\log r)^{\beta^{(1)}_\alpha}} \le \theta\right) &\ge 1-\theta^{-k'}, \\ \limsup_{\theta\to\infty} \mathbf{P}\left(\theta^{-1} \le \frac{\mathbb{E}_{\rho_\alpha^{\mathrm{dec}}}\left[\tau^d_{A^d_r}\right]}{r^{w_\alpha}(\log r)^{\beta^{(1)}_\alpha}} \le \theta\right) &\ge 1-\theta^{-k'}. \end{aligned} \tag{26}$$

A transition density upper bound follows by [KM08, Proposition 3.1a)] applied to the ball B^R_r (note that this only requires an effective resistance bound between points, and not to the boundary of a ball). The arguments used to prove [KM08, Equation (1.25)] on page 19 of that paper also apply to give the tail decay and annealed transition density upper bound. For the quenched result, we can apply Borel-Cantelli along the subsequence 2^n and use the fact that $p_{2n}(\rho^{\mathrm{dec}}_\alpha, \rho^{\mathrm{dec}}_\alpha)$ is decreasing in n (this follows from a spectral decomposition, for example).

The transition density lower bounds follow from the same arguments used in [KM08, Proofs of Proposition 3.2 and Proposition 3.3(2)], along with the arguments used in the proof of Proposition 1.4 on page 18 of that paper, but using the set A^R_r in place of B^R_r. The quenched lower bound again follows from Borel-Cantelli along the subsequence 2^n and monotonicity of $p_{2n}(\rho^{\mathrm{dec}}_\alpha, \rho^{\mathrm{dec}}_\alpha)$.

To obtain the lower bounds on $d^{\mathrm{dec}}_g(\rho^{\mathrm{dec}}_\alpha, X_n)$, note that, by Cauchy-Schwarz and a similar argument to [CHN, Equation (26)], we have for any $B \subset \mathcal{T}^{\mathrm{dec}}_\alpha$ that

$$\begin{aligned} \mathbb{P}_{\rho^{\mathrm{dec}}_\alpha}(X_n \in B) = \sum_{v\in B} \frac{\mathbb{P}_{\rho^{\mathrm{dec}}_\alpha}(X_n = v)}{\sqrt{\deg v}}\sqrt{\deg v} &\le \left(\sum_{v\in B} \frac{\mathbb{P}_{\rho^{\mathrm{dec}}_\alpha}(X_n = v)^{\,2}}{\deg v} \cdot \sum_{v\in B} \deg v\right)^{\frac12} \\ &= \left(\sum_{v\in B} \frac{\mathbb{P}_{\rho^{\mathrm{dec}}_\alpha}(X_n = v)\,\mathbb{P}_v\big(X_n = \rho^{\mathrm{dec}}_\alpha\big)}{\deg \rho^{\mathrm{dec}}_\alpha} \cdot \sum_{v\in B} \deg v\right)^{\frac12} \\ &\le \big(p_n(\rho^{\mathrm{dec}}_\alpha, \rho^{\mathrm{dec}}_\alpha)\mu(B)\big)^{\frac12}. \end{aligned} \tag{27}$$

Taking $B = B_{\theta^{-1} n^{d^{\mathrm{dis}}_\alpha} (\log n)^{\beta^{(3)}_\alpha}}$, (25) therefore follows from Proposition 5.6, (23) and (27). This argument also implies a tail bound polynomial in θ, so for any $\delta > 0$ we can take $\theta > 0$ so that

$$\liminf_{n\to\infty} \mathbf{P} \times \mathbb{P}_{\rho^{\mathrm{dec}}_\alpha}\left(\theta^{-1} \le \frac{1 + d^{\mathrm{dec}}_g(\rho^{\mathrm{dec}}_\alpha, X_n)}{n^{d^{\mathrm{dis}}_\alpha}(\log n)^{\beta^{(3)}_\alpha}}\right) > \delta,$$

which gives the result on $\mathbf{E}\big[\mathbb{E}_{\rho^{\mathrm{dec}}_\alpha}\big[d^{\mathrm{dec}}_g(\rho^{\mathrm{dec}}_\alpha, X_n)\big]\big]$. The quenched lower bound on $\sup_{k\le n} d^{\mathrm{dec}}_g(\rho^{\mathrm{dec}}_\alpha, X_n)$ follows from (25) and a Borel-Cantelli argument along the subsequence $r_n = 2^n$.

For the upper bounds on $d^{\mathrm{dec}}_g(\rho^{\mathrm{dec}}_\alpha, X_n)$, assume first that n is even, and set $r = n^{d^{\mathrm{dis}}_\alpha}(\log n)^{\beta^{(3)}_\alpha}$. Note that it follows from the proof of Proposition 5.6(ii) (which still hold if $f^v_\alpha < 1$) that there exists $a \in (0,1), b > 0$ such that $\mathbf{P}\Big(\frac{A_{r\theta^a} \setminus B_{r\theta}}{A_{r\theta^a}} \ge \theta^{-b(1-a)}\Big) \le \theta^{-c}$. Now taking $B = A_{r\theta^a} \setminus B_{r\theta}$, we get from (27) that (since $s_\alpha = d^{\mathrm{dis}}_\alpha d^{\mathrm{vol}}_\alpha$):

$$\begin{aligned} \mathbf{P} \times \mathbb{P}_{\rho^{\mathrm{dec}}_\alpha}(X_n \in B) \le{}& \mathbf{P}\big(p_n(\rho^{\mathrm{dec}}_\alpha, \rho^{\mathrm{dec}}_\alpha) \ge n^{s_\alpha}\theta^c\big) + \mathbf{P}\big(\mu(A_{r\theta^a}) \ge n^{-s_\alpha}\theta^d\big) \\ &+ \mathbf{P}\left(\frac{A_{r\theta^a} \setminus B_{r\theta}}{A_{r\theta^a}} \ge \theta^{-b(1-a)}\right) + \theta^{\frac{c+d-b(1-a)}{2}}. \end{aligned} \tag{28}$$

In particular, we can choose a and then c and d small enough that $c + d - b(1-a) < 0$. Still taking $r = n^{d^{\mathrm{dis}}_\alpha}(\log n)^{\beta^{(3)}_\alpha}$, we therefore have from (26) and (28) that there exists $k > 0$ such that

$$\mathbf{P} \times \mathbb{P}_{\rho^{\mathrm{dec}}_\alpha}\left(\frac{d^{\mathrm{dec}}_g(\rho^{\mathrm{dec}}_\alpha, X_n)}{n^{d^{\mathrm{dis}}_\alpha}(\log n)^{\beta^{(3)}_\alpha}} > \theta\right) \le \mathbf{P} \times \mathbb{P}_{\rho^{\mathrm{dec}}_\alpha}\Big(\tau^d_{A^d_{r\theta^a}} \le r^{w_\alpha}(\log r)^{\beta^{(1)}_\alpha}\Big) + \mathbf{P} \times \mathbb{P}_{\rho^{\mathrm{dec}}_\alpha}(X_n \in A_{r\theta^a} \setminus B_{r\theta}) \le \theta^{-k}.$$

When d_g^{dec} is the graph metric, the result for odd n follows immediately for all n since $d_g^{\mathrm{dec}}(X_n, X_{n+1}) = 1$. If d_g^{dec} is some other metric, we can still easily extend the result to odd times since

$$\begin{aligned}
&\limsup_{n\to\infty} \mathbf{P}\Big(d_g^{\mathrm{dec}}(X_n, X_{n+1}) \geq \theta n^{d_\alpha^{\mathrm{dis}}} (\log n)^{\beta_\alpha^{(3)}}\Big) \\
&\leq \limsup_{n\to\infty} \mathbf{P}\left(\sup_{v\in T_\alpha^\infty : G(v)\cap B_{\theta n^{d_\alpha^{\mathrm{dis}}}(\log n)^{\beta_\alpha^{(3)}}} \neq \emptyset} \mathsf{Diam}(G(v)) \geq \theta n^{d_\alpha^{\mathrm{dis}}} (\log n)^{\beta_\alpha^{(3)}}\right) \\
&\leq \theta^{-s_\alpha^v}.
\end{aligned}$$

(This can be proved formally using similar arguments to those used to prove (16), and considering the backbone graphs separately). The polynomial tail bound in θ also implies the result for $\mathbf{E}\big[\big(\mathbb{E}_{\rho_\alpha^{\mathrm{dec}}}\big[d_g^{\mathrm{dec}}(\rho_\alpha^{\mathrm{dec}}, X_n)\big]\big)^{\gamma_3}\big]$. □

Theorem 1.6 follows from the $\mathbf{P}$-almost sure bounds on $p_n(\rho_\alpha^{\mathrm{dec}}, \rho_\alpha^{\mathrm{dec}})$.

8 Examples

In terms of exponents, the main results of the previous sections are that we established the following exponents:

1. The **fractal dimension** is equal to $\frac{\alpha(s_\alpha^d \wedge 1)}{(\alpha-1)(f_\alpha^v \wedge 1)}$.
2. The **spectral dimension** is equal to $\frac{2\alpha(s_\alpha^R \wedge 1)}{\alpha(s_\alpha^R \wedge 1)+(\alpha-1)(f_\alpha^v \wedge 1)}$.
3. The **displacement exponent** is equal to $\frac{(s_\alpha^R \wedge 1)(\alpha-1)(f_\alpha^v \wedge 1)}{(\alpha-1)(f_\alpha^v \wedge 1)(s_\alpha^d \wedge 1)+\alpha(s_\alpha^R \wedge 1)(s_\alpha^d \wedge 1)}$ (this is also the inverse of the **walk dimension**).

Below, we give the values of these exponents for several examples of interest.

8.1 Trees

By inserting an appropriate "star" graph at every vertex, or simply repeating the arguments employed in the previous section directly for trees, we recover some results for random walks on critical Galton-Watson trees with offspring distribution satisfying (3), conditioned to survive. We do not go into the details, but in this setting Assumption 1.4 is effectively satisfied with $d = R = \infty, v = 1$ for $\lambda > 2$ (though to make this rigorous, it easier just to repeat the arguments directly with these trees in mind). Random walks on these trees were studied by Croydon and Kumagai in [CK08], and we recover the exponents they establish there, as given in Table 1.

We note that it is possible to improve the polynomial tails of Section 5.2 and [CK08, Proposition 2.6] by adapting the argument used for looptrees in [BS15, Lemma 3.2], and obtain that, for T_α^∞,

$$\mathbf{P}\Big(\mu(B(\rho, r)) \leq r^{\frac{1}{\alpha-1}} \lambda^{-1}\Big) \leq Ce^{-c\lambda^{\frac{\alpha-1}{\alpha}}}.$$

For more detailed results on the tree case, see [CK08] and [Kor17, Proposition 6].

Inserted graph	Range	Volume	Spectral dim	Displacement
Star (tree)	all	$\frac{\alpha}{\alpha-1}$	$\frac{2\alpha}{2\alpha-1}$	$\frac{\alpha-1}{2\alpha-1}$
Loop (looptree)	all	α	$\frac{2\alpha}{\alpha+1}$	$\frac{1}{\alpha+1}$
β-stable trees	$\frac{\beta}{\beta-1} \geq \frac{1}{\alpha-1}$	$\frac{\alpha}{\alpha-1}$	$\frac{2\alpha}{2\alpha-1}$	$\frac{\alpha-1}{2\alpha-1}$
	$\frac{\beta}{\beta-1} < \frac{1}{\alpha-1}$	$\frac{\beta\alpha}{\beta-1}$	$\frac{2\beta\alpha}{\beta-1+\beta\alpha}$	$\frac{\beta-1}{\beta-1+\beta\alpha}$
Finite variance dissections, if $\mu(k,\infty) = o(k^{\frac{-\alpha}{\alpha-1}})$	$\alpha \geq \frac{3}{2}$	$\frac{\alpha}{\alpha-1}$	$\frac{2\alpha}{2\alpha-1}$	$\frac{\alpha-1}{2\alpha-1}$
	$\alpha \leq \frac{3}{2}$	2α	$\frac{4\alpha}{2\alpha+1}$	$\frac{1}{2\alpha+1}$
β-stable dissections*	$\beta \geq \frac{1}{\alpha-1}$	$\frac{\alpha}{\alpha-1}$	$\frac{2\alpha}{\alpha-1}$	$\frac{\alpha-1}{2\alpha-1}$
	$\beta \leq \frac{1}{\alpha-1}$	$\alpha\beta$	$\frac{2\alpha\beta}{\alpha\beta+1}$	$\frac{1}{\alpha\beta+1}$
Critical Erdös-Rényi (n)	$\alpha \geq \frac{3}{2}$	$\frac{\alpha}{\alpha-1}$	$\frac{2\alpha}{2\alpha-1}$	$\frac{\alpha-1}{2\alpha-1}$.
	$\alpha \leq \frac{3}{2}$	2α	$\frac{4\alpha}{2\alpha+1}$	$\frac{1}{2\alpha+1}$
Critical Erdös-Rényi (n^β)	$\alpha \geq \beta$	$\frac{\alpha}{\alpha-1}$	$\frac{2\alpha}{2\alpha-1}$	$\frac{\alpha-1}{2\alpha-1}$
	$\frac{\beta+2}{2} \leq \alpha \leq \beta$	$\frac{\beta}{\alpha-1}$	$\frac{2\beta}{\beta+\alpha-1}$	$\frac{\alpha-1}{\alpha-1+\beta}$
	$\alpha \leq \frac{\beta+2}{2}$	2	$\frac{4}{3}$	$\frac{1}{3}$
Sierpinski triangle	$\alpha \geq \frac{\log 10-\log 3}{\log 2}$	$\frac{\alpha}{\alpha-1}$	$\frac{2\alpha}{2\alpha-1}$	$\frac{\alpha-1}{2\alpha-1}$
	$\frac{\log 3}{\log 2} \leq \alpha < \frac{\log 10-\log 3}{\log 2}$	α	$\frac{2\alpha \log 2}{\alpha \log 2+\log 5-\log 3}$	$\frac{\log 2}{\alpha \log 2+\log 5-\log 3}$
	$\alpha < \frac{\log 3}{\log 2}$	$\frac{\log 3}{\log 2}$	$\frac{2 \log 3}{\log 5}$	$\frac{\log 2}{\log 5}$
Complete graph	all	$\frac{2}{\alpha-1}$	$\frac{4}{\alpha+1}$	$\frac{\alpha-1}{\alpha+1}$
2d box	all	2	$\frac{4}{\alpha+1}$	$\frac{1}{\alpha+1}$

Table 1: Quenched exponents for the models considered below. *Here only Assumption 1.1 is satisfied.

8.2 Looptrees

By inserting deterministic loops at each vertex, we also recover the discrete looptree model that was considered more thoroughly in [BS15]. In this case, $d = R = v = 1$. In fact, in this case it is also possible to repeat the arguments of [Arc20a] used for the continuum case to get stronger tail decay, and show that

$$\mathbf{P}(\mu(B(\rho, r)) \geq r^\alpha \lambda) \leq C\lambda^{\frac{1}{\alpha+1}} e^{-c\lambda^{\frac{\alpha-\varepsilon}{\alpha+1}}}$$

In terms of the lower tail decay, the following bound was shown in [BS15, Equation (3.23)] (we can drop the extra δ present there since we are neglecting slowly-varying corrections to the offspring distribution):

$$\mathbf{P}\big(\mu(B(\rho, r)) \leq r^\alpha \lambda^{-1}\big) \leq Ce^{-c\lambda^{\frac{1}{\alpha}}}.$$

As shown in [BS15], it follows from this tail decay that Assumption 1.4 is satisfied, so that both the annealed and quenched spectral dimensions are $\frac{2\alpha}{\alpha+1}$, and the displacement exponent is $\frac{1}{\alpha+1}$. See [BS15] for more details.

8.3 Inserting trees

Since we have good control on volumes in trees, we could also insert a separate Galton-Watson tree conditioned to have n leaves at each vertex of degree n. To establish the volume exponents in this case, note that since the number of edges of a tree is one less than the total number of vertices, we can use [Kor12, Proposition 1.6] to deduce that, if T is an unconditioned Galton-Watson tree with offspring distribution $\hat{\xi}(x) \sim cx^{-\beta}$ as $x \to \infty$ for some $\beta \in (1, 2]$, and $l(T)$ is its number of leaves, then (applying Lemma 4.2 and an LDP):

$$\mathbf{P}(l(T) = n, |T| \geq \lambda n) = \sum_{p \geq \lambda n} \frac{1}{p} \mathbf{P}\big(W'_{p-n} = n-1\big)\, \mathbf{P}(S_p = n) \leq \frac{C}{\lambda n} \left(e^{-\lambda^\beta} + o(n^{\frac{-1}{\beta}})\right) e^{-c\lambda n},$$

where W' is a random walk started from zero with jump distribution $\eta(i) = \frac{p_{i+1}}{1-p_0}$ for $i \geq 1$, and S_p is a sum of p independent $\mathsf{Bernoulli}(p_0)$ random variables. Using also the asymptotic of [Kor12, Theorem 3.1(ii)] that $\mathbf{P}(l(T) = n) \sim cn^{-(1+\frac{1}{\beta})}$ as $n \to \infty$, we deduce that

$$\mathbf{P}(|T| \geq \lambda n \mid l(T) = n) \leq Ce^{-\lambda^{\beta}} + Ce^{-c\lambda n} = o(e^{-c\lambda}).$$

Clearly also

$$\mathbf{P}\left(|T| \leq \lambda^{-1} n \;\middle|\; l(T) = n\right) = 0$$

for all $\lambda \geq 2, n \geq 1$, so we deduce that $v = 1, s_\alpha^v = \alpha - 1$, and $f_\alpha^v = \alpha$.

To bound $\mathbf{P}\left(\mathsf{Diam}(T) \geq \lambda n^{1-\frac{1}{\beta}} \;\middle|\; l(T) = n\right)$, we first bound the quantity

$$\mathbf{P}\left(l(T) \in [n, 2n] \;\middle|\; \mathsf{Diam}(T) \geq \lambda n^{1-\frac{1}{\beta}}\right)$$

by decomposing along the Williams' spine, which we know has length at least $\frac{1}{2}\lambda n^{1-\frac{1}{\beta}}$. By Proposition 4.5, we know that for any vertex v on the Williams' spine within distance $\frac{1}{4}\lambda n^{1-\frac{1}{\beta}}$ of the root, there exists a constant c such that for all $x \leq \lambda n^{\frac{1}{\beta}}$, $\mathbf{P}(\deg v \geq x) \geq cx^{-(\beta-1)}$, independently for each such v. Therefore, by coupling the degree of v with an independent random variable Y satisfying $\mathbf{P}(\deg v \geq x) \geq cx^{-(\beta-1)}$ for all $x \geq 0$, we get that

$$\mathbf{E}\left[e^{-\theta(\deg v - 2)}\right] \leq \mathbb{E}\left[e^{-\theta Y}\right] + \mathbf{P}\left(Y \geq \lambda n^{\frac{1}{\beta}}\right),$$

where the latter term corresponds to a "worst-case" scenario on "seeing the difference" between Y and $\deg v$. In particular, if $\theta = cn^{\frac{-1}{\beta}}$ then by Lemma 3.3

$$\mathbf{E}\left[e^{-\theta(\deg v - 2)}\right] \leq 1 - c'n^{\frac{-(\beta-1)}{\beta}} + \lambda^{\frac{-1}{\beta-1}} n^{\frac{-(\beta-1)}{\beta}} \leq \exp\{-c''n^{\frac{-(\beta-1)}{\beta}}\}. \tag{29}$$

Therefore, letting $v_1, v_2, \ldots v_{\frac{1}{4}\lambda n^{1-\frac{1}{\beta}}}$ denote the vertices on the Williams' spine within distance $\frac{1}{4}\lambda n^{1-\frac{1}{\beta}}$ of the root, $(T_{i,j})_{i=1}^{\deg v_j - 2}$ denote the subtrees emanating from all of the non-spinal offspring of vertex v_j, $l(T_i)$ denote the number of leaves in each T_i, using the asymptotic of [Kor12, Theorem 3.1(ii)] that $\mathbf{P}(l(T) = n) \sim cn^{-(1+\frac{1}{\beta})}$ as $n \to \infty$, and then taking $\theta = n^{-1}$ and using (29) we have that

$$\mathbf{P}\left(l(T) \leq 2n \;\middle|\; \mathsf{Diam}(T) \geq \lambda n^{1-\frac{1}{\beta}}\right) \leq \mathbf{E}\left[\exp\left\{-\theta \sum_{j=1}^{\frac{1}{4}\lambda n^{1-\frac{1}{\beta}}} \sum_{i=1}^{\deg(v_j)-2} l(T_{i,j})\right\}\right] e^{2\theta n} \leq Ce^{-c\lambda}.$$

To recover the desired bound, we then use monotonicity and Bayes' Law to write

$$\mathbf{P}\left(\mathsf{Diam}(T) \geq \lambda n^{1-\frac{1}{\beta}} \;\middle|\; l(T) = n\right) \leq \mathbf{P}\left(\mathsf{Diam}(T) \geq \lambda n^{1-\frac{1}{\beta}} \;\middle|\; l(T) \in [n, 2n]\right) \leq Ce^{-c\lambda}.$$

Since resistance is equal to the graph distance on trees, this verifies the remaining conditions of Assumption 1.4 with $R = d = \frac{\beta}{\beta-1}$, so that $s_\alpha^d = s_\alpha^R = \frac{\beta(\alpha-1)}{\beta-1}$, so substituting these we deduce the results in Table 1.

Note that we would expect the same results if we inserted a tree with n vertices in total, rather than n leaves, at a vertex of degree n, since the leaves asymptotically make up a constant proportion of the mass of the tree.

8.4 Outerplanar maps: inserting dissected polygons

Let P_n be a convex polygon inscribed in the unit disc whose vertices correspond to the n^{th} roots of unity. A *dissection* of P_n is obtained from P_n by inserting a collection of chords that make up distinct diagonals of P_n: see Figure 3.

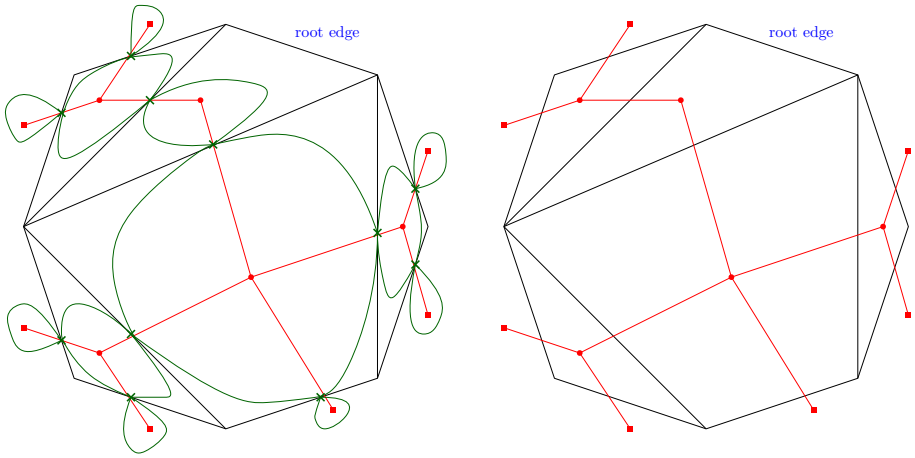


Figure 3: A dissection and its inscribed tree and looptree.

If ν is a critical probability measure on the set $\{0, 2, 3, 4, \ldots\}$, we can define a Boltzmann measure on dissections of the n-gon, $\mathbb{P}_n^\nu$, by setting

$$\mathbb{P}_n^\nu(D) \propto \prod_{F \in \mathsf{Faces}(D)} \nu_{\deg f - 1}.$$

For convenience we will assume that the support of ν is the entirety of the set $\{0, 2, 3, 4, \ldots\}$. The measure $\mathbb{P}_n^\nu$ is then well-defined under sensible assumptions on the tail of ν (see [Kor14, Section 1]).

Letting D_n^ν denote a random Boltzmann dissection sampled according to $\mathbb{P}_n^\nu$, D_n^ν is now a natural candidate for decoration at a vertex of degree n in T_α^∞. We will view D_n^ν as a metric space (rather than as an embedding in the plane) by giving each edge of D_n^ν length 1. To establish exponents for the diameter and two-point function of D_n^ν, we will use a bijection between dissections of P_n and trees with n vertices, as illustrated in Figure 3. It is shown in [Kor14, Proposition 1.4] that, if T_n is the tree obtained from D_n^ν in this way, then T_n has the law of a Galton-Watson tree with offspring distribution ν, conditioned on having n leaves.

The main observation that will allow us to control the diameter and two-point function of D_n^ν is that D_n^ν looks a lot like $\mathsf{Loop}(T_n)$, as pointed out in [CK14, Section 4.3]. We do not give the details, but Figure 3 suggests a natural way to define a correspondence between D_n^ν and $\mathsf{Loop}(T_n)$. This straightforwardly enables us to verify Assumption 1.4(V′). However, the control on $\mathsf{Diam}(T_n)$ and $\mathsf{Diam}_{res}(T_n)$ is not necessarily as strong as we'd like, with

$$\mathbf{P}\left(\mathsf{Diam}(D_n^\nu) \geq n^{\frac{1}{\beta}}\lambda \;\middle|\; l(T_n) = n\right) \asymp \mathbf{P}\left(\mathsf{Diam}(\mathsf{Loop}(T_n)) \geq n^{\frac{1}{\beta}}\lambda \;\middle|\; l(T_n) = n\right) \asymp \lambda^{-\beta}$$

in the case of stable tails, with $\mathsf{Diam}_{res}(T_n)$ experiencing the same asymptotics. Note that Assumption 1.1 is always satisfied, but Assumption 1.4 is only satisfied in the finite variance case when $\nu(k, \infty) = o(k^{\frac{-\alpha}{\alpha-1}})$. In this case $d = R = 2$, and $v = 1$, so we get the results in the table.

Using Theorem 7.5, we also give the *typical* displacement exponent for when ν has β-stable tails. In this case, we have that $d = R = \beta, v = 1$, so that $s_\alpha^d = s_\alpha^R = \beta(\alpha - 1)$, $f_\alpha^v = \alpha$, which give the claimed results.

8.5 Critical Erdös-Rényi

Motivated by the example of the IIC on the UIHPT, we can also consider a model where we insert a connected component of an Erdös-Rényi graph in the critical window, by which we mean the graph $G(n, p)$ such that $p = \frac{1}{n} + \frac{t}{n^{\frac{4}{3}}}$ for some $t > 0$ (see e.g. [Gol20, Section 2] for an introduction to this model and the critical window).

It is well-known that, at criticality, a connected component of $G(n, p)$ looks roughly like a critical Galton-Watson tree with an $O(1)$ number of "surplus" edges. Heuristically, this can be explained as follows: let $\mathcal{C}$

denote a connected component of $G(n,p)$, and let $v_0 \in \mathcal{C}$. We consider the "exploration tree" rooted at v_0, constructed as follows: first let v_0 be the root. Then consider all vertices connected to v_0 and let these form the next generation of the tree. The number of such vertices is $\mathsf{Binomial}(n-1,p)$; denote this number M_1. Then, given a vertex v_1 in generation one, we can repeat this process to find all the *new* neighbours of v_1, and define these to be the offspring of v_1: the number of offspring is therefore $\mathsf{Binomial}(n-1-M_1,p)$. We can repeat this process inductively to explore the cluster in a depth-first way: this will produce a spanning tree of the cluster, and as long as the total number of vertices explored remains small compared to n, it is fairly accurate to approximate the offspring distribution of this tree by a $\mathsf{Binomial}(n-1,p)$ distribution. At any stage, there is a small probability that a given vertex v also has some neighbours that correspond to vertices that have already been discovered, so that in order to reconstruct $\mathcal{C}$ from its spanning tree we must add a few extra edges.

To avoid ambiguities, for this construction we will fix $t>0$, set $p_n = \frac{1}{n^{\frac{3}{2}}} + \frac{t}{n^2}$ and let G_n have the law of the largest component of $G(n^{\frac{3}{2}}, p_n)$ conditioned on having n vertices (by [Ald97, Corollary 2], n is therefore on the natural scale to be the size of the largest cluster of $G(n^{\frac{3}{2}}, p_n)$). Using the tree viewpoint, we can relate the volume, two-point function and diameter of critical connected Erdös-Rényi graphs to give the following results.

Proposition 8.1. *Take G_n as above. Then there exist constants $c, c' \in (0,\infty)$ such that for all $n \geq 1, \lambda \geq 1$:*

(i) $\mathbf{P}(\mathsf{Diam}(G_n) \geq \lambda\sqrt{n}) \vee \mathbf{P}(\mathsf{Diam}_{res}(G_n) \geq \lambda\sqrt{n}) \leq e^{-c\lambda^2}$.

(ii) $\mathbf{P}\big(d^U(G_n) \geq \sqrt{n}\big) \wedge \mathbf{P}\big(R^U(G_n) \geq \sqrt{n}\big) \geq c > 0$.

(iii) $\mathbf{P}(\mu(G_n) \geq \lambda n) \leq Ce^{-c\lambda^{\frac{1}{3}}}$.

(iv) $\mathbf{P}(\mu(G_n) \geq n-1) = 1$.

Proof. We just sketch the proof. For part (i) with the graph distance, the result follows by repeating the proof of the height bound of [AB19, Theorem 1.1] (it does not quite follow directly since our exploration tree is not quite a critical Galton-Watson tree, but we are close enough that the proof still works, and being slightly subcritical is intuitively helpful for this bound anyway since this corresponds to more of a condensation regime). This also gives the resistance result since the resistance is upper bounded by the graph distance. For part (ii), we first condition on having zero surplus, which has strictly positive probability in the limit. In this case the exploration tree is again close to a critical Galton-Watson tree (its law is not tilted, since if the surplus is zero we do not have to count the options for where we can add extra edges), and the resistance is equal to the graph distance. The result then follows since the offspring distribution of the tree is close to $\mathsf{Poisson}(1)$, which corresponds to a uniform labelled Galton-Watson tree, and in this case the it is known that on rescaling by $\sqrt{n}$ the two-point function satisfies $\mathbf{P}\big(d^U(T_n) = \lfloor\lambda\sqrt{n}\rfloor\big) \sim \frac{\lambda}{\sqrt{2n}} e^{-\frac{\lambda^2}{2}}$ as $n \to \infty$ (e.g. see [FDP06]).

To control the volume (i.e. number of edges) we control the surplus. To do this, note that given a vertex v_k in the exploration tree, there can only be extra edges joining within the same generation, or to an adjacent generation (otherwise this disrupts the generation structure of the tree). Therefore, if v_k is a vertex of the tree and D_m denotes the m^{th} generation of the tree, we can introduce a $\mathsf{Binomial}(|D_{|v_k|}| + |D_{|v_k|-1}|, p_n)$ random variable which we denote S_{v_k}, and the total surplus is upper bounded by summing these over all vertices in the tree. Then, again after taking care of the necessary details that our tree is not quite a critical Galton-Watson tree, we have by [ABDJ13, Theorem 1.1] that $\mathbf{P}(\sup_m |D_m| \geq \sqrt{n}\lambda^p) \leq \lambda^{-2p}$. Then, $\mathbf{P}\Big(\mathsf{Binomial}(2n^{\frac{3}{2}}\lambda^p, \frac{1}{n^{\frac{3}{2}}} + \frac{t}{n^2}) \geq \lambda\Big) \leq ce^{-c\lambda^{1-p}}$. We take $p = \frac{1}{3}$. On the complement of these events, the surplus is less than λ, so $\mu(G_n) \leq n + \lambda$, which gives part (iii). □

We therefore deduce that $d = R = 2, v = 1$, and the fundamental exponents take the following values:

$$f^v_\alpha = \alpha, \qquad s^d_\alpha = s^R_\alpha = 2(\alpha - 1).$$

Rather than forcing all vertices of the inserted critical graph to be boundary vertices, we could also consider inserting an independent copy of $G(n^\beta)$ at a vertex of degree n, for some $\beta \geq 1$ and uniformly choosing n distinct vertices to be boundary vertices. In this case, it follows from Proposition 8.1 that $d = r = \frac{2}{\beta}$ and $v = \beta$, so that

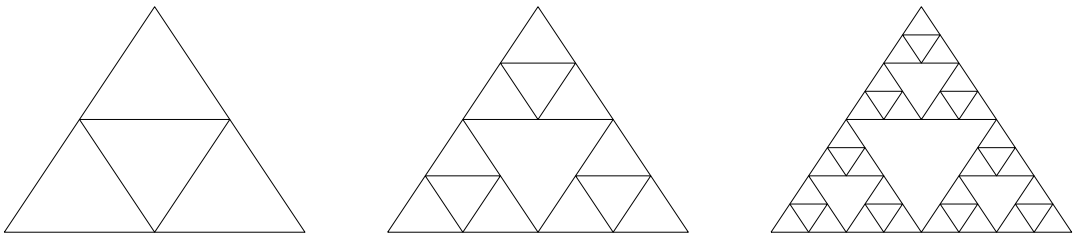
$$f^v_\alpha = \frac{\alpha}{\beta}, \qquad s^d_\alpha = s^R_\alpha = \frac{2(\alpha-1)}{\beta}.$$

Note that if $\beta > 2$, the local geometry always dominates and we never see the tree geometry.

Remark 8.2. *We have not written the details, but one would expect the same result on taking a critical configuration model in place of the Erdös-Rényi graph. We also anticipate that we could insert a β-stable graph, as considered in [GHS20, CKG20] and we would get the same results as for inserting β-stable trees by making similar arguments to the Erdös-Rényi example considered above.*

8.6 Sierpinski triangle

In order to gain insight into the effect of inserting fractal-type graphs, one could also consider the exponents obtained when inserting a Sierpinski triangle. Letting T^Δ_n be the n^{th} level approximation to the Sierpinski triangle as defined in [Bar98, Section 2] (also depicted in Figure 4), it is always the case that the boundary length of T^Δ_n is equal to $3 \cdot 2^n$: therefore, if $m \in (3 \cdot 2^n, 3 \cdot 2^{n+1})$, one would have to do appropriate "surgery" to the graph G_{n+1} in order to define an appropriate version of "the Sierpinski triangle with boundary length m". We will not do this is explicitly here, and just give the appropriate volume bounds for the level n approximation T^Δ_n.



(a) T^Δ_1 (b) T^Δ_2 (c) T^Δ_3

Figure 4: Sierpinski triangles

We can use the self-similarity of the Sierpinski triangle to study resistances, volumes and diameters of T^Δ_n as well, and give the (deterministic) results for these in Table 2. The effective resistance bound can be obtained using the $\Delta - Y$ transformation (e.g. [LP16, 2.3.III]): see also [Bar98, Section 2] for more explicit computations. Here we assume that 1 and 2 are the labels of two distinct extremal corners of T^Δ_n.

Boundary	**Volume**	$d(1,2)$	$R_{\mathbf{eff}}(1,2)$	**Diameter**	**Resistance diam**
$3 \cdot 2^n$	3^{n+1}	2^n	$\frac{2}{3}\left(\frac{5}{3}\right)^n$	$c_3 2^n$	$c_4\left(\frac{5}{3}\right)^n$
m	$c_1 m^{\frac{\log 3}{\log 2}}$	$\frac{1}{3}m$	$c_2 m^{\frac{\log 5 - \log 3}{\log 2}}$	$c'_3 m$	$c'_4 m^{\frac{\log 5 - \log 3}{\log 2}}$

Table 2: Volumes and distances in the n^{th} level Sierpinski gasket, in terms of m and n, where m is the boundary length.

We can also (crudely) bound the diameters using the bound for the distances between extremal corners in the table: since to go from any point $x \in T^\Delta_n$ to any other point $y \in T^\Delta_n$ we have to pass through at most two triangles at a specific level $m \leq n$ (once on the "way up" from x, then once on the "way down" to y), we get that

$$\mathsf{Diam}(T^\Delta_n) \leq 2\frac{2}{3}\sum_{m=0}^{n} 2^m \leq c_3 2^n, \qquad \mathsf{Diam}_{\mathrm{res}}(T^\Delta_n) \leq 2\frac{2}{3}\sum_{m=0}^{n}\left(\frac{5}{3}\right)^m \leq c_4\left(\frac{5}{3}\right)^n.$$

Since the graphs are deterministic, the functions giving the polynomial tail decay in Assumption 1.4 are all zero for sufficiently large λ, so we have that $d = 1$, $R = \frac{\log 2}{\log 5 - \log 3}, v = \frac{\log 3}{\log 2}$, and obtain the following exponents:

$$s_\alpha^d = \alpha - 1, \qquad s_\alpha^R = \frac{(\alpha-1)\log 2}{\log 5 - \log 3}, \qquad f_\alpha^v = \frac{\alpha \log 2}{\log 3},$$

which lead to the claimed results.

8.7 Supercritical Erdös-Rényi , the complete graph, or the 2-dimensional box

As well as critical Erdös-Rényi, one could also insert supercritical Erdös-Rényi graphs, as well as the complete graph. It is well-known that, if $G(n,p)$ is the Erdös-Rényi graph on n vertices and $p = \frac{\lambda}{n}$ for some $\lambda > 1$, then the largest connected component has order n vertices, order n^2 edges, and diameter of order $\log n$ (e.g. see [RW10, Theorem 1.1]). The complete graph on n vertices similarly has n vertices, order n^2 edges, diameter 1 and resistance diameter of order $\frac{1}{n}$.

To fit these models into the framework of this paper, we therefore effectively want to take $d = R = \infty$, and $v = 2$. Some care is needed to check that we can really do this, but we can dominate $\log n$ by n^δ for some sufficiently small δ, and this also gives us very good control on the tail decay required for Assumption 1.1 (D). Additionally, in sufficiently supercritical regimes resistance will actively stochastically decrease with n which is clearly different to the assumptions of this paper; however, since "most" vertices are of low degree it is clear that asymptotically resistance should still add up in the same way.

To define this model in the supercritical case, we let G_n be the largest connected component of $G(Cn,p)$, conditioned to have n vertices, where $C \geq 1$ is an appropriately chosen constant. Alternatively, we can let G_n be the complete graph on n vertices.

Setting $d = R = \infty$ (effectively) and $v = 2$ therefore gives the results in the penultimate line of Table 1.

The same intuition works on inserting a two-dimensional lattice box with side lengths $\frac{1}{4}n$: the volume of the box is of order n^2, the graph distance across the box of order n, and the resistance of order $\log n$, which can be stochastically dominated by n^δ for all $\delta > 0$. We effectively want to take $R = \infty, d = 1, v = 2$, so that $s_\alpha^d = \alpha - 1$, $s_\alpha^R = \infty$, $f_\alpha^v = \frac{\alpha}{2}$.

A Appendix: proof of results in Section 3

In this section we give proofs of the technical lemmas in Section 3. The first result is used in the proof of Lemma 3.1.

Lemma A.1. *Let $(X_i)_{i=1}^n$ be i.i.d. such that $\mathbb{P}(X_1 > x) \sim cx^{-\beta}$, for some $\beta \in (0,1]$, and let $T^{(k)} = \inf\{i \geq 1 : X_i > k\}$ (or equal to infinity if this set is empty). Set*

$$S^{(k)} = \sum_{i=1}^{T^{(k)}-1} X_i.$$

(i) If $\beta \in (0,1)$, then there exist $c, C \in (0,\infty)$ such that $\mathbb{P}\big(S^{(k)} \geq \lambda k\big) \leq Ce^{-c\lambda}$.

(ii) If $\beta = 1$, then there exist $c, C \in (0,\infty)$ such that $\mathbb{P}\big(S^{(k)} \geq \lambda k \log k\big) \leq Ce^{-c\lambda}$.

Proof. (i) The proof is essentially the same as the argument for a similar result on [CK14, p. 25]. Let $S_n = \sum_{i=1}^n X_i$, $H_0 = 0$, and $H_k = \inf\{n \geq 0 : S_n > k\}$ be the hitting time of k for the random walk $(S_n)_{n\geq 1}$. Then, if $S^{(k)} \geq k\lambda$ it must be the case that $H_{k\lambda} \leq T^{(k)}$. Therefore, for any $A \geq 1$ we can write

$$\mathbb{P}\Big(S^{(k)} \geq k\lambda\Big) = \mathbb{P}\Big(H_{kA\lambda} \leq T^{(k)}\Big)$$

$$\leq \mathbb{P}\Big(H_{2kA} \leq T^{(k)}, H_{4kA} \leq T^{(k)}, H_{6kA} \leq T^{(k)}, \ldots, H_{2\lfloor \frac{1}{2}\lambda \rfloor kA} \leq T^{(k)}\Big)$$
$$\leq \prod_{i=1}^{2\lfloor \frac{1}{2}\lambda \rfloor} \mathbb{P}\Big(H_{2ik} \leq T^{(k)} \;\Big|\; H_{2(i-1)k} \leq T^{(k)}\Big)$$

Since there are no jumps exceeding k before time $T^{(k)}$, it follows that $S_{H_{2iAk}} \leq (2iA+1)k$ for all $i \geq 1$. Moreover, since $T^{(k)}$ has a geometric distribution, it therefore follows from the memoryless property that for all $i \geq 1$:

$$\mathbb{P}\Big(H_{2iAk} \leq T^{(k)} \;\Big|\; H_{(2iA-1)k} \leq T^{(k)}\Big) \leq \mathbb{P}\Big(H_{2iAk} \leq T^{(k)} \;\Big|\; H_{2(iA-1)k} \leq T^{(k)}\Big) \leq \mathbb{P}\Big(H_{Ak} \leq T^{(k)}\Big).$$

Therefore, the exponential decay will follow once we can show that $\mathbb{P}\big(H_{Ak} \leq T^{(k)}\big)$ can be bounded below 1 uniformly in k. To show this, we use Wald's equality to write:

$$\mathbb{P}\Big(H_{Ak} \leq T^{(k)}\Big) = \mathbb{P}\Big(S^{(k)} \geq Ak\Big) \leq \mathbb{E}\Big[S^{(k)}\Big]\, k^{-1} \leq A^{-1} c_1 \mathbb{E}[X_1 \mid X_1 < k]\, k^{\beta} k^{-1} \leq A^{-1} c_2$$

for all $k \geq 1$, where c_1 and c_2 are explicit constants depending on the tails of X_1. We can therefore choose $A > 1$ so that the right hand side above is strictly less than 1.

(ii) If $\beta = 1$, we can use the same proof but we consider $\mathbb{P}\big(S^{(k)} \geq A\lambda k \log k\big)$ for sufficiently large A, and use that $\mathbb{E}[X_1 \mid X_1 < k] \leq c \log k$ in the final line.

□

We will also need the following lemmas for the infinite mean case. The result should be standard; however we couldn't find a specific proof in the literature, so have provided one for completeness.

Lemma A.2. *[Lemma 3.1]. Let $(X_i)_{i\geq 1}$ be i.i.d. and such that $\mathbb{P}(X_i \geq x) \sim cx^{-\beta}$ for some $\beta \leq 1$. Let $S_n = \sum_{i=1}^n X_i$.*

(i) If $\beta < 1$, then there exists a constant $c' < \infty$ such that for each $n \geq 2$,

$$\mathbb{P}\Big(S_n \geq n^{\frac{1}{\beta}}\lambda\Big) \leq c'\lambda^{-\beta}$$

as $n \to \infty$.

(ii) If $\beta = 1$, then there exists a constant $c' < \infty$ such that for each $n \geq 2$,

$$\mathbb{P}\Big(S_n \geq \lambda n^{\frac{1}{\beta}} \log n\Big) \leq c'\lambda^{-\beta}$$

as $n \to \infty$.

(iii) If $\beta > 1$, then for any $\varepsilon > 0$ there exists a constant $c' < \infty$ such that for each $n \geq 2$,

$$\mathbb{P}(S_n \geq \lambda n) \leq c'\lambda^{-(\beta-\varepsilon)}$$

as $n \to \infty$. This bound still holds if the $(X_i)_{i\geq 1}$ are not independent.

Proof. We first give the proof in case (i); the proof is the same in case (ii) and we just import an extra log term from the application of Lemma A.1. The proof is no doubt standard and this particular formulation follows a similar strategy to the analysis on [Dur10, Chapter 3, p.160]. Decompose S_n as the sum $\hat{S}_n + \tilde{S}_n$, where

$$\hat{S}_n = \sum_{i=1}^n X_i \mathbb{1}\{X_i \geq n^{\frac{1}{\beta}}\}, \qquad \tilde{S}_n = \sum_{i=1}^n X_i \mathbb{1}\{X_i < n^{\frac{1}{\beta}}\}.$$

We condition on the number of terms of $\hat{S}_n$, which is stochastically dominated by $\mathsf{Binomial}(n, 2cn^{-1})$ for large enough n.

Given that $N_n := |\{i \le n : X_i \ge n^{\frac{1}{\beta}}\}| = m$, $\tilde{S}_n$ can be dealt with using Lemma A.1: since we have $m+1$ copies of the sum considered there, it is necessary for one such copy to be at least $\frac{\lambda n^{\frac{1}{\beta}}}{2(m+1)}$ in order that $\tilde{S}_n \ge \frac{\lambda n^{\frac{1}{\beta}}}{2}$. Similarly, to control $\hat{S}_n$, note that it is necessary that at least one term of $\hat{S}_n$ is at least $\frac{\lambda n^{\frac{1}{\beta}}}{2m}$ in order that $\hat{S}_n \ge \frac{\lambda n^{\frac{1}{\beta}}}{2}$. Moreover, $\mathbb{P}\left(X_1 \ge \frac{\lambda n^{\frac{1}{\beta}}}{2m} \,\middle|\, X_1 \ge n^{\frac{1}{\beta}}\right) \le 4^\beta m^\beta \lambda^{-\beta}$ for all sufficiently large λ. We can therefore write:

$$\begin{aligned}
\mathbb{P}\left(S_n \ge \lambda n^{\frac{1}{\beta}}\right) &\le \mathbb{P}\left(\hat{S}_n \ge \frac{\lambda n^{\frac{1}{\beta}}}{2}\right) + \mathbb{P}\left(\tilde{S}_n \ge \frac{\lambda n^{\frac{1}{\beta}}}{2}\right) \\
&\le \sum_{m=0}^{n} \binom{n}{m} (2cn^{-1})^m (1-2cn^{-1})^{n-m} \\
&\qquad \times \left[\mathbb{P}\left(\hat{S}_n \ge \frac{\lambda n^{\frac{1}{\beta}}}{2} \,\middle|\, N_n = m\right) + \mathbb{P}\left(\tilde{S}_n \ge \frac{\lambda n^{\frac{1}{\beta}}}{2} \,\middle|\, N_n = m\right)\right] \\
&\le \sum_{m=0}^{n} \binom{n}{m} (2cn^{-1})^m (1-2cn^{-1})^{n-m} \left[m^{\beta+1} 4^\beta \lambda^{-\beta} + Cme^{-\frac{\lambda}{2(m+1)}}\right] \\
&\le \sum_{m=0}^{n} \frac{1}{m!} e^{-2c} \left(\frac{2c}{1-2cn^{-1}}\right)^m \left[Cm^{\beta+1}\lambda^{-\beta}\right] \\
&\le \lambda^{-\beta} \sum_{m=0}^{\infty} \frac{Cm^{\beta+1}}{m!} \left(\frac{2c}{1-2cn^{-1}}\right)^m \\
&\le C\lambda^{-\beta}.
\end{aligned}$$

In case (iii), we can apply Hölder's inequality with $p = \beta - \varepsilon$ to get that

$$\mathbb{P}\left(\sum_{i=1}^{n} X^{(i)} \ge \lambda n\right) \le \mathbb{E}\left[\left(\sum_{i=1}^{n} X^{(i)}\right)^p\right] n^{-p}\lambda^{-p} \le n^p \mathbb{E}\left[\left(X^{(1)}\right)^p\right] n^{-p}\lambda^{-p} \le c\lambda^{-(\beta-\varepsilon)}.$$

□

We also have a bound for the lower tails.

Lemma A.3. *[Lemma 3.2]. Let $(X_i)_{i\ge1}$ be i.i.d. and such that $\mathbb{P}(X_i \ge x) \sim cx^{-\beta}$ for some $\beta \le 1$. Let $S_n = \sum_{i=1}^n X_i$.*

(i) If $\beta < 1$, then there exists a constant $c > 0$ such that for each $n \ge 2$,

$$\mathbb{P}\left(S_n \le n^{\frac{1}{\beta}}\lambda^{-1}\right) \le e^{-c\lambda^\beta}$$

as $n \to \infty$.

(ii) If $\beta = 1$, then there exists a constant $c > 0$ such that for each $n \ge 2$,

$$\mathbb{P}\left(S_n \le \lambda^{-1} n \log n\right) \le e^{-c\lambda(\log\lambda)^{-1}}.$$

(iii) If $\beta > 1$, then there exists a constant $c > 0$ such that for each $n \ge 2$,

$$\mathbb{P}\left(S_n \le \lambda^{-1} n \log n\right) \le e^{-c\lambda}.$$

Proof. (i)

$$\mathbb{P}\left(S_n \le n^{\frac{1}{\beta}}\lambda^{-1}\right) \le c'\lambda^{-\beta} \le \mathbb{P}\left(\nexists i \le n : X_i \ge n^{\frac{1}{\beta}}\lambda^{-1}\right) \le (1-n^{-1}\lambda^\beta)^n \le e^{-\frac{1}{2}\lambda^\beta}$$

for all sufficiently large n.

(ii) Since by [Ber19, Equation (1.3)] there exists $A > 0, c < 1$ such that $\mathbb{P}(S_n \le A \log n) \to c$, and since $(X_i)_{i=1}^{\infty}$ are non-negative, we can write

$$\mathbb{P}\big(S_n \le \lambda^{-1} n \log n\big) \le \mathbb{P}\big(S_{A^{-1}n\lambda^{-1}\log\lambda} \le \lambda^{-1} n \log n\big)^{A\lambda(\log\lambda)^{-1}} \le c^{A\lambda(\log\lambda)^{-1}}.$$

(iii) Similarly to above, we have

$$\mathbb{P}\big(S_n \le \lambda^{-1} n\big) \le \mathbb{P}\big(S_{A^{-1}n\lambda^{-1}} \le \lambda^{-1} n\big)^{A\lambda} \le c^{A\lambda}.$$

□

Lemma A.4. *[Lemma 3.3]. Let X be a random variable, and suppose that $\mathbb{P}(X \ge x) \sim cx^{-\beta}$ as $x \to 0$ for some $\beta > 0$. Then there exists a constant $c \in (0, \infty)$ such that $\mathbb{E}\big[e^{-\theta X}\big] \sim c\theta^{-\beta}$ as $\theta \to \infty$.*

Proof. This is a standard Tauberian theorem, for example see [Ber96, p.10]. □

Lemma A.5. *[Lemma 3.4]. Let $\beta > 0$, let X be a random variable with $\mathbb{P}(X > x) \sim cx^{-\beta}$ as $x \to \infty$, and let f be a function of X such that there exists $z > 0$ and a function $p(\lambda)$ such that $\mathbb{P}(f(X) \ge \lambda n^z \mid X = n) \sim p(\lambda)$ for all sufficiently large n, where $p(\lambda) = o(\lambda^{\frac{-\beta}{z}})$ as $\lambda \to \infty$. Then, there exist $c, C \in (0, \infty)$ such that*

$$ck^{-\frac{1}{z}\beta} \le \mathbb{P}(f(X) \ge k) \le Ck^{-\frac{1}{z}\beta}$$

Proof. This is just a computation. (In this paper we mainly apply the result when X is a positive integer, but clearly the same proof works for continuous random variable). Since $O(\lambda^{\frac{-(\beta+\varepsilon)}{z}})$, we have for all sufficiently large k that

$$\begin{aligned}
\mathbb{P}(f(X) \ge k) \le \mathbb{P}\Big(X \ge k^{\frac{1}{z}}\Big) + \sum_{x=k}^{k^{\frac{1}{z}}} \mathbb{P}(X = x)\, \mathbb{P}\big(f(X) \ge kx^{-z}X^z\big) &\le ck^{-\frac{\beta}{z}} + \sum_{x=0}^{k^{\frac{1}{z}}} cx^{-(\beta+1)} p(kx^{-z}) \\
&\le ck^{-\frac{\beta}{z}} + \sum_{x=0}^{k^{\frac{1}{z}}} cx^{-(\beta+1)} k^{\frac{-(\beta+\varepsilon)}{z}} x^{\beta+\varepsilon} \\
&= c'k^{-\frac{\beta}{z}}.
\end{aligned}$$

The proof is the same when $m < \frac{\beta}{z}$:

$$\mathbb{P}(f(X) \ge k) \le \sum_{x=k}^{\infty} \mathbb{P}(X = x)\, \mathbb{P}\big(f(X) \ge kx^{-z}X^z\big) \le \sum_{x=0}^{\infty} cx^{-(\beta+1)} k^{-m} x^{zm} = c'k^{-m}.$$

The lower bound is even simpler: we write

$$\mathbb{P}(f(X) \ge k) \ge \mathbb{P}\Big(X \ge k^{\frac{1}{z}}\Big)\, \mathbb{P}\Big(f(X) \ge k \;\Big|\; X \ge k^{\frac{1}{z}}\Big) \ge ck^{\frac{-\beta}{z}} p(1).$$

If instead $m < \frac{\beta}{z}$:

$$\mathbb{P}(f(X) \ge k) \ge \mathbb{P}(X \ge 1)\, \mathbb{P}(f(X) \ge k \mid X \ge 1) \ge ck^{-m}.$$

(If $m = \frac{\beta}{z}$, we similarly acquire a logarithmic correction to the tail decay, but we won't need this case). □

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